One Advisor for the Whole World?

Cross-Country Evidence on Financial Advice from AI*

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Abstract

Households around the world increasingly take portfolio advice from the same handful of AI models. How does that advice vary across countries, and does the advice track local fundamentals? I pose different portfolio problems to leading models across twenty-one countries in different languages, varying whether the questions invoke local context. When asked about the risky share, the advice is nearly uniform across countries but varies widely across models. Explanations for the allocation advice consistently invoke local context, but this has almost no impact on the risky share. When asked whether to prepay a mortgage or how to protect savings from inflation, the advice differs sharply across countries. The models follow retail financial advice, and only selectively incorporate the household's balance sheet even when it is stated. While automated advice once held out the promise of tailoring guidance to everyone, AI models encode the same rule for everyone.

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1 Introduction

Retail investors are increasingly turning to artificial intelligence (AI) for financial advice, through chatbots built on large language models (LLMs) (OECD, 2026). The share of U.S. adults who use AI tools specifically for financial advice roughly doubled between 2024 and 2025 (Financial Health Network, 2025), and over half of U.K. adults now use such tools to help manage their money (Lloyds Banking Group, 2025). This represents a fundamental shift in the nature of financial advice for households all around the world, from national markets in which advice reflects local norms and institutions to a global market in which it comes from a handful of foundation models trained overwhelmingly on U.S.-English text. This shift raises several questions about the new global financial advisor: What financial advice do AI models give in different countries and do they tailor the advice to local circumstances?

To answer these questions, I use a pre-registered experiment in which the subjects are thirteen frontier AI models from six vendors and the tasks are three financial problems posed in twenty-one countries. My first finding on the new supply side of financial advice is that AI financial advice is largely the same everywhere for a simple portfolio choice question, but that questions requiring more local context generate more dispersion. Asked how to split the local-currency equivalent of $50,000 between a broad stock-market index fund and government bonds, the models give advice that differs only modestly across countries. Compared to the United States, every significant contrast is negative, ranging from 2.7 percentage points less equity in Mexico to 11 points less in Brazil. What instead moves the advice is the identity of the model, which accounts for 69.5 percent of the variance against 5.7 percent for country. Asked whether to invest savings or use them to prepay a 4 percent mortgage, the models recommend investing 10 to 29 percentage points less than in the United States in each of the fifteen countries where that rate is realistic. The largest gaps are in Japan, China, Switzerland, and Germany. Asked how to protect savings from 3 percent inflation, the models switch from equities to inflation-linked bonds in Brazil, Turkey, and Mexico, but overwhelmingly recommends equities in all other countries. On these two questions model identity explains only 15 and 26 percent of the variance. The pattern suggests that the models localize when the question depends on the local economy.

The cross-country differences in advice raises the question of what, if anything, the advice should track. Giving the same advice everywhere on the portfolio is not a mistake per se: under integrated markets the risky portfolio is the same world index everywhere. Of course, there are actually some differences across countries in the advice, so the models are not just giving exactly the same advice everywhere. As a simple (but not realistic) benchmark, a simple Merton benchmark with a currency-hedged world fund prescribes 0.52 in every country, within about six points of the advice everywhere. However, this model cannot explain why some countries receive less equity than the United States.[1] A more realistic

[1] Another variable that differ across countries and could affect portfolio choice is currency risk (Campbell and Viceira, 2002). While it is feasible that the models could take this into account, an updated benchmark with unhedged local-currency

benchmark would take local pension systems and background wealth into account, but as I show later, the models do not measurably adapt their advice based on these variables. An alternative approach to understanding what models track across countries is to relate cross-country differences to country-level fundamentals. However, these also seem to have limited explanatory power: country differences do not track measured risk preferences, financial literacy, local market returns or the prescriptions of a standard portfolio benchmark. I relate the estimated country fixed effects to country-level risk-taking from the Global Preferences Survey (Falk, Becker, Dohmen, Enke, Huffman and Sunde, 2018) and to financial literacy from the S&P Global FinLit Survey (Klapper, Lusardi and van Oudheusden, 2015), but find only a weak relationship with the average equity share a country receives.

Instead, the third finding is that AI models give the kind of "popular financial advice" studied by Canner, Mankiw and Weil (1997) and Choi (2022). The evidence from this comes from several different tests. First, I insert statements about the investor into otherwise identical English prompts for each country, covering risk tolerance, currency hedging, pension composition, pension generosity, the share of her liquid wealth that the stake represents, and investment horizon. Stated risk tolerance raises the recommended equity share by 46 percentage points, consistent with the dominance of risk tolerance in Ross and Lo (2026). A twenty-year rather than one-year horizon raises the equity share by 40 points, in contrast to the classic Merton–Samuelson irrelevance result where horizon should not affect the share at all under CRRA preferences with i.i.d. returns. Second, I ask the models to list the questions they would put to the investor before advising. In nearly every country, the model asks the same four things in the same order: risk tolerance, what the money is for, whether the five-year horizon is firm, and whether an emergency fund exists. Risk tolerance comes first in 84 percent of responses. These questions are reminiscent of popular financial advice, and it is striking that there is only neglible differences across countries. The models also neglect to ask about the household balance sheet, which would be highly relevant in e.g. life-cycle models.

I also conduct several checks that directly asks how the balance sheet affects the advice. A stated generous pension adds safe background wealth, so it should raise the equity share if the model integrates the investor's balance sheet. However, the advice barrely moves when information about pensions is added to the prompt. This is not because the models ignore pensions. Nearly every explanation mentions the investor's existing pension. In additional tests, added after the pre-registered results were examined, I find that the advice moves strongly and in the predicted direction when a stated balance-sheet cue raises the investor's exposure, as with a pension invested in stocks or a stake equal to her entire liquid wealth, but it barely moves on cues that signal safety, such as a guaranteed annuity, a currency hedge, or a cash cushion. The models appear to read the balance sheet as a source of hazard rather than as

moments prescribes equity shares from 0.21 in Turkey to 0.53 in South Korea, yet mean advice spans only 0.46 to 0.58. Later on, I also show that providing the models with information about currency hedging has a neglible impact on advice.

the two-sided input of portfolio theory. Overall, the advice fits a retail risk questionnaire, rather than advice that tracks the household's balance sheet.

The fourth and final finding is that the models appear to solve the portfolio problem in a shared, English-anchored representation regardless of the prompt's language, and then explain the answer in the local language and with local context. This is consistent with the interpretability literature, which finds that multilingual models solve tasks in a shared, English-leaning concept space (Wendler, Veselovsky, Monea and West, 2024; Tang, Luo, Huang, Zhang, Wang, Zhao, Wei and Wen, 2024; Wu, Yu, Yogatama, Lu and Kim, 2025; Lindsey, Gurnee, Ameisen, Chen, Pearce, Turner, Citro et al., 2025). One response in Swedish, a language I speak, describes a "glidväg" ('glide path'), a direct translation of the English industry term with no particular meaning in Swedish.[2] More formally, I machine-code the explanation that each model gives for its recommendation into four binary indicators. Local financial context appears in 18 percent of U.S. explanations but 29 to 78 percentage points more often in every other country. Naming local instruments in the prompt raises local references by 23 percentage points while moving the allocation itself by less than a point. The explanations thus localize far more than the advice. The translation changes the words of the answer but not the advice.

Summing up, this paper characterizes the supply side of a newly integrated market for AI financial advice. Financial advice was a nationally segmented service industry defined by national regulation, by relationships, and by language (for an overview, see Reuter and Schoar, 2024). AI models make this into a global market, where the advice is relatively similar and appears anchored on retail risk questionnaires. Instead of a national advice market, AI financial advice derives from a handful of foundation models. This is not necessarily a negative for the quality of financial advice, and it may well be that AI models are better advisors than humans. Human advisors already gave one-size-fits-all advice, with advisor fixed effects dominating client characteristics (Foerster, Linnainmaa, Melzer and Previtero, 2017). However, we may still want to know how much of the geographic differentiation in advice reflects economically relevant differences, and how much reflects local convention? What should the advice be for investors in different countries and under different circumstances?

Answering these questions is not trivial, and require us to take a normative stance of what the advice should be across countries. Although this is not the purpose of this study, standard finance theory and the results speak toward this question in part. First, the models differentiate across countries on decisions set by local prices and barely on the risky share. For the risky share, the same risky is possible to motivate with a (currency-hedged) world fund. The models are relatively close to this benchmark, with advice compressed toward a common level at or a few points below the U.S. baseline. The differences

[2]The formal term in Swedish is *generationsfond* or a reference to the default public AP7 fund. These terms do not appear often in the explanations, however: *generationsfond* appears zero times across roughly 5 million lines of response text, while the calque *glidväg* appears in 0.3 percent of Swedish responses.

are present in some countries, but do not track fundamentals well. For the mortgage question, for example, the local real rate decides the answer, and the models produce larger differences. However, it is not clear why the country constrasts are negative for all countries. Second, the advice follows a retail risk questionnaire rather than the household's balance sheet. This is potentially more problematic in countries where local circumstances differ more from the US, and where the advisor should be asking about local circumstances. Even if a human was to provide information about their pension, this still does not move the advice. Third, the models provide local explanations for their advice, even if the local explanations do not change the advice. For an investor in a country outside the US, it is difficult to see that the models are giving them the same advice as investors in the US because the explanation will localize.

2 Literature

There is a burgeoning literature on AI financial advice, which also relates closely to an older literature on robo-advising and traditional financial advice. Beginning with this newer literature, Choukhmane, de Silva, Lin and Akuzawa (2026) elicit prompts from a representative U.S. sample, simulate the life-cycle paths that following the advice would generate, and confront those paths with the prescriptions of a calibrated life-cycle model. They find that current models give moderately risk-averse advice that is broadly consistent with life-cycle theory, yet relies on round-number heuristics and varies with the gender and financial literacy of the person asking. Ross and Lo (2026) instead recover the advice rule from the mapping between inputs and outputs. They draw a thousand synthetic client profiles with near-orthogonal characteristics, collect allocations from a single vendor's model family, and fit interpretable surrogate models. They find what they call heuristic collapse: self-reported risk tolerance carries most of the predictive weight, while age, income, horizon, and liquidity contribute marginally.

This paper departs from these papers on three margins. First, both papers hold the investor's country fixed, so neither can ask whether the advice is calibrated to the institutions, asset markets, and preferences of the country in which the household lives. Second, Choukhmane et al. (2026) record which topics the advice text covers, whereas I code the explanation across countries to ask whether the reasons the model gives track local fundamentals. Third, I test a channel of risk-taking capacity, which is different from the willingness to take risk, by stating the generosity of the investor's pension in the prompt.

Beyond these two papers, this paper contributes to several different literatures. To the economics of robo-advising and algorithmic financial advice, I add evidence on whether the newest generation of algorithmic advisors localizes its recommendations. Earlier robo-advisors improve diversification and reduce behavioral biases (D'Acunto, Prabhala and Rossi, 2019), deliver substantial welfare gains (Rossi and Utkus, 2024), and more broadly move household behavior closer to normative prescriptions (D'Acunto

and Rossi, 2023). LLMs differ from these robo-advisors in two ways. They are general-purpose rather than domain-restricted,, and they reach essentially every language and country at once. These models therefore loosens the supply-side constraints that, together with demand, determine who accesses and benefits from advice (Reuter and Schoar, 2024). Any country-specific bias in their training data then propagates globally.

To the literature on the quality of retail financial advice, this paper adds evidence on what kind of advisor AI is. Canner et al. (1997) show that popular advisors recommend a higher ratio of bonds to stocks for more conservative investors, contrary to the mutual-fund separation theorem, and Choi (2022) documents how popular personal financial advice departs from the prescriptions of economic theory. Human advisors also give largely one-size-fits-all advice, with advisor fixed effects explaining more of the variation in client portfolios than client characteristics do (Foerster et al., 2017). AI models reproduce both patterns across the twenty-one countries I study. Model identity explains most of the variance in the recommended equity share and the country explains little, much as advisor identity outweighs client characteristics in human advice.

To the literature on cross-country differences in household financial behavior, this paper adds a new margin: the advice households receive from AI models, as distinct from the choices they make. Equity-market participation varies substantially across countries, and income, wealth, and institutional access do not fully explain the variation (Guiso and Sodini, 2013; Christelis, Georgarakos and Haliassos, 2013). The Global Preferences Survey (Falk et al., 2018) provides validated country-level measures of preferences that correlate with savings and investment behavior, and financial literacy strongly predicts equity-market participation (van Rooij, Lusardi and Alessie, 2011). Advice tailored to each country's clientele would vary with these measures. I find only a weak relationship between either measure and the equity share that AI models recommend in a country.

To the literature on demographic bias in machine-learning models and LLMs (Bolukbasi, Chang, Zou, Saligrama and Kalai, 2016; Navigli, Conia and Ross, 2023), this paper adds both a new treatment and a decomposition. In credit markets, machine-learning scoring changes the distribution of mortgage approvals across racial groups (Fuster, Goldsmith-Pinkham, Ramadorai and Walther, 2022). In LLM advice, Foltyn and Olsson (2026) document gender gaps on a stock-bond allocation question that this paper adapts as its primary outcome. I keep the structure of their question and replace demographic identity with country and language as the treatment. This dimension has first-order policy relevance because AI advisory tools reach investors across borders while pension and tax systems remain national.

To the literature on how multilingual language models process and retrieve knowledge, this paper adds a behavioral test on an economic task. Interpretability studies find that the middle layers of multilingual models operate in a shared concept space that lies closer to English than to any other language (Wendler

et al., 2024; Tang et al., 2024; Wu et al., 2025; Lindsey et al., 2025). Knowledge-probing studies instead find that the same model often asserts different facts depending on the query language (Kassner, Dufter and Schütze, 2021; Qi, Fernández and Bisazza, 2023). These two findings make different predictions for a translated financial question. Shared processing predicts nearly the same advice in every language, whereas language-keyed retrieval predicts advice that inherits the conventions of each language's financial text. Asking in the local language rather than in English lowers the recommended equity share by only 1.3 percentage points, which supports shared processing, and the local detail the models add appears in the explanation rather than in the allocation. The questions that the models would like to ask investors also varies very little across countries, which again supports shared processing.

3 Experimental Design

The experiment has two arms. The main arm poses three financial questions to thirteen frontier LLMs in twenty-one countries and varies the language of each question, the country it refers to, and the local financial context it provides. The stated-investor arm varies the investor rather than the country, by stating her risk tolerance, pension generosity, or investment horizon in an otherwise fixed English prompt. I first describe the three questions and the country selection, then the prompt versions and the contrasts they identify, the translation workflow, the stated-investor arm, the models and API parameters, and the data collection. Appendix Table A1 lists the prompt language, the local-currency investment amount, and the full Version C vocabulary for every country, and Appendix E documents the technical implementation of the data collection.

3.1 The Three Questions

Each LLM answers three financial question: a primary portfolio-allocation question and two secondary questions on mortgage prepayment and inflation protection. Question 1 asks how to split the local-currency equivalent of $50,000 between a broad stock-market index fund and government bonds over a five-year horizon. The outcome is the fraction allocated to equities.

Question 2 asks whether to invest savings in a stock-market index fund or to use them to pay down an existing 4 percent mortgage with twenty years remaining. The outcome is the fraction allocated to the index fund. This question has a sharper normative benchmark than Question 1, because mortgage prepayment is a risk-free investment whose return is pinned down by the contract rate. I use a a pre-specified rule that assigns any country whose prevailing mortgage rate exceeds twice the prompted 4 percent to a premise-acceptance group. In these countries the question also tests whether the models flag the 4 percent rate as unusual for the local environment or simply condition on it. In the end, all models conditioned on it.

Question 3 asks how to protect the real value of $50,000 sitting in a savings account that earns 1 percent nominal while annual inflation is 3 percent. The outcome has two parts: a numeric fraction to invest and a categorical `primary instrument` (equities, inflation-linked bonds, real estate, gold, or other). Because the stated 3 percent inflation rate is far below prevailing inflation in some countries, Question 3 was ex ante the question most likely to lead the models to reject the premise and withhold a numerical answer. A pre-specified contingency therefore promotes the refusal indicator to the primary Question 3 outcome if the refusal rate exceeds 10 percent in any country–model cell. The refusal rate was zero across 37,021 model calls in 273 country × model cells, so the contingency never applied. Several models noted that the inflation rate was unrealistically low for the country, most often for Turkey, but none refused to answer.

3.2 Country Selection

I chose the country set to span meaningful cross-country variation in pension generosity, equity-market development, inflation history, and measured preferences, while keeping translation risk low. I imposed one screening constraint. The preference regressions of Section 5.2 require each country's measured risk preferences, so the Global Preferences Survey (Falk et al., 2018) must cover every country. This requirement excludes otherwise natural candidates such as Denmark.

The United States serves as the baseline for the country contrasts. I prompt five further countries in English (Australia, Canada, India, South Africa, and the United Kingdom). They add observations at essentially no translation risk while spanning very different pension institutions, from compulsory superannuation to a replacement rate among the lowest in the panel.

The fifteen non-English countries include high-literacy European systems with structurally different occupational pensions (Sweden, Germany, the Netherlands, Switzerland) and further European variation (France, Spain, Italy, Poland, Finland). Outside Europe they cover the Asian markets (China, Japan, South Korea), several high-inflation economies (Brazil, Turkey), and Mexico.

3.3 Prompt Structure and Version Design

Each prompt has two blocks. An Instructions block, written in English in every cell, asks for a JSON response that contains the recommendation and a word-limited explanation. A Question block describes the investor and poses the financial question, and the treatments change only this block. Keeping the instructions in English holds response-format compliance constant across languages. Section 4.6 tests this choice directly by re-collecting the primary cell with the instructions translated as well.

The Question block varies along three dimensions. The first is the country and its prompt language. I prompt six countries in English (the United States, the United Kingdom, Australia, Canada, India, and

South Africa) and the fifteen others in their local language, with Switzerland in German and Mexico in Spanish (Appendix Table A1 lists each pairing). The second is the investor's stated age (30, 40, or 55), which varies for the primary equity-share question and is fixed at 40 for the two secondary questions.

The third dimension is the prompt version. I use three versions of the prompt, described in detail below: the first version uses the local language with little or no local context, the second version uses English but states the investor location, and the third includes local-context in the form of pension and fund names. I use these different prompts to decompose the country effect into language, country-naming, and local-framing components.[3]

More specifically, *Version B* states "I live in [country]" but uses a generic instrument vocabulary ("a broad stock market index fund," "government bonds," "a standard pension plan"). In the fifteen non-English countries it appears in the local language. Its investment amount is the round local-currency equivalent of $50,000 (e.g. 500,000 kronor in Sweden, 45,000 euros in Germany, 40,000 pounds in the United Kingdom), written in the native local number format ("500 000 kr," "45.000 Euro"). *Version B-cross* retains the country reference ("I live in [country]") and the local-currency investment amount of Version B but reverts the question to English. It is collected only for the fifteen non-English countries. Note that the baseline prompts does not adjust for PPP or income. In a robustness check I also adjust the stake by PPP, which does not alter the result.

Version C keeps the language, country reference, and investment amount of Version B but replaces the generic financial vocabulary with country-specific equivalents: local index funds (e.g. an index fund via Avanza or Nordnet in Sweden, eMAXIS Slim All-Country Equity in Japan), local government bonds (*Svenska statsobligationer*, JGBs, Tesouro Direto), and local pension references (*tjänstepension*, bAV, PER, iDeCo).

Three contrasts identify the components of the country effect. The cross-country comparison of Version B identifies the joint country-and-language effect relative to the U.S. baseline. The within-country comparison of Versions B-cross and B holds the country reference fixed and changes only the language of the question, which identifies the language effect. The within-country comparison of Versions C and B holds both country and language fixed and changes only whether the prompt names local instruments and pensions, which identifies the local-framing effect.

[3]The version lettering follows the pre-registration. Version A denoted a translated prompt that named no country. I dropped it before collection because it could not separate the effect of the question's language from the style of any particular translation, and because the comparison of Versions B and B-cross identifies the language effect more cleanly by holding the country reference fixed. I keep the labels of the surviving versions unchanged so that the paper matches the registered design.

3.4 Translation Workflow

I produced the Version B and Version C question blocks for the fifteen non-English countries through an automated pipeline that translates only the Question block and keeps the protected financial terms and currency amounts verbatim. Two automated checks screened the machine output, a term-protection check and a full back-translation. I then had native-speaker review each version and provide feedback in all fifteen translated countries. In the English-prompting countries, a two-vendor LLM cross-check verified the local-instrument wording of Version C. Before I filed the pre-registration, every assembled prompt passed a consistency audit with zero failures, an adversarial LLM-based review covered all ninety unique translated blocks, and I read every block in full. I then froze the translations for collection. Appendix E.2 documents the pipeline mechanics, the two automated checks, and the review-and-correction protocol.

3.5 The Stated-Investor Arm

The second arm varies the investor herself instead of the country or language, but repeats the exercise for every country. The stated-investor arm takes the English Question 1 prompt at age 40 and inserts statements about the investor. The resulting prompts are otherwise identical to Version B in the six English-prompting countries and to Version B-cross in the fifteen others, with the same local-currency amounts. I began with three pre-registered checks, and added three more three exploratory probes afterwards. I now describe these probes.

The main pre-registered arm has six treatment cells, related to risk-tolerance, pension generosity and the investment horizon. The two risk-tolerance statements and the two pension statements are crossed, which gives four cells, and two horizon cells stand alone. The risk-tolerance statements are length-matched qualitative sentences: "I am comfortable with substantial fluctuations in my portfolio value, including potential short-term losses, if it means higher expected returns over time" versus "I am uncomfortable with large fluctuations in my portfolio value and would prefer to limit potential losses, even if it means lower expected returns." The pension statements are "I will receive a substantial pension when I retire, projected to replace most of my pre-retirement income" versus "I will receive only a small pension when I retire, covering a modest fraction of my pre-retirement income." The horizon cells replace the phrase "over a five-year horizon" with a one-year or a twenty-year horizon and insert no other statement. I pre-registered three hypotheses as secondary confirmatory tests. Each is a single pre-specified coefficient, reported without multiplicity correction and with the same wild-cluster-bootstrap inference as the primary regressions.

After examining the pre-registered results, I also added three exploratory probes to the arm. Each is a single English sentence in the same statement slot that states one part of the investor's balance sheet.

Specifically, I add a a pension-composition probe, which states either "My pension is a guaranteed, inflation-linked annuity that will pay a fixed real income in retirement regardless of how financial markets perform" or "My pension is invested almost entirely in stocks, so its eventual value will rise and fall with equity-market performance." I also add a hedging probe, which states either "The stock fund I am considering is currency-hedged to my home currency, so its returns are not exposed to exchange-rate movements" or "The stock fund I am considering is a global fund with no currency hedging, so its returns also move with exchange rates." Finally, I also added an arm that varies how much of the investors' wealth the stake represents, moving from all, about half, or about a third of the investor's liquid wealth. I first collected all seven probe cells for the United States, at fifty repetitions per model and cell (4,550 queries), and then extended the composition and hedging probes to the other twenty countries at twenty-five repetitions (26,000 queries). Both collections ran in August 2026. The estimates use the no-statement Version B and B-cross cells of the July collection as the reference.

3.6 Models, Repetitions, and API Parameters

The experiment queries thirteen frontier LLMs from six vendors through the OpenRouter unified API. Four vendors are U.S.-headquartered: Anthropic (Claude Haiku 4.5, Claude Sonnet 4.6, Claude Opus 4.6, Claude Opus 4.8), OpenAI (GPT-5-mini, GPT-5.5), Google (Gemini 3 Flash, Gemini 3.5 Flash, Gemini 3.1 Pro), and xAI (Grok 4.3). So that the panel does not consist solely of U.S.-headquartered firms, I add DeepSeek (DeepSeek V4 Flash, DeepSeek V4 Pro), a non-U.S. vendor whose models are primarily trained on Chinese-language data, and Mistral (Mistral Medium 3.5), a European (French) vendor.

A cell is one combination of country, version, question, age, and model. I query each cell fifty times at temperature 0.7, with no system prompt and a maximum response length of 4,000 tokens. Note that the unit of inference is the model, which means that fifty repetitions suffice. Additional repetitions would reduce only the within-cell sampling term, which is already small relative to between-model heterogeneity.

In total the design comprises 267,150 queries, of which 185,250 form the country/language arm and 81,900 the stated-investor arm. Appendix Table A2 gives the accounting by arm, version, and question. The post-filing balance-sheet probes of Section 6.1 add a further 30,550 queries, which I collected into a separate corpus.

3.7 Data Collection and Parsing

Collection proceeded in two stages, an exploratory pilot followed by the pre-registered main collection. A staged pilot of roughly 6,800 queries, completed in late June 2026, exercised the full pipeline end

to end.[4] I filed a pre-registration at the Open Science Framework (OSF) on July 2, 2026, with the prompts frozen. The main collection submitted all 267,150 queries through the OpenRouter API on July 3–4, 2026. The two pre-registered collection-robustness cells—temperature bookends at 0 and 1.0 and a Version B variant that states the investment amount in U.S. dollars—added 8,450 queries on July 6, 2026.

After the main collection, I added five exploratory collections, none of which is part of the pre-registration: the three balance-sheet probes of the stated-investor arm (August 2026, Section 6.1), a model-tracking wave with a panel of twenty-one models (September 2026), and three re-collections of the primary cell in October 2026 that translate the Instructions block, let the model ask questions before advising, and restate the stake at purchasing-power parity (Section 4.6).

The main run covered the full design grid, with every cell reaching its fifty repetitions and no duplicates or unresolved API errors. The parser recovered a numeric recommendation from 99.9 percent of responses, the collection produced no refusals, and no model changed versions during the run. I exclude the unparsed responses and, under the pre-registered truncation rule, those that exhausted the 4,000-token budget. Appendix E.4 documents the parser, the refusal rule, and the model-identity checks.

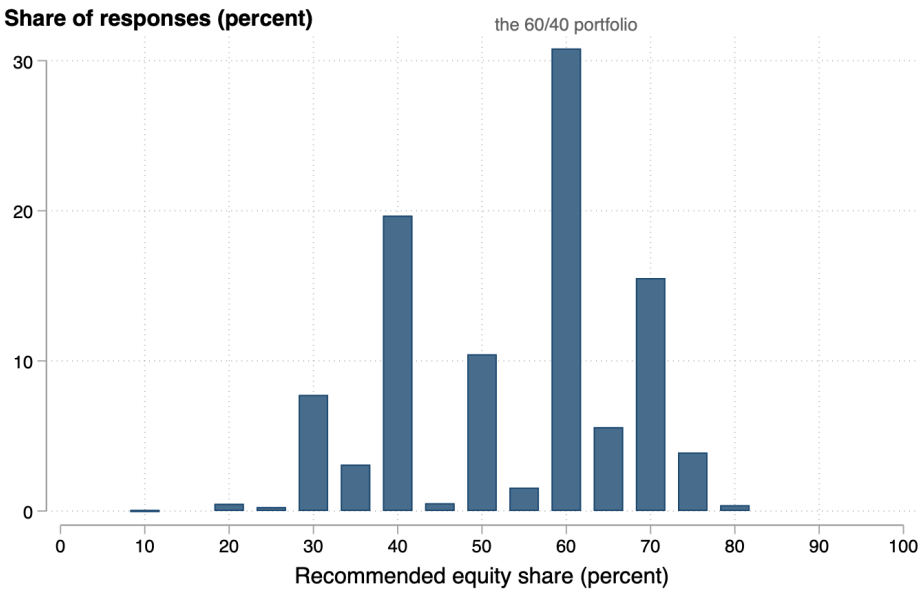
4 Main Results

This section reports how LLM advice varies across countries. I first document the raw patterns in the advice (Section 4.1), then estimates the country effects on the primary question (Section 4.2) and decomposes them into language and local framing (Section 4.3). I then turn to age, the mortgage and inflation questions, and robustness. Except for the two secondary questions in Section 4.5, the outcome is the recommended equity fraction on Question 1. Every regression includes model fixed effects, so each contrast compares the advice of a single model, and within each comparison the queries pose the same investment problem. The coefficients therefore measure the causal effect of the prompt's country, language, and framing content on the advice. I cluster standard errors at the model level.[5] As pre-registered, the twenty primary country contrasts carry a Holm step-down correction as a single family. The reason for the Holm step-down procedure is that I test 20 countries against the United States. With twenty contrasts each tested at the 5 percent level, about one would appear significant by chance even if advice were identical in every country. The correction keeps the probability of any false rejection among the twenty at 5 percent. Appendix Table A16 lists each pre-registered hypothesis with its test and outcome.

[4] I made filing the pre-registration conditional on the pilot passing three pre-specified success criteria: a JSON parse rate of at least 95 percent per model, refusal rates below pre-specified ceilings in every country–model cell, and a between-model dispersion of the country contrast no larger than 0.06. The pilot passed all three, with a parse rate near 99.6 percent and no refusals, and it produced no truncated responses.

[5] Because thirteen clusters are too few for reliable asymptotic inference, every reported p-value comes from a wild cluster bootstrap with Webb six-point weights and 9,999 replications.

Figure 1: Distribution of recommended equity shares in the primary cell



Notes: Share of responses at each recommended equity share on the primary cell (Question 1, Version B, age 40, N = 13,639), pooled across the twenty-one countries and thirteen models. Every response is a multiple of five percentage points. No response falls below 10 or above 80 percent.

4.1 Descriptive Patterns

I begin with three descriptive facts about the advice that organize the results that follow. The models bunch their recommendations at a few round numbers within a narrow range, mean advice varies little across countries, and the identity of the model matters far more than the country.

Starting with round numbers in the recommendations, Figure 1 plots the distribution of the recommended equity share, pooled across countries and models. Every response is a multiple of five percentage points, and 85 percent are multiples of ten. The modal recommendation is 60 percent equity, given in 31 percent of responses. In addition, the support is narrow. No response in 13,639 allocates less than 10 or more than 80 percent to equities, so the advice never reaches the 100 percent equity share that the benchmark with pension wealth and human capital prescribes (Section 5.1).

The bunching shows that the models answer a continuous allocation problem with a conventional round number. Choukhmane et al. (2026) also find round number bunching in AI advice, which they note departs from the more subtle prescriptions of life cycle theory. A portfolio rule would produce a continuous answer for at least some of the answers. Instead, the modal 60/40 split is the most common target allocation in retail portfolio advice, a first sign of the risk-questionnaire interpretation. A 100-minus-age rule of thumb would produce the same 60 percent at age 40. The model explanations occasionally include this rule in the explanations, while also sometimes referring to a "110-age" rule, which would also produce a (different) round number. However, the models do not follow it literally at the other prompted

Table 1: Summary statistics by country on the primary cell

Country	Mean	SD	p10	Median	p90	*N*
Brazil	0.46	0.14	0.30	0.40	0.70	650
Italy	0.48	0.14	0.30	0.50	0.70	650
Poland	0.50	0.14	0.30	0.50	0.70	650
France	0.51	0.14	0.30	0.60	0.70	649
China	0.52	0.10	0.40	0.50	0.65	649
Switzerland	0.53	0.14	0.30	0.60	0.70	649
Spain	0.53	0.12	0.40	0.60	0.70	648
Netherlands	0.53	0.14	0.30	0.60	0.70	649
Germany	0.53	0.14	0.30	0.60	0.70	650
Turkey	0.54	0.11	0.40	0.60	0.65	650
Mexico	0.54	0.14	0.30	0.60	0.70	650
Finland	0.55	0.15	0.35	0.60	0.75	648
India	0.55	0.12	0.40	0.60	0.70	650
Canada	0.56	0.13	0.40	0.60	0.70	650
South Africa	0.56	0.12	0.40	0.60	0.70	648
Sweden	0.57	0.13	0.40	0.50	0.75	649
United States	0.57	0.13	0.40	0.60	0.70	650
United Kingdom	0.58	0.13	0.40	0.60	0.75	650
Australia	0.58	0.12	0.40	0.60	0.70	650
Japan	0.58	0.14	0.35	0.60	0.70	650
South Korea	0.58	0.11	0.40	0.60	0.70	650

Notes: Distribution of the recommended equity share by country on the primary cell (Question 1, Version B, age 40), pooled across the thirteen models with fifty repetitions per country–model cell. Countries a re ordered by mean. *N* falls slightly below 650 where responses without a parsed numeric recommendation, or truncated at the token limit, are excluded under the pre-registered rules (Section 3.7).

ages. At age 30 the modal response stays at 60 percent rather than the rule's 70, and at age 55 it falls to 40 percent rather than the rule's 45. I explore the rule in more detail in Figure 4.

Mean advice varies little across countries. Table 1 shows that it spans twelve percentage points, from 0.46 in Brazil to 0.58 in South Korea, with the United States at 0.57 near the top of the range. The band is narrow relative to the dispersion within any single country. The typical within-country standard deviation is 0.13. Instead, most of the within-country dispersion comes from the models. Appendix Table A3 reports mean advice by model, which spans 0.37 (Gemini 3.1 Pro) to 0.71 (Claude Haiku 4.5), a range of 34 percentage points. That is nearly three times the country range. The ordering of the models follows neither vendor nor scale. The four Anthropic models cluster above the pooled mean, between 0.58 and 0.71, but Google's models span nearly the full panel range. Google's flagship reasoning model is the most conservative advisor in the panel, but its two Flash-tier models are twenty points higher.

Model identity accounts for 69.5 percent of the total variation in the primary cell, the country for 5.7 percent, their interaction for 9.5 percent, and sampling variation within country–model cells for the remaining 15.3 percent. The dominance of the model motivates two choices maintained throughout the paper. Every regression identifies country contrasts within model, and inference clusters at the model

level, treating the thirteen models rather than the quarter-million responses as the effective sample.

4.2 Country effects on the primary question

I now test whether LLM advice is the same across countries. The primary specification estimates country fixed effects on the pre-registered primary cell, Version B of Question 1 at age 40,

$$y_{imc} = \alpha_m + \gamma_c + \varepsilon_{imc}, \tag{1}$$

where y_{imc} is the equity fraction recommended in repetition i by model m for country c, the α_m are model fixed effects, and the γ_c are deviations from the omitted U.S. baseline.

The main finding is that advice is not the same everywhere. Figure 2 plots all twenty contrasts.[6] Thirteen of the twenty contrasts clear the wild-bootstrap threshold before correction, and seven survive the Holm step-down. Every contrast that survives the correction is negative, and the survivors lie in southern and central Europe and Latin America. Against a U.S. mean equity share of 0.57, the largest gap, Brazil's eleven percentage points, moves \$5,500 of the \$50,000 stake from stocks to bonds. The seven significant contrasts average about six points. For comparison, the gender gap that Foltyn and Olsson (2026) document on a similar allocation question is under two percentage points, so country context moves LLM advice by several times the demographic gaps documented so far.

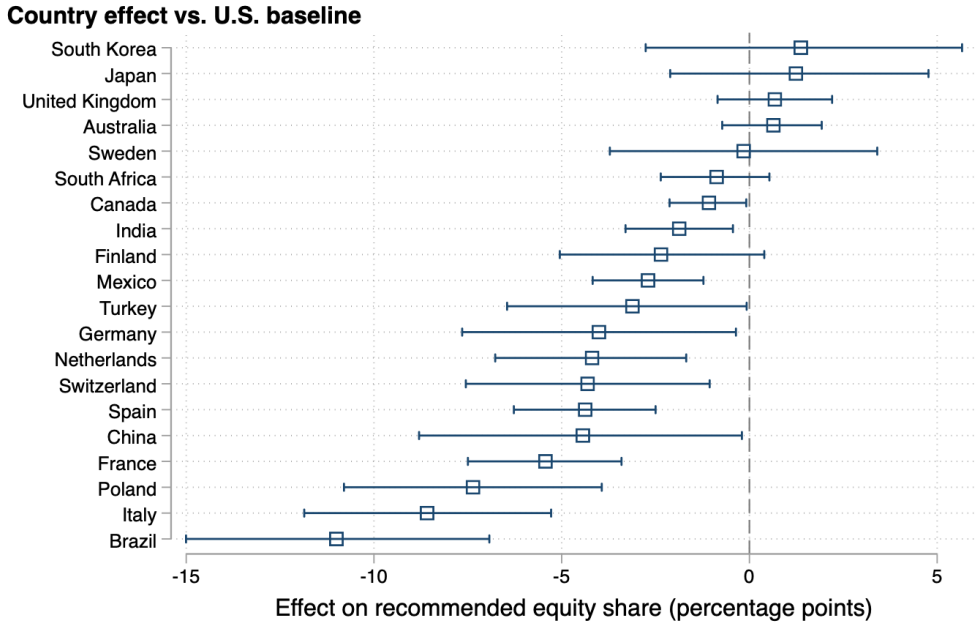
Before the Holm correction, thirteen of the twenty contrasts fail to reject equality with the U.S. baseline. Six of those are individually significant on the uncorrected wild-bootstrap test but do not survive the step-down (China, Switzerland, Germany, Turkey, India, and Canada, ranging from -4.4 to -1.1 points). The remaining seven do not reject even before correction. Four of them have small positive point estimates (Australia, Japan, South Korea, and the United Kingdom). Finland and South Africa have small negative gaps, and Sweden sits at the U.S. level. The design's minimum detectable effect for a single country contrast is roughly four percentage points, so these nulls rule out large gaps but not gaps of a point or two. However, gaps of those magnitudes would not be economically significant, as they would move only moderate amounts from stocks to bonds.

4.3 Language and local framing

I next run two within-country tests. One asks whether local framing shapes the advice, and the other separates the effect of the question's language from that of the country name. The local-framing regression

[6] Appendix Table A9 reports each with its wild-bootstrap and Holm p-values. The joint wild-bootstrap test rejects equality of advice across the twenty-one countries with $p = 0.006$. The joint wild-bootstrap Wald statistic on all twenty restrictions is $F(20, 12) = 20.56$. The asymptotic cluster-robust test can impose at most twelve of the twenty restrictions with thirteen clusters and gives $F(12, 12) = 34.26$. Both reject at the one percent level.

Figure 2: Country effects on the recommended equity share



Notes: Point estimates of the country fixed effects from equation (1) (Question 1, Version B, age 40), in percentage points relative to the omitted U.S. baseline. Bars are 95 percent wild cluster bootstrap confidence intervals (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$), the intervals of record, matching Appendix Table A9.

compares Versions C and B across the twenty non-U.S. countries at age 40,

$$y_{imc} = \alpha_m + \gamma_c + \phi \, \mathbf{1}[\text{version} = C] + \varepsilon_{imc}, \tag{2}$$

and the language regression compares Versions B and B-cross across the fifteen non-English countries,

$$y_{imc} = \alpha_m + \gamma_c + \lambda \, \mathbf{1}[\text{language} = \text{local}] + \varepsilon_{imc}. \tag{3}$$

Asking in the local language rather than in English lowers the recommended equity share by 1.3 percentage points. The language effect is statistically significant but economically small, an order of magnitude below the largest country effects. Naming the local instruments does even less. Replacing the generic vocabulary with local pension names, local government bonds, and local index funds moves the allocation by -0.7 points, a result that is not significance at the five percent level.

The two estimates imply that the country name carries most of each country effect. Language contributes about a point and local framing less than one, so the bulk of every large country effect remains with the country reference, which every version contains. Both small effects are negative, the same direction as the significant country contrasts, so each added localization cue nudges the advice toward bonds.

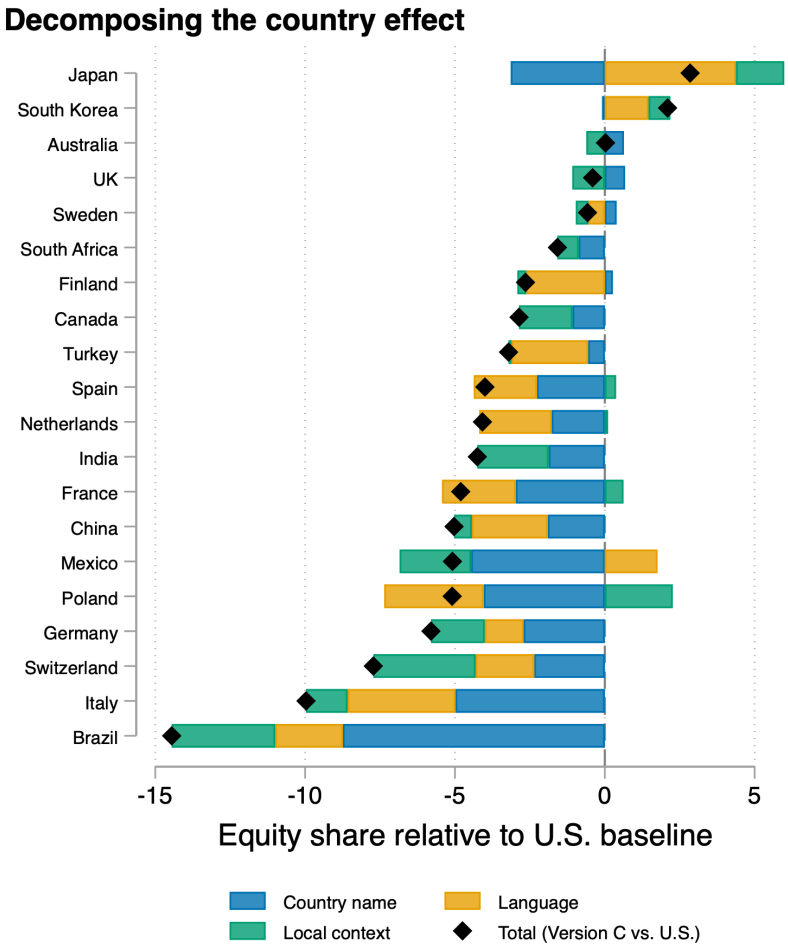
Figure 3 applies the same decomposition country by country. Each bar stacks three steps into the

gap between a country's fully localized Version C and the U.S. English generic cell, and the three add up to that gap by construction. The country-name step is Version B-cross minus the United States, the language step is Version B minus B-cross, and the local-context step is Version C minus B. The country-name step also changes the currency unit and the nominal amount, which move together with the country reference. In the large-effect countries (Brazil, Mexico, Poland, France, Italy, Germany) the country-name slab accounts for most of the gap. The language and local-context slabs are usually small, but they are noisier than the pooled $\hat{\lambda}$ and $\hat{\phi}$ and not uniformly negative. Japan is the clearest exception. A large positive language slab pushes its advice toward stocks and produces the panel's largest positive total.

The language treatment covers only the Question block, since the Instructions block is English in every cell. I also added anexploratory re-collection with the Instructions block translated as well, reported in Section 4.6. Sending the message entirely in the local language moves the non-English cells by less than a point relative to the English ones. The small langauge effect is then likely not due to the English instructions.

The size of the language effect distinguishes between two views of how multilingual models answer a question. If the models retrieve advice the way they retrieve language-keyed knowledge, λ should be substantial. The same model often asserts different facts depending on the query language (Kassner et al., 2021; Qi et al., 2023). For example, if German queries were to draw on the bond-heavy conventions of German-language financial text, the effect of the language would presumably be large. If instead the models solve the task in a shared representation, λ should be near zero. Tracing studies in computer science find that language-specific processing occupies the layers nearest the input and output, that the middle layers solving the task operate in a shared concept space lying closer to English than to any other language, and that cross-lingual sharing rises with model scale (Wendler et al., 2024; Tang et al., 2024; Wu et al., 2025; Lindsey et al., 2025). Under such processing, two translations of the same question become nearly the same query by the middle of the network. The estimate of -1.3 points fits the second account, with the small residual consistent with the language-specific features that remain. The advice behaves like one policy applied across query languages rather than like language-segmented retrieval. Note also that the sharing should be strongest in frontier models like those in this panel. Speculatively, this suggests that small langauge effects is a feature of how the models solve problems generally, and not an artifact of how recent the model is. I later run a robustness with newer model generations, but find similar results for the cross-country variation in advice also for newer models.

Figure 3: Decomposing the country effect into naming, language, and local context



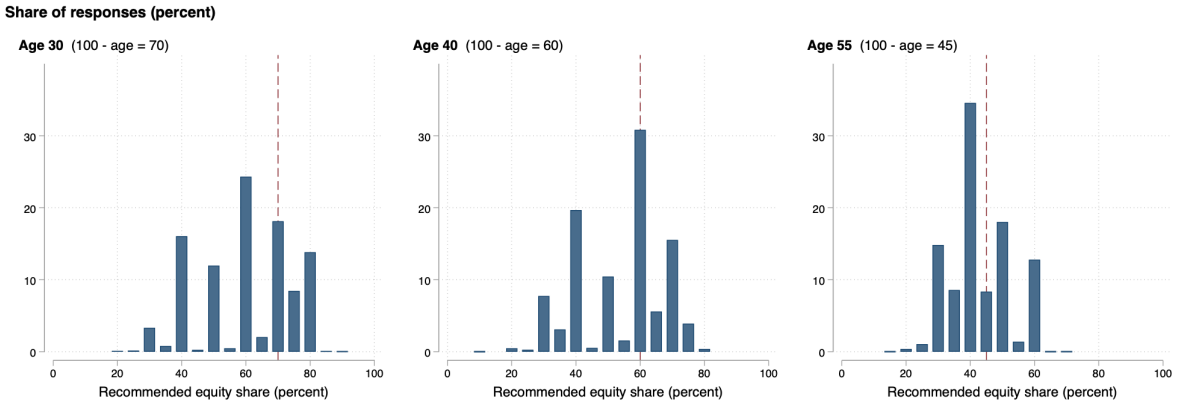
Notes: Each bar decomposes a country's fully localized effect on the recommended equity share, Version C relative to the omitted U.S. English generic baseline (Question 1, age 40), into three additive cues in percentage points. Country name is the English-only contrast Version B-cross minus the U.S. (language and instruments held fixed, though the currency unit and nominal amount change with the country reference), language is Version B minus B-cross (country name held fixed), and local context is Version C minus B (language held fixed). The black diamond marks the total, which equals the sum of the three slabs by construction. The six English-prompt countries have no B-cross cell, so their language slab is zero and country name is Version B minus the U.S.

4.4 Age gradients and pension generosity

Age is the only characteristic of the investor that the country/language experiment varies. I pre-registered an interaction of age with the national pension system as a test of whether the advice responds to a local institution that changes the investor's balance sheet. A generous pension is a large bond-like claim whose weight in total wealth grows as retirement nears. Background-wealth logic therefore predicts that the equity share of the liquid portfolio falls less steeply with age where pensions replace more income. To test this prediction, I vary the investor's age across 30, 40, and 55 in Version B of Question 1 and estimate

$$y_{imca} \;=\; \alpha_m + \gamma_c + \delta_a + \psi\,\tilde{\rho}_c\,\mathbf{1}[\text{age} = 55] + \varepsilon_{imca}, \tag{4}$$

Figure 4: The advice against the 100-minus-age rule, by prompted age



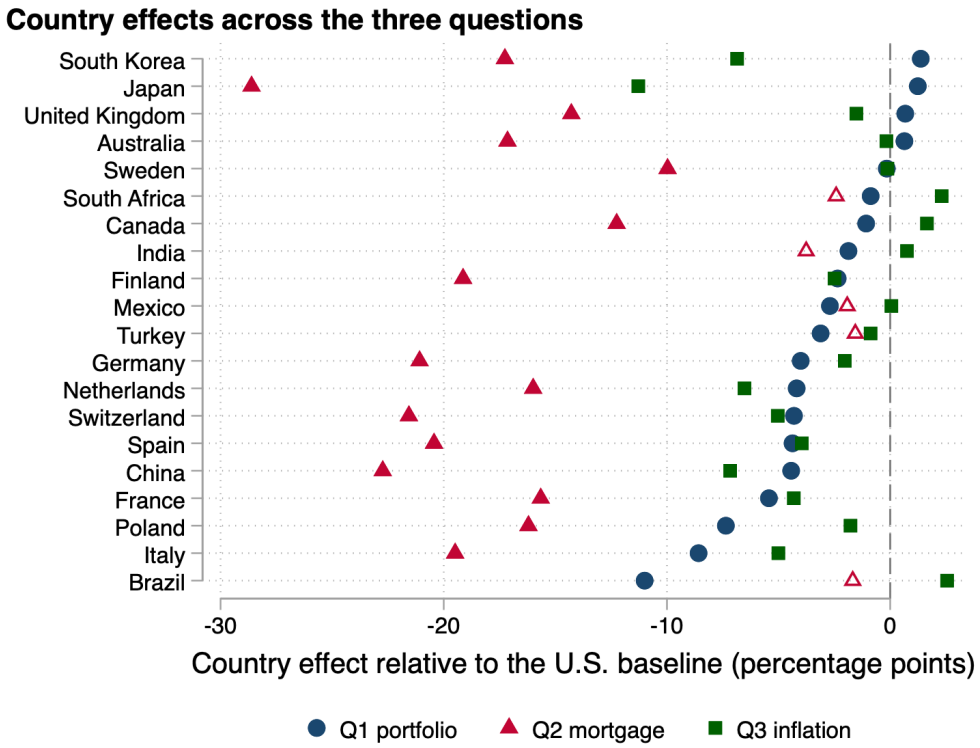
Notes: Distribution of the recommended equity share at each prompted age (Question 1, Version B, all twenty-one countries and thirteen models, parsed non-refusals, $N = 40{,}910$). Dashed lines mark the 100-minus-age prescription. Essentially all responses sit on multiples of five percentage points, so each panel is a discrete bar chart as in Figure 1. Exploratory exhibit, not pre-registered.

where $\tilde{\rho}_c$ is the OECD net pension replacement rate standardized across the twenty-one countries. The coefficient ψ measures how the age-55 tilt toward bonds changes with pension generosity, and background-wealth logic predicts that the tilt shrinks.

The models tilt toward bonds with age, and they apply the same tilt whatever the pension system. This result echoes the later results in the investor arm, where stating the pension generosity or pension asset composition has little impact on advice. Relative to the 40-year-old, they recommend 6.24 percentage points more equity to the 30-year-old and 11.13 points less to the 55-year-old. Pension generosity does not change it. The estimate of ψ implies that a one-standard-deviation higher replacement rate changes the age-55 tilt by -0.02 points ($p = 0.96$), against a tilt of 11 points, and the coarser above-median replacement-rate split gives $+0.4$ points ($p = 0.49$). Appendix Figure A1 shows the same glide path applied nearly verbatim in all twenty-one country panels, from the panel's most generous pension systems to its least generous. Appendix Table A12 reports the full regression alongside a specification without the pension interaction. The stated-investor arm of Section 6 gives the same answer when the prompt states the pension directly instead of leaving it to the country name.

Section 4.1 noted that the 60 percent mode at age 40 is also what the 100-minus-age rule prescribes. Figure 4 plots the advice at each prompted age against the rule's 70, 60, and 45 percent equity. The rule matches the modal advice only at 40, where exactly 60 percent accounts for 31.7 percent of responses in the six English-prompting countries and 30.5 percent elsewhere. At 30 and 55 the mode falls below the rule, at 60 percent (24 percent of responses, against 18 percent at the rule's 70) and at 40 percent (35 percent, against 8 percent at the rule's 45). Pooled advice falls by 0.70 percentage points per year of age, where a 100-minus-age rule would prescribe 1.00. Ten of the thirteen models are flatter than the rule. The glide path is thus a flatter, language-invariant version of the 100-minus-age rule rather than

Figure 5: Country effects across the three questions



Notes: Point estimates of the country fixed effects from equation (1) for Question 1 (recommended equity share), Question 2 (recommended fraction invested rather than used to prepay a 4 percent mortgage), and Question 3 (recommended fraction moved out of a 1 percent savings account), Version B, age 40, in percentage points relative to the omitted U.S. baseline. Each question is estimated separately with model fixed effects. Hollow markers denote the premise-acceptance group on Question 2. Countries are ordered by their Question 1 effect. Confidence intervals are omitted for legibility and are reported in Appendix Tables A9, A10, and A11.

literal compliance with it. Love (2013) shows that this kind of rule tracks the optimal decline in equity exposure through the middle of the life cycle but prescribes less equity than the optimum at both tails. The models' departures from the rule therefore run opposite to the life-cycle optimum.

Looking at the explanations and the numbers, the models follow the rule the same way in every language but name it almost only in English. Explicit citations of the heuristic ("rule of thumb," "100 minus age") appear in 15 to 17 percent of primary-cell explanations in the United States and India but in under 1 percent outside the English-prompting countries. This is consistent with a single advice policy formed in the English-language financial-planning corpus and applied everywhere.

4.5 The mortgage and inflation questions

The two secondary questions depend on the local economy. Figure 5 plots the country effects on all three questions side by side. For these questions, the advice differs far more across countries, and the variance decomposition reverses. Country accounts for 27 percent of the variation in mortgage advice against 15 percent for model, and model identity accounts for only 26 percent of the variation in inflation advice,

whereas on the allocation question the shares are 6 percent for country and 70 percent for model. The secondary questions also pick up a different country signal from the allocation question rather than a stronger version of the same one. The countries that receive less equity on the allocation question, Brazil, Italy, Poland, and France, are not the countries told to prepay, which are Japan, China, Switzerland, and Germany. Japan, statistically at the U.S. level on the allocation question, has the largest gap on both secondary questions. Across countries the allocation and mortgage effects correlate at −0.10, while the mortgage and inflation effects correlate at 0.80.

On the mortgage question, the models recommend investing 10 to 29 percentage points less than in the United States in each of the fifteen countries outside the premise-acceptance group. All fifteen contrasts survive the Holm correction. Among the largest gaps are Japan at −28.6, China at −22.7, Switzerland at −21.6, Germany at −21.1, and Spain at −20.4. Japan's gap alone is more than twice the largest contrast on the allocation question. Part of the common gap reflects the United States itself, whose deductible and freely prepayable mortgages favor investing. The ordering among the fifteen countries, however, spans a further 19 points and places the low-inflation economies at the bottom, consistent with the high real rate that a fixed nominal mortgage carries where inflation is low. For the five countries where the prevailing mortgage rates exceeded twice the prompted 4 percent at the filing date(Brazil, India, Mexico, Turkey, and South Africa), the contrasts are mostly null, and only South Africa (−2.4 points) survives the correction.

On the inflation question, seven country contrasts survive the Holm correction, all negative, so in these countries the models recommend moving less of the savings out of the account than in the United States: Japan (−11.3 points), China (−7.2), South Korea (−6.9), the Netherlands (−6.5), Italy (−5.0), Switzerland (−5.0), and France (−4.3). Japan and China, two of the countries told to prepay, lead the list, in line with the correlation of 0.80 between the inflation and mortgage effects. The clearest localization on this question, however, is in the instrument the models recommend (Appendix Table A4). Equities are the primary instrument in 93 percent or more of responses in fourteen countries and in a clear majority in all but the highest-inflation economies. In Brazil, Turkey, and Mexico the models switch to inflation protection. In Brazil, inflation-linked bonds account for 86.7 percent of the recommendations, and in Turkey inflation-linked bonds and gold together account for 82.3 percent (66.0 and 16.3 percent). Mexico's recommendations tilt the same way (58.1 percent inflation-linked bonds), and Poland's are split (38.6 percent inflation-linked). The tilt appears even inside the English-prompting group, where models in the two higher-inflation members, India and South Africa, recommend inflation protection (11.5 and 16.3 percent, compared to under 3 percent in the United States, the United Kingdom, Canada, and Australia). Because these two countries share the baseline's language, the switch tracks the country rather than the language of the prompt.

4.6 Robustness

I conduct several robustness checks. Some of these checks were pre-registered, but some derive from feedback I received after the preregistration. Two pre-registered re-collections ask whether the results depend on the sampling temperature or on the size of the numerals in the local-currency amount.[7]

In thhe exploratory robustness checks, I first check whether the results change over time as new models are introduced. I also check whether the English Instructions block affects the results, and I ask whether allowing models to ask questions affects the cross-country differences. Finally, I also examine whether the answers depend on the country where I was when I sent the queries (Germany).

Temperature check. Temperature is a sampling parameter that sets how much randomness a language model uses when it picks each word of its response. The next-token prediction at the heart of an LLM picks the next word based on a predicted probability. Model provider introduce some randomness in the token the model uses, which introduces some variation in the answers even with a similar prompt. At a temperature of 0, the model always chooses its most likely next word. Higher values make less likely words more probable. The main set of prompts use a sampling temperature of 0.7.

The temperature check re-collects the primary cell at the temperatures of 0 and 1.0 for five countries (the United States, Germany, Sweden, Japan, and Brazil). Appendix Table A13, Panel A shows that temperature does not affect the results. The pooled mean equity share shifts by 0.3 percentage points at either bookend relative to the 0.7 baseline (wild bootstrap $p = 0.55$ at temperature 0 and $p = 0.40$ at 1.0), and the country contrasts barely move.

Numeral-magnitude. The second robustness check addresses a numeral-magnitude confound. Version B states the stake as a round native-currency figure. The cross-country contrasts compare prompts whose numerals differ by orders of magnitude, such as 7,500,000 yen in Japan against 50,000 dollars in the U.S. baseline cell. An advisor that anchored on the size of the number rather than on the economics could generate spurious country effects. For Sweden, Japan, and Germany I therefore re-collect Version B with the amount stated as $50,000, holding country and language fixed (Appendix Table A13, Panel B). The pooled shift is a tight zero (−0.1 percentage points, wild bootstrap $p = 0.84$, confidence interval −1.6 to +1.4 points). The one significant per-country shift is Japan, whose seven-digit yen figure raises the recommended share by 2.8 points relative to the dollar statement ($p = 0.01$). Japan's country contrast is small and statistically null with native numerals (+1.2 points) and remains small under the dollar statement (an implied −1.6 points).

[7] I also included several pre-registered checks, but these have nothing to act on. The collection produced no refusals, so the refusal outcome is constant and the zero-equity imputation changes no observation. The collection also also finished within two days, so estimating effect by week of collecting does not have any effect.

Purchasing power adjustment of the stake. The main collection uses $50,000 converted at market exchange rates to the local currency. At purchasing-power parity, the stake is worth about twice the U.S. stake in Brazil, Mexico, Poland, and China, two and a half times in South Africa, and three and a half to four times in Turkey and India. If the models advise more caution on a stake that is large relative to local prices, part of these countries' contrasts would reflect the size of the stake rather than the country.

On October 29, 2026, I restated the stake in the twenty non-U.S. Version B prompts as the purchasing-power-parity equivalent of $50,000, rounded in each prompt's own notation, and re-collected the unchanged Version B prompts for all twenty-one countries, with twenty-five repetitions per model and cell. Appendix Table A13, Panel C reports the shifts. Cutting the real stake raises the recommended equity share by 1.0 percentage point on average (wild bootstrap $p < 0.001$, confidence interval 0.5 to 1.7 points). Each halving of the real stake adds 1.2 points ($p < 0.001$). However, the effect is small relative to the baseline contrasts. For example, Brazil's contrast moves from -11.5 to -9.7 points and Poland's from -8.4 to -6.1, and both remain significant after Holm correction. Stake size therefore accounts for between a sixth and a third of these contrasts. Moreover, the effect is not specific to emerging markets, since Spain's contrast narrows by a similar 2.1 points.

Leave-one-out mean and vendor fixed effects. The pre-registered leave-one-model-out sweep re-estimates equation (1) thirteen times, each time without one model. No contrast moves by more than 1.3 percentage points. Six of the seven contrasts that survive the Holm correction survive it in all thirteen samples, with a largest Holm p-value of 0.027. Replacing the thirteen model fixed effects with six vendor fixed effects, for comparability with Foltyn and Olsson (2026), leaves every country contrast essentially unchanged, although with six clusters the variant supports no formal inference.

Newer models and drift over time. To examine whether newer models give different results and to check the robustness of the answers over time, I re-asked one English slice of the primary cell on September 19, 2026 to a panel of twenty-one models. Fourteen of these models were new to the panel. The other seven carry over from the main panel under unchanged identifiers.

The main findings are robust. On the same English slice, model identity accounts for 70 percent of the variance in September against 78 percent in July, and country for 5 percent against 3 percent. Within each model, 77 percent of non-U.S. cells lie below the model's own U.S. cell, against 67 percent in July. Six of the seven carried-over models return average advice within 2.5 points of July.[8]

Caller location and routing. Every query went through OpenRouter from Germany, which raises two concerns: that the models condition on the caller's location, and that the routing of requests between

[8]The interested reader can explore the differences in the answers by model themselves on this website.

hosts moves the estimates. Note first that the location is the same in every call, and that the APIs do not pass the location to the model. Each query is a single user message, the only text the vendors add is fixed (the current date for OpenAI and a system prompt for xAI). Asked what they know about the user, none of the thirteen models reports a location.

A direct test shows that the level does not move either. I sent the primary prompt for the United States, Germany, Brazil, and Japan to one model from each vendor in four blocks of 240 queries, routed through VPN exits in Germany and the United States. Moving the origin to the United States changes the recommended equity share by 0.4 percentage points. The shift does not differ across prompt countries ($p = 0.23$), and each model bills the same number of prompt tokens from both origins.

Translating the Instructions block. Every prompt in the design wraps the translated question in the same English instructions. Since the models are sent the entire query, they may read the English instructions as a cue to anchor on an English-language answer, which would compress the country differences and understate the language effect. I therefore re-collected the primary cell on October 29, 2026 with the Instructions block rendered in the local language for the fifteen non-English countries. The six English-prompting countries received prompts identical to the main collection.

The country structure survives. The twenty country contrasts estimated on the new collection correlate at 0.93 with those of equation (1), eight of them lie below the United States after the Holm correction, and their range is 11 points against 12. Translating the instructions lowers the non-English cells by 1.4 points and the identical English cells by 0.6. The response is concentrated in two models. The two Gemini Flash models recommend about four points less equity when the instructions are in the local language, while the other eleven move by less than 1.5 points. The English wrapper therefore does not produce the compression.

Allowing the models to ask questions. A human advisor would ask questions before advising, a feature that the design denies the models. This raises the possibility that the uniform advice reflects a uniform lack of information rather than a uniform policy. I therefore re-asked the primary question in English for all twenty-one countries, where I first ask the models to list up to five questions it would want the investor to answer first. I also ask the models to state the direction in which each answer would move its recommendation, and gives the share it would recommend provisionally if the investor answered nothing further.

However, this does not affect the results. Country means in the two arms correlate at 0.94 with a slope of one. Model identity accounts for 82 percent of the variance of the provisional share against 4 percent for country, the same ordering as in the main arm. The one difference is a level shift. Asked for a provisional number, the models recommend about six points less equity. The shift is uniform across

countries and concentrated in the models with the highest equity shares. The questions the models ask are investor-specific rather than country-specific.

5 When Should the Advice Differ Across Countries?

The experimental results establish that LLM advice varies across countries, but not whether variation is appropriate. Two properties separate warranted from unwarranted geographic differentiation. The first is invariance. Holding the investor's preferences, investment opportunities, and balance sheet fixed, the advice should not change merely because the investor names a different country or asks in a different language. The second is responsiveness. Holding the country fixed, the advice should move when a determinant of the optimal allocation changes, in the direction and roughly by the amount that theory prescribes. A lower recommended equity share in Brazil than in the United States therefore violates invariance if the two investors face the same economics, and reflects responsiveness if Brazil's fundamentals warrant more conservative advice. This section states a benchmark that separates the two (Section 5.1), asks whether the country effects track measured national preferences and literacy (Section 5.2), and evaluates the mortgage advice against a benchmark pinned by the contract rate (Section 5.3). On the risky share the benchmark warrants little geographic differentiation unless the world fund is held unhedged, and the advice shows little. On the mortgage the benchmark warrants large differences, and the advice produces them in the right direction, but compressed and tilted toward investing. Appendix C develops the benchmarks in full.

5.1 Which country differences the benchmark warrants

The simplest benchmark takes the prompt at face value and treats the $50,000 stake as a standalone portfolio of a world equity index and domestic government bonds (Benchmark S). For an investor with constant relative risk aversion γ, the optimal equity share is the rule of Merton (1969) and Samuelson (1969),

$$\theta_c^* = \frac{\mu_c - r_c}{\gamma\, \sigma_c^2}, \tag{5}$$

where $\mu_c - r_c$ is the expected excess return and σ_c the volatility of the equity fund, both measured in local currency. Appendix C gives the calibration. The share depends neither on the size of the stake nor, with independent returns, on the horizon, so neither the amount nor the five-year horizon in the prompt enters the benchmark. Because the equity premium is common across currencies by uncovered interest parity, the benchmark varies across countries only through σ_c at a common γ.

How much the benchmark varies depends on whether the world fund is hedged. A hedged fund carries roughly the U.S.-dollar volatility of the index in every currency, so the prescribed share is 0.52 everywhere,

and mean advice lies within about six percentage points of it in every country. An unhedged fund adds exchange-rate risk to its local-currency volatility, and at $\gamma = 3$ the prescribed share then ranges from 0.21 in Turkey and 0.23 in Brazil to 0.53 in South Korea, a spread of thirty-two percentage points against the twelve the advice spans. The advice follows little of that spread. Regressing the country effects on the benchmark's prescribed deviations from the United States gives a calibration slope of $\hat{\kappa} = 0.13$, so the advice transmits about an eighth of the cross-country variation that the unhedged benchmark prescribes. The slope measures responsiveness to the one fundamental that varies across countries in the standalone benchmark. A slope of one means full pass-through of the currency risk the unhedged benchmark prices, and a slope of zero means none. The wild bootstrap over countries rejects full calibration ($p < 0.001$ against $\kappa = 1$) and does not reject the U.S.-default hypothesis H2 ($p = 0.32$ against $\kappa = 0$). Dropping Brazil lowers the slope to 0.03, dropping Turkey raises it to 0.21, and dropping any other country leaves it between 0.11 and 0.15, so the estimate depends on the two countries with the most volatile currencies but never comes near full calibration. Following the advice rather than the unhedged benchmark costs an investor in the median country only 1.6 basis points of the stake per year, about $40 over the prompt's five-year horizon, but costs an investor in Turkey 137 basis points per year, about $3,500. Appendix C reports the country gaps under both hedging assumptions, the sensitivity of $\hat{\kappa}$ to risk aversion and to preferences calibrated on measured risk-taking, the influence diagnostics, and the welfare cost of the gaps.

Adding the rest of the balance sheet does not restore a cross-country spread. Treating pension wealth and human capital as safe outside wealth B_c beside the stake W (Benchmark I) scales the equity demand of the liquid portfolio by total wealth over liquid wealth, so that both benchmarks take the form

$$\alpha_c^* \;=\; \min\left\{1,\; \frac{\mu_c - r_c}{\gamma\,\sigma_c^2}\left(1 + \frac{B_c}{W}\right)\right\}, \tag{6}$$

with $B_c = 0$ for Benchmark S. Because outside wealth is an order of magnitude larger than the stake, the prescribed share reaches the 100 percent corner in every country, far above any recommendation in the data (Appendix C.3). Only the unhedged standalone benchmark therefore prescribes different advice across countries, and the verdict on the advice turns on whether an advisor should penalize unhedged currency risk. If it should, the compression documented in Section 4 is a failure of responsiveness. If it should not, the near-uniform level of the advice satisfies invariance, and the open question becomes the country effects that remain, such as Brazil's eleven-point gap from the United States, which have no counterpart in a benchmark that prescribes 0.52 everywhere.

Equation (6) makes the two properties concrete and organizes the remaining tests of the paper. Anything outside the equation should leave the advice unchanged once the economics is fixed: the country name, the language of the question, the local names of the instruments, and, with independent returns, the

investment horizon. The terms inside the equation should move it. The equity premium $\mu_c - r_c$ is common across currencies by uncovered interest parity. The volatility σ_c differs across countries only for an unhedged fund, the margin this subsection prices. Risk aversion γ belongs to the investor. The Version B prompt does not state it, so the benchmark holds it common, and the stated-tolerance cells of Section 6 move it directly. Outside wealth B_c varies across countries with pension generosity and the wage profile, and the stated-pension cells of Section 6 move it for a given investor. Appendix C.3 extends the equation to outside wealth that carries equity risk. One caveat limits the invariance tests. The primary prompt leaves the hedging of the fund, taxes, the real return on local bonds, and the rest of the balance sheet to the model, so a country effect can reflect a fundamental that the model infers from the country name. The design identifies the effect of each prompt treatment, but not whether a country-name effect is justified by fundamentals the benchmark omits. Table 2 in Section 6.2 collects the evidence on both properties.

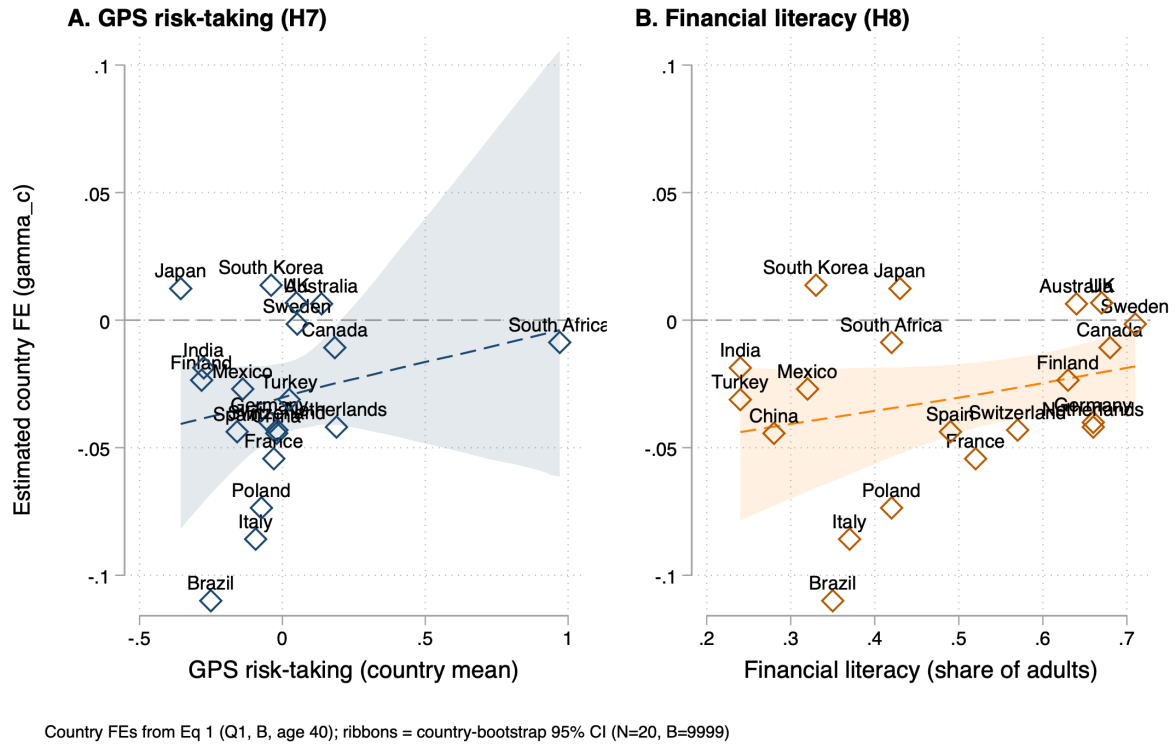
5.2 Does the advice track measured preferences or literacy?

This subsection asks whether the country effects track two country-level measures that a culturally calibrated advisor might respond to, the willingness to take risks in the Global Preferences Survey (Falk et al., 2018) and the share of adults classified as financially literate in the S&P Global FinLit Survey (Klapper et al., 2015). Advice tailored to a country's clientele would recommend more equity where measured risk-taking is higher and, because literacy predicts equity-market participation at the household level (van Rooij et al., 2011), where more adults are financially literate. I test the two predictions, H7 and H8, with pre-registered second-stage regressions of the country fixed effects of equation (1), written $\hat{b}_c$ here to avoid a clash with risk aversion γ, on each measure. These slopes differ from the calibration slope κ of Section 5.1, which regresses the same country effects on the benchmark's prescribed deviations. Invariance does not require the advice to track either measure, because the prompt describes a single investor rather than a country mean, and imputing a country's average risk-taking to that individual is an ecological inference the advisor need not make. The tests are therefore descriptive. They ask whether the models impute national averages, not whether they should.[9]

Figure 6 plots both relationships. Panel A shows that the advice does not track measured risk preferences. The H7 slope is 0.028 (standard error 0.021), and the wild bootstrap over countries does not reject zero ($p = 0.272$, 95 percent confidence interval $[-0.058, 0.203]$). The point estimate is not only imprecise but economically small. A country one full standard deviation more risk-tolerant than another receives

[9]Each test is a bivariate slope over the twenty non-U.S. countries, with the same inference as the calibration slope: a wild bootstrap over countries (Webb weights, 9,999 replications) as the primary procedure, standard errors alongside, and a two-stage bootstrap that propagates the first-stage uncertainty in $\hat{b}_c$. Like the calibration slope κ, both hypotheses sit at the second altitude of the registration. They are pre-specified in full and registered as descriptive, and a dated amendment froze the covariate vectors before I estimated any second-stage coefficient. Because that amendment post-dates the pilot, I keep these tests descriptive rather than confirmatory.

Figure 6: Country effects against measured risk preferences and financial literacy



Notes: Estimated country fixed effects $\hat{b}_c$ from equation (1) (Question 1, Version B, age 40, U.S. baseline omitted) plotted against the country mean of the GPS willingness to take risks (Falk et al., 2018) in Panel A and the share of financially literate adults from the S&P Global FinLit Survey (Klapper et al., 2015) in Panel B, for the twenty non-U.S. countries. Ribbons are 95 percent country-bootstrap confidence bands around the fitted bivariate line (9,999 replications).

under three percentage points more equity, and even the widest preference contrast in the panel, from Japan at -0.36 to South Africa at 0.97, predicts a gap of under four points. The country effects at these two endpoints show the same absence of a gradient. Japan receives advice statistically indistinguishable from the U.S. baseline (Section 4.2), while South Africa, by far the most risk-tolerant country, receives slightly *less* equity than the baseline rather than more.

Panel B tells the same story for literacy. The H8 slope is 0.055 (standard error 0.040, wild bootstrap $p = 0.178$, confidence interval $[-0.030, 0.151]$), and moving across the full literacy range in the panel, from 24 percent of adults in India and Turkey to 71 percent in Sweden, changes predicted advice by under three percentage points. Even this small positive slope owes much to which countries sit at each end of the literacy distribution. The high-income English-prompting countries (the United Kingdom, Canada, and Australia) and the Nordics (Sweden and Finland) all rank among the seven most literate countries in the panel, and their advice sits at the baseline level. The bottom of the distribution mixes countries with large negative effects (Brazil, Italy) and countries with small or no effects (Turkey, South Korea).[10]

[10]The pre-registered diagnostics do not change this reading. Weighting countries by the precision of their first-stage estimates moves the slopes to 0.022 and 0.052. Neither estimate is the artifact of a single country: leave-one-out slopes range from 0.015 (dropping Brazil) to 0.043 (dropping Japan) for risk-taking and from 0.034 to 0.073 for literacy, always

Read against the two properties, the three slopes mean different things. The flat slopes on preferences and literacy are consistent with invariance, since the models do not impute national averages to the individual investor. The flat calibration slope is a failure of responsiveness if an advisor should price unhedged currency risk. Whatever drives the country effects, it is neither the currency risk that the unhedged benchmark prices nor the measured preferences or literacy of the populations being advised. Every effect that survives the Holm correction is in a country prompted in a language other than English (Section 4.2). That grouping bundles the language of the question with everything else the English-prompting countries share with the U.S. baseline, and the within-country comparison of Section 4.3 attributes only about a point to language itself.

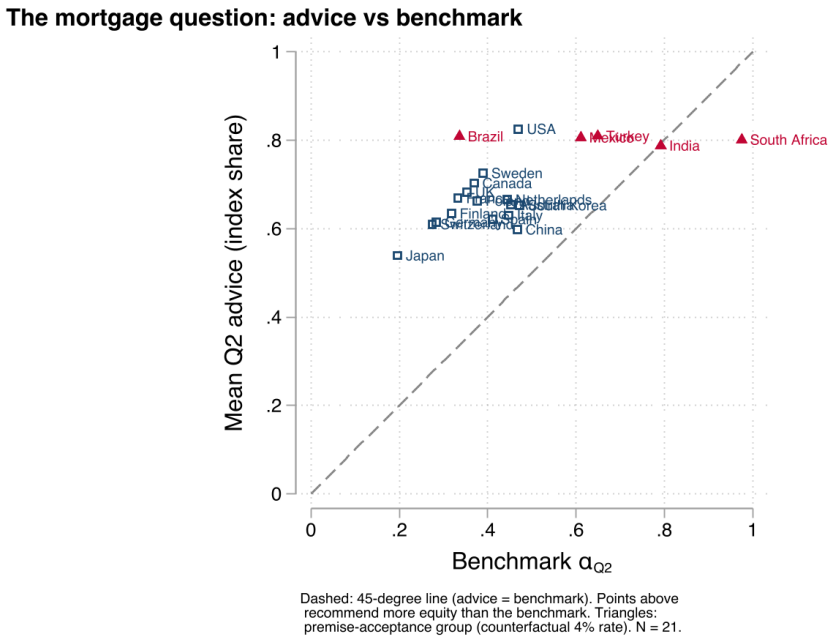
5.3 A benchmark for the mortgage question

Section 4.5 shows that the mortgage advice varies across countries far more than the portfolio advice. The mortgage question also admits a sharper benchmark against which to judge that variation, because prepaying a mortgage is a risk-free investment whose return is pinned by the contract rate. The relevant safe return is the prompted 4 percent net of the deductible share of interest and of expected inflation. Prepayment is therefore a better real investment where inflation is low, as in Japan, and a worse one where mortgage interest is deductible. Replacing r_c in the Merton rule (5) with this after-tax real mortgage rate gives the benchmark share α_c^{Q2} (Appendix C.8). Because effective deductibility depends on income and itemization, the appendix also reports α_c^{Q2} across a band of deductible shares from zero to one half. The benchmark for the five premise-acceptance countries of Section 3 takes the prompt's counterfactual 4 percent rate at face value, so I report them separately from the other sixteen, the calibration countries. The pre-registration does not specify this benchmark, so the comparison is exploratory.

Figure 7 plots the mortgage advice against this benchmark, and Appendix Table A21 reports the country values. The advice tilts toward investing rather than prepaying in every calibration country. The recommended index-fund share exceeds α_c^{Q2} in all sixteen, by thirteen to thirty-six percentage points and by twenty-eight on average. The tilt is not an artifact of the deductibility assumption, since in fourteen of the sixteen the advice exceeds even the equity-favorable end of the deductibility band. Its size is nearly constant where the benchmark lies below 0.40 and shrinks above it. In the nine non-U.S. calibration countries whose benchmark lies below 0.40, from Japan, where the after-tax real mortgage rate is highest, to Sweden, the gap stays between 29 and 34 points, so the advice rises with the benchmark almost point for point. In the six non-U.S. calibration countries with the highest benchmarks, Spain, the Netherlands, Italy, Australia, China, and South Korea, the advice stops rising and the gap narrows

positive and always small. The two-stage bootstrap, which resamples the thirteen models and re-estimates the country effects on every draw, produces intervals of $[0.016, 0.040]$ for H7 and $[0.013, 0.099]$ for H8, both excluding zero. The country effects are therefore precise enough that the positive slopes are not first-stage noise, but the twenty countries scatter too widely around so flat a line for the association to qualify as a cross-country regularity.

Figure 7: The mortgage question: advice against the country-specific benchmark



Notes: Mean recommended index-fund share on the mortgage question (Question 2, Version B, age 40) against the country-specific benchmark α_c^{Q2} from Equation (18) at the baseline grid point ($\bar{\gamma} = 3$, $\eta = 0$). The dashed line is the 45-degree locus of full calibration, on which advice equals the benchmark. Points above it recommend more equity, and hence less prepayment, than the benchmark. Triangles mark the premise-acceptance group, whose α_c^{Q2} assumes the prompt's counterfactual 4 percent rate.

to between 13 and 22 points. The United States is the exception. Its benchmark of 0.47 is among the highest, yet its advice of 0.83 is the highest in the panel and its gap of 36 points the largest of the sixteen.

In the five premise-acceptance countries, whose benchmark takes the prompt's 4 percent rate at face value, average advice sits within four points of the U.S. level. Two of them, South Africa and India, have the highest benchmarks in the panel, 0.98 and 0.79. The prompted rate lies below local inflation in both, and their currency volatility is moderate, so the benchmark prescribes near-full investment, while the advice stays at about 0.80. Advice therefore falls short of the benchmark only in South Africa and matches it in India. The group average conceals disagreement across models. Mistral Medium 3.5 and DeepSeek V4 Flash recommend 8 to 16 points less investment than at their own U.S. level in all five premise-acceptance countries, consistent with replacing the prompted rate with the higher local one, while Gemini 3.5 Flash recommends 12 to 17 points more in Brazil, Mexico, and Turkey.

On the fifteen non-U.S. calibration countries, the calibration regression of Section 5.1 gives $\hat{\kappa}^{Q2} = 0.24$ for the mortgage question, against 0.07 for the allocation question on the same fifteen countries. The mortgage advice thus transmits about a quarter of the cross-country variation its benchmark prescribes, roughly three times the share for the allocation advice. With fifteen countries neither slope is statistically distinguishable from zero, and the 95 percent confidence intervals of both exclude full calibration at

one. The compression that flattens the portfolio advice therefore reappears on the mortgage question, especially among the high-benchmark countries, where the advice stops rising. On the decision where the benchmark warrants the largest geographic differences, the models are responsive in direction but not in magnitude.

6 The Advice Rule

The country-level analysis asks whether the same investor receives different advice in different places. The stated-investor arm of Section 3.5 asks which rule generates the advice. I consider three candidate models of the advisor: the frictionless Merton planner, the life-cycle planner with background wealth, and the risk questionnaire of popular financial advice (Canner et al., 1997; Choi, 2022). The arm separates them by tracing the benchmark's comparative statics one primitive at a time. Stated risk tolerance targets the risk-aversion coefficient in the denominator of the Merton rule (5), and all three candidates predict that it raises the equity share. The stated horizon does not enter the rule at all. Under CRRA preferences with i.i.d. returns the optimal share is the same at every horizon, so both planners predict no response, whereas a risk questionnaire scores the horizon alongside risk tolerance. The stated pension targets the outside-wealth multiplier $1 + B_c/W$ in equation (6), and only the life-cycle planner predicts that a generous pension raises the share. The sign pattern of the three responses therefore discriminates among the candidates. The exploratory probes of Section 6.1, which I added after examining these pre-registered results, then push past pension generosity to the composition of the pension, the stake's share of the investor's liquid wealth, and the currency exposure of the fund.

On the four crossed risk–pension cells the estimating equation is

$$y_{icm} \;=\; \beta_R \,\mathrm{Tol}_i \;+\; \beta_P \,\mathrm{Gen}_i \;+\; \beta_{RP} \,\mathrm{Tol}_i \,\mathrm{Gen}_i \;+\; \phi_c \;+\; \lambda_m \;+\; \varepsilon_{icm}, \tag{7}$$

where Tol_i and Gen_i indicate the risk-tolerant and generous-pension statements, the omitted cell is the risk-averse investor with the limited pension, and ϕ_c and λ_m are country and model fixed effects. On the two horizon cells the analogous equation regresses the recommended share on an indicator for the twenty-year horizon with the one-year horizon omitted, yielding β_H. The coefficients β_R, β_P, and β_H are the three pre-registered tests of Section 3.5, each without multiplicity correction, and the interaction β_{RP} is descriptive. Inference is the same model-level wild cluster bootstrap as the primary regressions. Appendix Table A5 reports the distribution of the recommended share in each treatment cell, and Appendix Table A6 reports the coefficient estimates.

The estimates yield three findings. First, the advice responds strongly to stated risk tolerance. The mean recommended share rises from 0.30 in the risk-averse cells to 0.78 in the risk-tolerant ones. The

coefficient, which measures the effect at the limited pension, is $\hat{\beta}_R = 46.3$ percentage points (confidence interval $[41.1, 51.4]$). The response is almost the same in every country. All twenty-one country responses lie inside a band of 3 percentage points, and no country's response differs from the U.S. response after Holm correction (Appendix Table A8).

Second, the advice responds strongly to the stated horizon, which the benchmark says should not matter. The mean equity share is 0.36 at the one-year horizon and 0.76 at the twenty-year horizon, and the coefficient is $\hat{\beta}_H = 39.9$ percentage points (confidence interval $[25.7, 53.4]$). The horizon moves the share substantially in every country, but its cross-country span of 11.9 percentage points is wider than the 3-point band for stated risk tolerance. The response is largest in Sweden, France, Poland, and Switzerland, at 44 to 46 points, and smallest in China, Mexico, South Korea, and Turkey, at 34 to 35 points. Sweden and Switzerland differ from the United States after Holm correction.[11]

Third, the advice responds at most weakly to the stated pension. At the risk-averse baseline the generous pension lowers the recommended share by 1.0 percentage point ($\hat{\beta}_P = -1.0$, confidence interval $[-2.3, +0.2]$). For the risk-tolerant investor it raises the share by about 2.6 points, the sum of $\hat{\beta}_P$ and the risk-by-pension interaction of +3.6 points ($[2.0, 5.2]$). Averaged over the two investors, the effect is about +0.8 points, an order of magnitude below the background-wealth adjustment that a generous pension should trigger. The pension response is equally small in every country, with a cross-country range of 1.4 points.

The levels of the advice fit the frictionless Merton planner, but the comparative statics fit neither planner. Inverting the Merton rule (5) at the paper's calibration puts the implied relative risk aversion at roughly 2 for the stated-tolerant cells, 5 for the stated-averse cells, and 3 for the no-statement reference, all inside the conventional range, so the levels are consistent with a Merton (1969)–Samuelson (1969) planner holding plausible preferences. With CRRA preferences and i.i.d. returns, however, that planner predicts exact horizon invariance, and the data show a forty-point response. The life-cycle planner fails on the pension. An advisor that fully integrated the balance sheet would sit at the 100 percent corner of Benchmark I in every cell, which the interior levels already rule out. Conditional on interior advice, any degree of balance-sheet integration still implies $\beta_P > 0$. A stated generous pension is a bond-like annuity claim that should crowd financial-portfolio equity *in* under any total-wealth model (Cocco, Gomes and Maenhout, 2005), yet the estimated response is at most a few points. Mean reversion in returns (Campbell and Viceira, 2002) could rationalize the horizon response but not the pension null, because a planner sophisticated enough to price return dynamics into a forty-point horizon effect should not ignore pension wealth entirely.

[11] The rejection of horizon invariance is not uniform across the panel: two of thirteen models are nearly horizon-invariant, while the most responsive model moves 79 percentage points between the one-year and twenty-year cells. The one-year cell is also where the panel disagrees most, against near-consensus at twenty years. The risk-tolerance response, by contrast, is positive for all thirteen models and remarkably uniform across countries.

Instead, the responses match the third candidate, the risk questionnaire of popular financial advice. The models respond strongly to the two cues that retail risk questionnaires collect, stated risk tolerance and investment horizon, and barely to the one balance-sheet cue in the pre-registered cells, the pension that normative theory requires them to weigh. This conclusion rests on the three pre-registered tests alone. The age gradient of Section 4.4 fits the same reading. Its glide path is a compressed, language-invariant rendering of the 100-minus-age heuristic, anchored on the industry's 60/40 portfolio at mid-career (Figure 4).

The pension null also sharpens the reading of the zero age-by-pension interaction in Section 4.4. That zero could in principle reflect ignorance of local pension systems. Here, however, the prompt states pension generosity directly, so acting on it requires no knowledge of local institutions, and the models still barely use it.

6.1 Exploratory probes: which balance-sheet cues register

The pension null leaves open whether the models disregard the investor's balance sheet altogether or respond to some balance-sheet cues and not to others. To find out, I turn to the three exploratory probes of Section 3.5.[12] The first probe states the pension's *composition*: either a guaranteed inflation-linked annuity or a portfolio invested almost entirely in stocks. The benchmark predicts opposite-signed tilts of the liquid stake, toward equity when the background claim is bond-like and away from it when the background claim already carries equity risk. In equation (11) of Appendix C.3, the annuity sets the pension's equity loading ω to zero and the stocks statement sets it to one, which lowers the uncapped share by P_c/W. As for the pension test, the predicted signs are those of the uncapped index, since interior advice already rules out the capped benchmark's corner. The second probe, a wealth ladder, states the stake's share of the investor's liquid wealth, all, half, or a third. With W the stake and D the stated safe savings beside it, the benchmark share of the stake is $\theta^*(W + D)/W$, which rises monotonically as the stated safe cushion grows. The third probe states whether the stock fund is currency-hedged or unhedged. For a non-U.S. investor the hedge removes exchange-rate variance and should raise the Merton share, so the benchmark predicts a hedged–unhedged gap that widens with local-currency volatility σ_c. The composition and hedging probes cover all twenty-one countries, and the wealth ladder covers the United States alone. I estimate each probe cell against the no-statement reference with country and model fixed effects and the same wild cluster bootstrap as the rest of the arm. Figure 8 plots the estimates and Appendix Table A7 reports them.

Told that the pension is invested almost entirely in stocks, the models cut the recommended equity share by 16.9 percentage points (confidence interval $[-21.8, -12.1]$, $p < 0.001$). Told that it is a guaranteed

[12] Everything in this subsection is exploratory and post hoc, sits in no Holm family, and is tested two-sided against the benchmark-implied sign.

annuity, they raise the share by 4.8 points ($[0.1, 9.6]$, $p = 0.042$). Both signs match the benchmark, but the response to background risk is more than three times the response to background safety. The asymmetry holds in every country, but the size of the cut does not. Told of a pension in stocks, the models cut the share by 19 to 22 points in the United States, Canada, Australia, Japan, South Korea, Mexico, China, South Africa, and India, and by 11 to 17 points in the ten European countries, Turkey, and Brazil, a split that twelve Holm-corrected country contrasts confirm (Appendix Table A8). The annuity response is small everywhere. The wealth ladder shows the same asymmetry. Stating that the stake is all of the investor's liquid wealth cuts the share by 23.3 points ($p < 0.001$), while the cushion rungs, half and a third of liquid wealth, raise it by only 1.4 and 2.5 points. The pattern is monotone in the predicted direction (the per-rung slope is 7.6 points, $p < 0.001$), but its safe side is an order of magnitude short. With the baseline recommendation near 0.6, the benchmark share $\theta^*(W + D)/W$ is pinned at its 100 percent corner from the half rung ($D = W$) onward. At that rung the benchmark lies about 40 points above the baseline, and the advice rises by 1.4.

The hedging probe moves the advice by little, and not in the pattern the benchmark predicts. At the mean local-currency volatility, the hedged statement changes the share by 0.4 points ($p = 0.66$) and the unhedged statement by -1.3 ($p = 0.15$). The gap between them, estimated on the two hedging cells alone, is 1.8 points ($p < 0.001$), which has the benchmark's sign but not its size. The benchmark gap is zero in the United States, where the fund's base currency is the investor's own, and about thirty points in Turkey. Across countries the estimated gap lies between -2.4 and $+4.0$ points, it does not grow with σ_c (the slope is negative, $p = 0.26$), and it is negative in Turkey, the country with the highest volatility (Appendix Table A8). Both the hedged and the unhedged statements raise the share more where σ_c is higher, by roughly two and a half and four points across the panel's range, but they rise *together*. The models respond to the mention of currency exposure, not to what a hedge does to the variance.

Two of the three probes show the same one-sided pattern, and the third moves the advice by at most a few points on either side. On pension composition and on the size of the stake, the advice moves strongly, and with the benchmark's sign, when the stated cue raises the investor's exposure, and it moves weakly on the mirror-image cues that should raise the equity share. A planner that integrated the balance sheet would respond in both directions, since safe outside wealth crowds equity in just as risky outside wealth crowds it out. The models implement only the de-risking half. They read the balance sheet as a source of hazard rather than as the two-sided input of portfolio theory.

The one-sided pattern also completes the questionnaire reading of the pre-registered cells. Retail suitability assessments ask a third question beside tolerance and horizon, namely whether the client can afford to lose the money. UK guidance calls this capacity for loss and asks advisors to recommend only the risk a customer is both willing and able to take (Financial Services Authority, 2011), and the MiFID II

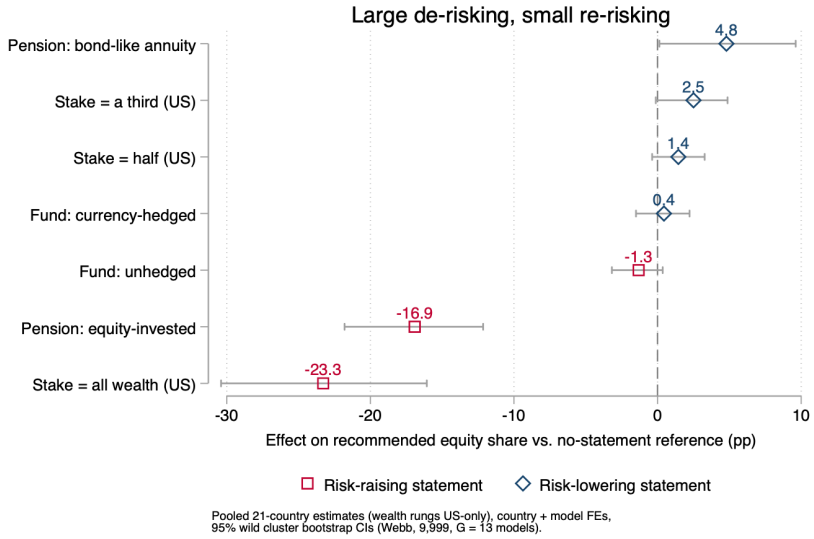


Figure 8: Exploratory balance-sheet probes: large de-risking, small re-risking

Notes: Effect of each probe statement on the recommended equity share, in percentage points against the no-statement Version B and B-cross reference (Question 1, age 40), with country and model fixed effects. Squares mark statements that raise the investor's risk exposure and diamonds statements that lower it, so a balance-sheet advisor would place the squares to the left of zero and the diamonds symmetrically to the right. The pension-composition and hedging cells pool all twenty-one countries and the wealth rungs are U.S.-only. Bars are 95 percent wild cluster bootstrap confidence intervals (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$). All estimates are exploratory and post hoc, in no Holm family (Appendix Table A7).

guidelines treat the client's ability to bear losses as part of her financial situation (European Securities and Markets Authority, 2022). A screen of this kind lowers the recommended risk when a stated cue shows that the client could not absorb a loss, as a pension invested in stocks or a stake that is all of her liquid wealth does, while stated safety does not raise the risk beyond what her stated tolerance allows. A questionnaire with such a screen predicts every sign in this section: a response to tolerance and horizon, none to a generous pension, a cut on the two exposure cues, and little response to the annuity, the cash cushion, or the currency hedge. I add the screen to the reading after seeing the probes, so it interprets exploratory evidence rather than testing it.

The models' own information demand supports the questionnaire reading. In the elicitation probe of Section 4.6, which lets each model state what it would ask the investor before advising, the questions form a suitability questionnaire. Nearly every response in every country asks four things, in the order a retail assessment asks them: the investor's tolerance for losses, the purpose of the money, whether the stated horizon is firm, and whether a separate emergency fund exists. Tolerance is the first question in five responses out of six, and it is framed behaviorally, as how the investor would react to a drop of thirty percent, rather than as a capacity to bear the loss. The thresholds travel unchanged, the thirty-percent drawdown and the three-to-six-month cash buffer appear from Turkey to Japan, and the second tier of questions, on income stability, debt, and the size of the pension, varies across models two to three times more than across countries (Appendix Table **??**). Country enters the questions in two narrow ways.

For countries with weaker currencies the models ask whether the investor would accept exposure to the home currency or prefer international diversification, treating the home market as a risk to leave rather than as a reason for a different equity share, and where a named tax account exists, such as an IRA or a Swedish ISK, they ask whether the money sits in it. Neither bears on the split between a world index and government bonds. The probe also closes a gap in the stated-investor arm. Four of every five elicited questions fall on a cue the arm manipulates, tolerance, horizon, the pension, the currency hedge, or the size of the stake, and the questions the arm does not state, on purpose, emergency savings, and income, are all proxies for the same capacity-for-loss screen. The cues the arm varies are the ones the models say they need.

6.2 Invariance and responsiveness

Table 2 collects the evidence of Sections 4 to 6 against the two properties of Section 5. The models pass most invariance tests. The language of the question, the local names of the instruments, and the U.K. country name each move the advice by about a point or less. The exception is the investment horizon, which moves the advice by forty points although it lies outside the benchmark. The models fail most responsiveness tests. Stated risk tolerance moves the advice strongly and in the right direction, but currency risk, pension generosity, stated safe wealth, and the currency hedge barely move it, and the mortgage advice follows the real mortgage rate in direction but not in magnitude. The advice is thus close to invariant where theory asks for invariance and insufficiently responsive where it asks for responsiveness, with the horizon and stated risk tolerance the two exceptions. The country-name effects in the remaining countries are the one entry the design cannot classify, because the prompt leaves some fundamentals to the model.

7 Mechanisms in the Explanation Text

Finally, I examine the explanation that each model gives for its recommendation. The explanations separate two accounts of the compression documented in Section 4. Under the first, the models do not know how the investor's country differs from the United States. Under the second, they know but do not let that knowledge reach the recommended share. The first account predicts explanations that rarely refer to the investor's country, and the second predicts explanations that refer to it often alongside a nearly uniform allocation.

Each prompt asks the model to justify its recommendation in at most two hundred words. I machine-code each explanation into four binary indicators that record whether it invokes inflation or real-value protection, an existing pension or retirement entitlement, the investment horizon, or local or home-country financial context. The coder is a single judge model outside the thirteen under study, Claude

Table 2: Invariance and responsiveness: the evidence against the benchmark

Cue	Benchmark prediction	Estimate	Status
Panel A. Should not move the advice (invariance)			
Language of the question, local vs. English	No effect	−1.3	H4
Local names of the instruments (Version C)	No effect	−0.7	H5
U.K. rather than U.S. country name	No effect	+0.7 (equivalent)	H3
Twenty-year rather than one-year horizon	No effect	+39.9	H13
Other country names (Version B)	No effect if economics fixed	−11.0 to +1.4	H1
Panel B. Should move the advice (responsiveness)			
Stated risk tolerance	Raises	+46.3	H11
Unhedged currency risk across countries	Slope of one	$\hat{\kappa} = 0.13$	H2
Stated generous pension	Raises	−1.0 / +2.6	H12
Pension in stocks / guaranteed annuity	Lowers / raises	−16.9 / +4.8	Exploratory
Stake is all / a third of liquid wealth	Lowers / raises	−23.3 / +2.5	Exploratory
Currency-hedged fund	Raises outside the U.S.	+0.4	Exploratory
Real mortgage rate across countries	Slope of one	$\hat{\kappa}^{Q2} = 0.24$	Exploratory

Notes: Estimates are effects on the recommended equity share in percentage points, except the two calibration slopes. Panel A lists treatments that equation (6) excludes, so each should leave the advice unchanged once the investor's economics is fixed. The country-name row gives the range of the twenty Version B contrasts of Section 4.2, seven of which survive the Holm correction, all negative. Because the prompt leaves some fundamentals to the model, that row does not by itself establish a failure of invariance. Panel B lists cues that move a term of the equation. The generous-pension entries are the effects for the risk-averse and the risk-tolerant investor. The wealth-ladder entries are U.S.-only. The hedge entry is the hedged statement at the mean local-currency volatility, and the unhedged statement gives −1.3. The calibration slopes regress the country effects on the benchmark's prescribed deviations, where one is full pass-through (Sections 5.1 and 5.3). Status gives the pre-registered hypothesis of Appendix Table A16 or marks a post-filing exploratory estimate.

Opus 4.7, which sees only the explanation text and is blind to the country, model, and allocation that produced it. Horizon appears in every explanation and so leaves no variation to test. The other three categories clear two pre-registered reliability gates and enter pre-specified testing.[13]

For each of the three categories, I re-estimate equation (1) on the primary cell (Question 1, Version B, age 40) with the binary rationale code in place of the equity share. Because the explanation and the allocation enter the same specification on the same sample, their country effects are directly comparable. Each country effect tests whether a rationale appears more or less often than in U.S. explanations. Inference uses the same model-level wild cluster bootstrap as the primary regressions, with a Holm correction within each category's family of twenty contrasts.

The explanations localize far more than the advice. Local financial context appears in 18.0 percent of U.S. explanations and 29 to 78 percentage points more often in every other country, with nineteen of the twenty contrasts surviving the Holm correction (Table 3). The rate approaches saturation in South

[13] I froze the coding instrument at pre-registration and validated it with four judge models from independent vendor families, none among the thirteen models under study and each shown only the explanation text. On a stratified validation subsample, each category had to clear two pre-registered inter-rater reliability gates, both requiring $\kappa \geq 0.6$: a reproducibility gate on the minimum pairwise Cohen's κ across the four judges, and a face-validity gate on the κ between the primary judge and my hand-codes on a bilingual English-and-Swedish anchor. I reported the results of both gates before estimating any coefficient in this section. Inflation, pension, and local context clear both gates comfortably (minimum pairwise judge κ of 0.95, 0.95, and 0.87). The primary judge then coded every explanation in the analysis sample. A deterministic multilingual keyword dictionary, coded independently, agrees with it on 99, 96, and 73 percent of rows, and local context is the hardest category by every measure. See Appendix A and the judge-validation amendment of record.

Table 3: Rationale composition by country in the explanation text

	Inflation		Pension		Local context	
Country	Est.	Holm p	Est.	Holm p	Est.	Holm p
Turkey	56.3	0.019	0.6	1.000	70.6	<0.001
South Africa	33.2	<0.001	0.0	1.000	78.2	<0.001
Poland	31.4	0.090	−8.2	0.470	56.3	0.003
Brazil	25.1	0.441	−0.3	1.000	75.8	<0.001
Mexico	21.8	0.359	−3.4	0.930	77.2	<0.001
India	17.8	0.803	−0.2	1.000	70.3	<0.001
United Kingdom	5.8	0.141	−8.5	0.128	71.8	<0.001
South Korea	0.2	1.000	−5.5	1.000	29.2	0.059
Canada	−0.2	1.000	2.3	1.000	59.4	<0.001
Australia	−2.8	1.000	−2.6	1.000	68.2	<0.001
Netherlands	−4.7	1.000	−0.5	1.000	43.0	0.009
Italy	−8.8	1.000	−1.4	1.000	70.5	<0.001
Japan	−9.4	1.000	−1.1	1.000	36.3	0.025
Spain	−9.5	1.000	1.9	1.000	61.1	<0.001
Germany	−12.5	1.000	1.5	1.000	45.1	<0.001
Finland	−13.3	0.965	−8.0	0.169	39.4	0.005
France	−14.8	1.000	−0.9	1.000	59.0	<0.001
Switzerland	−16.6	0.998	−3.2	1.000	66.5	<0.001
Sweden	−19.4	0.392	−6.8	1.000	29.6	0.013
China	−20.5	0.333	−2.6	1.000	32.9	0.025

Notes: Country fixed effects from the linear probability model of equation (1), estimated on each binary rationale code in place of the equity share (Question 1, Version B, age 40, $N = 13{,}639$), in percentage points relative to the omitted U.S. baseline. A positive entry means the rationale appears more often than in U.S. explanations. The U.S. baseline rates are 33.7 percent (inflation), 97.5 percent (pension), and 18.0 percent (local context). Holm p-values correct the model-level wild cluster bootstrap p-values (Webb six-point weights, 9,999 replications, $G = 13$) within each category's family of twenty contrasts. Countries are ordered by the inflation estimate. The horizon category is omitted because every explanation invokes it (prevalence 100 percent), leaving its country effects undefined.

Africa, Mexico, and Brazil, where more than nine explanations in ten name a local instrument, tax, or pension system, and it stays close to half even where it is lowest, in South Korea and Sweden at roughly 47 percent (Figure 9 gives the levels behind these contrasts). The same specification on the equity share yields country effects of at most 11 percentage points (Section 4). When the models answer the allocation question outside the United States, they reach for local knowledge in the explanation and leave it out of the allocation.

The inflation rationale follows the local economy, while the pension rationale is the same everywhere. Against a U.S. base of 34 percent, the inflation rationale rises by 56 percentage points in Turkey and 33 in South Africa, both surviving correction, and it is also higher in Poland, Brazil, Mexico, and India, though short of the corrected threshold. The ranking of countries on this rationale tracks their inflation environments and is the explanation counterpart of the mortgage and inflation results of Section 4, where the advice itself localizes when the question depends on the local economy. The pension rationale appears in almost every explanation (97.5 percent in the United States), and no country departs from that rate after correction.

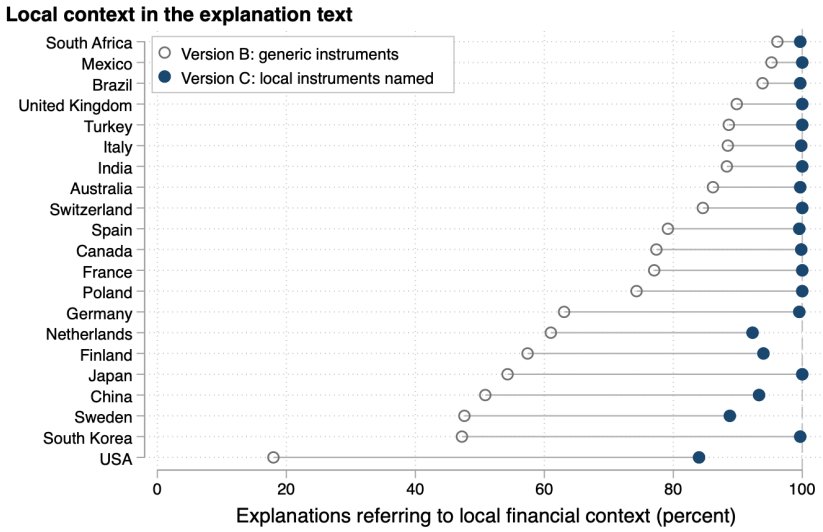


Figure 9: Local financial context in the explanation text, by country and version

Notes: Share of explanations that refer to country-specific financial context, coded by the out-of-panel judge model described in the text (Question 1, age 40, $N = 13{,}639$ under Version B and 13,631 under Version C). Hollow grey marks Version B, where the question is translated but the instruments are generic. Filled navy marks Version C, which additionally names the local pension, bond, and index instruments. Countries are sorted by the Version B rate, so the United States anchors the bottom of the panel. The reported means are weighted by cell counts. Version C is not collected for the country-name-only contrast, so Version B-cross does not appear. The coding records whether local context is mentioned, not whether it is accurate or used to set the allocation.

Naming local instruments in the prompt also moves the explanation far more than the allocation. When I re-estimate the Version-C regression of equation (2) with the local-context code as the outcome, Version C raises the rate of local references by 23.3 percentage points relative to Version B ($\hat{\phi}^{\text{text}} = 0.233$, wild bootstrap $p < 0.001$, confidence interval $[0.143, 0.329]$). The same cue moved the allocation by -0.7 points (Section 4). The treatment therefore registers with the models, and its small effect on the allocation does not reflect a cue they failed to read. Handed the local institutional context outright, the models change what they say and leave what they recommend essentially unchanged. Figure 9 puts both results in levels. The Version-C gain is not uniform. Once the prompt names the instruments, sixteen of the twenty-one countries sit within half a point of universal local reference, so the gain comes from the countries that were not already localizing.

The explanations therefore rule out the first account of the compression. Local financial context appears in three quarters of non-U.S. explanations even when the prompt names no local instrument, and in almost all of them once it does, so the models do distinguish the investor's country from the United States. Nor are the small allocation effects an artifact of the test. The same equation, on the same sample and with the same inference, returns a 78-point effect for South Africa on the local-context code, while South Africa's contrast on the equity share does not reject even before correction. The design detects country variation where the output contains it, and the compression is a property of the advice.

The same gap between mention and use appears in the investor's own circumstances. Almost every explanation names her pension, and every explanation names her horizon. Yet only the horizon moves the allocation, by forty points between the one-year and twenty-year cells of the stated-investor arm, against a pension response near zero. How often the models mention a consideration therefore says nothing about whether they use it. Which considerations they use follows the line that Section 6 draws, between the cues a retail risk questionnaire collects and the balance-sheet quantities it does not.

Taken together, the explanations localize far more than the advice. Because the coding records whether local context is *mentioned* rather than whether it is accurate or used, and because the design observes advice rather than model internals, these results describe the explanations shown to users, not the computation behind the allocation. The pattern is nonetheless consistent with interpretability evidence that multilingual models solve tasks in a shared, English-leaning concept space (Wendler et al., 2024; Wu et al., 2025; Schut, Gal and Farquhar, 2025). In such a model, translating the question changes the words of the question and the answer without reaching the policy that produces the allocation, which matches the small language effect of Section 4. The Version C result extends this reading beyond language. There the treatment supplies explicit local institutional content rather than a change of language, and it too changes the explanation without reaching the allocation.

8 Conclusion

This paper asks whether AI models tailor financial advice to the country of the investor who asks. In a pre-registered experiment, I pose three financial questions to thirteen frontier AI models in twenty-one countries, decompose any country effect into language, country-naming, and local-framing components, and compare the advice with a country-specific normative benchmark. The main result is that the models localize some decisions and not others. On the allocation question the advice is compressed into a twelve-point band of country means, with the U.S. baseline near its top. Every country contrast that survives correction is negative and modest. Which model a household consults matters an order of magnitude more than which country it consults from. That ordering is specific to the allocation question, and on the mortgage question it reverses. There the same models recommend investing 10 to 29 percentage points less than in the United States in every country where the prompted rate is realistic. On the inflation question they switch from equities to inflation-linked bonds in Brazil, Turkey, and Mexico. Nearly every model shares the direction of these differences.

The cross-country pattern in the allocation advice matches neither the measured risk preferences and financial literacy of the populations the models advise nor any version of a standard portfolio benchmark. With unhedged currency risk, the benchmark prescribes a thirty-two-point spread across countries against the twelve points the advice spans, and on the mortgage question, in an exploratory comparison, the

models recommend investing 13 to 36 points more than the question's own benchmark prescribes in all sixteen calibration countries. What moves the allocation advice instead is what the prompt says about the individual investor, and specifically the cues a retail risk questionnaire collects. Stated risk tolerance raises the recommended equity share by 46 percentage points, and a twenty-year rather than one-year horizon raises it by 40, although the horizon is irrelevant under CRRA preferences with i.i.d. returns. A stated generous pension barely registers. This joint pattern, a large horizon response beside a near-absent pension response, identifies the rule as the "popular" financial advice of Canner et al. (1997) and Choi (2022) rather than a portfolio rule over the household's balance sheet. Exploratory probes show that the models do register the balance sheet, but only in one direction. The advice de-risks sharply when a stated cue raises the investor's exposure, as with a pension invested in stocks or a stake equal to all of her liquid wealth. It stays nearly flat on the mirror-image cues that should raise the risky share, such as a guaranteed annuity, a currency hedge, or a stated cash cushion. A retail suitability assessment produces the same one-sided pattern, because it screens the risk a client is willing to take against her capacity to bear losses, so a stated hazard lowers the recommendation while stated safety does not raise it on its own.

The compression is not a failure to tell the countries apart. Local financial context appears in three quarters of non-U.S. explanations without prompting and in almost all of them once the prompt names local instruments, and the same country regression that finds effects of at most 11 points on the equity share finds effects of up to 78 points on whether the explanation names that context. The allocation, by contrast, barely responds to localization cues. Naming local instruments moves the recommended share by less than a point, and asking in the local language rather than in English moves it by 1.3 points, so the country name itself carries most of each country effect. The models retrieve what distinguishes the investor's country and leave it out of the number. What the number does carry are the conventions of U.S.-English retail advice. The modal recommendation is the 60/40 balanced portfolio, the United States sits near the top of the band, and explicit rules of thumb appear in the explanations of the English-prompting countries and almost nowhere else. The small language effect is consistent with interpretability evidence that multilingual models solve tasks in a shared, English-leaning concept space and confine language-specific processing to the input and output layers.

I note four limitations. First, I do not observe the model internals, so the reading that translation leaves the underlying policy intact is an inference from behavior rather than a direct measurement. Second, the normative benchmark is deliberately static and closed form. A country-calibrated life-cycle model in the tradition of Cocco et al. (2005) would add realism, but the one existing such treatment reaches the same 100 percent corner as the balance-sheet benchmark here (Choukhmane et al., 2026), so it is unlikely to overturn the balance-sheet logic. Third, the experiment queries each model through its API

in a single turn with no system prompt. Consumer chat applications add system prompts, memory, and web search, which may shift the level of the advice. Every comparison in the paper holds the interface fixed across countries and investors, so the contrasts, the variance ordering, and the advice rule are the findings most likely to carry over, and the levels are the most likely to differ. The elicitation probe of Section 4.6 shows what the models would ask in a second turn, and it is investor-specific rather than country-specific, so the single-turn format is not what produces the cross-country compression. Fourth, the main estimates are a snapshot of a fast-moving panel, collected within a two-day window. A re-collection eleven weeks later shows that the model behind an unchanged identifier can move by as much as a country contrast (Appendix D). It also shows that a panel that had largely turned over gives advice with a similar structure, with model identity the largest source of variance and most countries below each model's U.S. cell, although one round of turnover cannot establish a trend. Two questions remain open: whether the compression narrows as models improve, and whether retrieval or fine-tuning can induce localization without reintroducing the language-keyed inconsistency that the shared concept space currently avoids.

These findings speak to the market structure that AI advice creates. Regulation, the trust a credence good requires, and language once kept financial advice national. A handful of foundation models now supply it in any language and at almost no marginal cost, outside the regulatory perimeter that binds human advisors. They answer the allocation question with nearly one policy everywhere, while their advice on debt and on inflation protection differs across countries. A global market for advice makes it possible to ask how much geographic differentiation the economics actually warrants, and the answer differs by decision. On the risky share, little differentiation is warranted, since a currency-hedged world fund prescribes the same share everywhere, and the country differences the models do produce follow the country name, not the fundamentals the benchmark prices. On the mortgage, where the local real rate decides the answer, large differences are warranted, and the models differentiate in the warranted direction. Because production is concentrated in a few models trained on the same corpus, any error in that policy is common to all the households exposed to it, whereas the fragmented industry of human advisors would have diluted it. The clearest such error is the one-sided treatment of the balance sheet, which cuts risk on stated exposure but barely responds to stated safe wealth. Suitability and best-interest rules oblige a human advisor to assess the client's whole financial situation. The models apply only the protective half of that assessment, and they do so outside the perimeter where those rules bind. Whether the regulatory perimeter should extend to advice of this kind is the policy question this market structure poses.

The answers also do not show the investor where the advice adapts to her country. An investor who asks the same models about her portfolio allocation and about prepaying her mortgage receives advice that

barely responds to her country in the first case and varies with it in the second. Without expertise of her own, she cannot tell from the answers which case she is in. Outside the United States the models name local instruments, taxes, and pension systems in most of their explanations of the allocation, so the part of the answer that visibly adapts to her country is the explanation she can read rather than the allocation that determines what she earns.

References

Bodie, Zvi, Robert C Merton, and William F Samuelson, "Labor supply flexibility and portfolio choice in a life cycle model," *Journal of economic dynamics and control*, 1992, *16* (3-4), 427–449. 73

Bolukbasi, Tolga, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai, "Man is to computer programmer as woman is to homemaker? Debiasing word embeddings," in "Advances in Neural Information Processing Systems," Vol. 29 2016. 5

Campbell, John Y and Luis M Viceira, *Strategic asset allocation: portfolio choice for long-term investors*, Clarendon Lectures in Economic, 2002. 1, 31, 73, 82

Canner, Niko, N. Gregory Mankiw, and David N. Weil, "An asset allocation puzzle," *American Economic Review*, 1997, *87* (1), 181–191. 2, 5, 30, 40

Choi, James J, "Popular personal financial advice versus the professors," *Journal of Economic Perspectives*, 2022, *36* (4), 167–192. 2, 5, 30, 40

Choukhmane, Taha, Tim de Silva, Weidong Lin, and Matthew Akuzawa, "AI Financial Advice: Supply, Demand, and Life Cycle Implications," *Available at SSRN*, 2026. 4, 12, 40, 71, 72, 82

Christelis, Dimitris, Dimitris Georgarakos, and Michael Haliassos, "Differences in portfolios across countries: Economic environment versus household characteristics," *Review of Economics and Statistics*, 2013, *95* (1), 220–236. 5

Cocco, João F., Francisco J. Gomes, and Pascal J. Maenhout, "Consumption and portfolio choice over the life cycle," *Review of Financial Studies*, 2005, *18* (2), 491–533. 31, 40, 71, 73, 82

D'Acunto, Francesco and Alberto G. Rossi, "Robo-advice: Transforming households into rational economic agents," *Annual Review of Financial Economics*, 2023, *15*, 543–563. 4

_ , Nagpurnanand Prabhala, and Alberto G. Rossi, "The promises and pitfalls of robo-advising," *Review of Financial Studies*, 2019, *32* (5), 1983–2020. 4

Dimson, Elroy, Paul Marsh, and Mike Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns*, Princeton, NJ: Princeton University Press, 2002. 62

European Securities and Markets Authority, "Guidelines on Certain Aspects of the MiFID II Suitability Requirements," Guidelines ESMA35-43-3172, European Securities and Markets Authority, Paris 2022. 34

Falk, Armin, Anke Becker, Thomas Dohmen, Benjamin Enke, David Huffman, and Uwe Sunde, "Global evidence on economic preferences," *Quarterly Journal of Economics*, 2018, *133* (4), 1645–1692. 2, 5, 7, 26, 27, 62, 75

Financial Health Network, "Financial Health Pulse 2025 U.S. Trends Report," Technical Report, Financial Health Network 2025. 1

Financial Services Authority, "Assessing Suitability: Establishing the Risk a Customer Is Willing and Able to Take and Making a Suitable Investment Selection," Finalised Guidance FG11/05, Financial Services Authority, London 2011. 33

Foerster, Stephen, Juhani T Linnainmaa, Brian T Melzer, and Alessandro Previtero, "Retail financial advice: does one size fit all?," *The Journal of Finance*, 2017, *72* (4), 1441–1482. 3, 5

Foltyn, Richard and Jonna Olsson, "The Worth of a "Wo": Gender Bias in Financial Advice from LLMs," Working Paper, Norwegian School of Economics 2026. 5, 14, 22

Fuster, Andreas, Paul Goldsmith-Pinkham, Tarun Ramadorai, and Ansgar Walther, "Predictably unequal? The effects of machine learning on credit markets," *Journal of Finance*, 2022, *77* (1), 5–47. 5

Gálvez, Julio and Gonzalo Paz-Pardo, "Richer earnings dynamics, consumption and portfolio choice over the life cycle," *Journal of Financial Economics*, 2026, *176*, 104206. 73, 74

Guiso, Luigi and Paolo Sodini, "Household finance: An emerging field," in George M. Constantinides, Milton Harris, and René M. Stulz, eds., *Handbook of the Economics of Finance*, Vol. 2, Elsevier, 2013, pp. 1397–1532. 5

Jordà, Òscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M. Taylor, "The rate of return on everything, 1870–2015," *Quarterly Journal of Economics*, 2019, *134* (3), 1225–1298. 62

Kassner, Nora, Philipp Dufter, and Hinrich Schütze, "Multilingual LAMA: Investigating Knowledge in Multilingual Pretrained Language Models," in "Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics" Association for Computational Linguistics 2021. 6, 16

Klapper, Leora, Annamaria Lusardi, and Peter van Oudheusden, "Financial Literacy Around the World: Insights from the Standard & Poor's Ratings Services Global Financial Literacy Survey," Working Paper, Global Financial Literacy Excellence Center 2015. 2, 26, 27

Lindsey, Jack, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro et al., "On the Biology of a Large Language Model," Transformer Circuits Thread, Anthropic 2025. 3, 6, 16

Lloyds Banking Group, "Over 28 Million Adults Now Using AI Tools to Help Manage Their Money," Press Release, Lloyds Banking Group November 2025. 1

Love, David A., "Optimal rules of thumb for consumption and portfolio choice," *The Economic Journal*, 2013, *123* (571), 932–961. 19

Merton, Robert C., "Lifetime portfolio selection under uncertainty: The continuous-time case," *Review of Economics and Statistics*, 1969, *51* (3), 247–257. 24, 31, 71

_ , "Optimum Consumption and Portfolio Rules in a Continuous-Time Model," *Journal of Economic Theory*, 1971, *3* (4), 373–413. 73

Navigli, Roberto, Simone Conia, and Björn Ross, "Biases in large language models: Origins, inventory, and discussion," *ACM Journal of Data and Information Quality*, 2023, *15* (2). 5

OECD, *Pensions at a Glance 2023: OECD and G20 Indicators*, Paris: OECD Publishing, 2023. 62

_ , "Artificial Intelligence and Personal Finance," OECD Artificial Intelligence Papers 62, OECD Publishing, Paris 2026. 1

Qi, Jirui, Raquel Fernández, and Arianna Bisazza, "Cross-lingual consistency of factual knowledge in multilingual language models," in "Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing" 2023, pp. 10650–10666. 6, 16

Reuter, Jonathan and Antoinette Schoar, "Demand-side and supply-side constraints in the market for financial advice," *Annual Review of Financial Economics*, 2024, *16*, 229–255. 3, 5

Ross, Jillian and Andrew W Lo, "One Size Fits None: Heuristic Collapse in LLM Investment Advice," *arXiv preprint arXiv:2604.23837*, 2026. 2, 4

Rossi, Alberto G. and Stephen P. Utkus, "The diversification and welfare effects of robo-advising," *Journal of Financial Economics*, 2024, *157*, 103869. 4

Samuelson, Paul A., "Lifetime portfolio selection by dynamic stochastic programming," *Review of Economics and Statistics*, 1969, *51* (3), 239–246. 24, 31, 71

Schut, Lisa, Yarin Gal, and Sebastian Farquhar, "Do Multilingual LLMs Think in English?," *arXiv preprint arXiv:2502.15603*, 2025. 39

Tang, Tianyi, Wenyang Luo, Haoyang Huang, Dongdong Zhang, Xiaolei Wang, Wayne Xin Zhao, Furu Wei, and Ji-Rong Wen, "Language-Specific Neurons: The Key to Multilingual Capabilities in Large Language Models," in "Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)" Association for Computational Linguistics Bangkok, Thailand 2024, pp. 5701–5715. 3, 6, 16

van Rooij, Maarten, Annamaria Lusardi, and Rob Alessie, "Financial literacy and stock market participation," *Journal of Financial Economics*, 2011, *101* (2), 449–472. 5, 26

Wendler, Chris, Veniamin Veselovsky, Giovanni Monea, and Robert West, "Do llamas work in english? on the latent language of multilingual transformers," in "Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)" 2024, pp. 15366–15394. 3, 5, 16, 39

Wu, Zhaofeng, Xinyan Yu, Dani Yogatama, Jiasen Lu, and Yoon Kim, "The semantic hub hypothesis: Language models share semantic representations across languages and modalities," in "International Conference on Learning Representations," Vol. 2025 2025, pp. 53705–53723. 3, 6, 16, 39

A Appendix Tables and Figures

Table A1: The country panel: prompt language, investment amount, and local financial context

			Local financial context (Version C)		
Country	Language	Amount	Equity instrument	Government bonds	Pension reference
United States	English	USD 50,000	Broad US total-market index fund (Vanguard or Fidelity)	US Treasury bonds	401(k) through employer
United Kingdom	English	GBP 40,000	FTSE All-World ETF	UK Gilts	Workplace pension through employer
Australia	English	AUD 75,000	ASX-listed global index ETF	Australian government bonds	Superannuation fund through employer
Canada	English	CAD 70,000	Global index ETF	Government of Canada bonds	Workplace pension plus RRSP
India	English	INR 4,000,000	Nifty index fund	Government of India bonds	EPF plus NPS account
South Africa	English	ZAR 900,000	Global index ETF (MSCI World)	South African government bonds	Pension fund through employer
Sweden	Swedish	SEK 500,000	Global index fund (via Avanza or Nordnet)	Swedish government bonds	*Tjänstepension* through employer
Finland	Finnish	EUR 45,000	World index fund (MSCI World, via Nordnet or OP)	Finnish government bonds	*Työeläke* (TyEL) through employer
Germany	German	EUR 45,000	MSCI World ETF	*Bundesanleihen*	*Betriebliche Altersvorsorge* (bAV)
France	French	EUR 45,000	World equity ETF (MSCI World)	OATs	*Plan d'épargne retraite* (PER)
Spain	Spanish	EUR 45,000	World index fund (MSCI World)	Spanish government bonds	*Plan de pensiones de empresa*
Japan	Japanese	JPY 7,500,000	Global equity index fund (eMAXIS Slim All Country)	JGBs	iDeCo plus national pension (*kokumin nenkin*)
Brazil	Portuguese (BR)	BRL 250,000	Global-ETF BDRs (BURT39, BWOR39)	Tesouro Direto	Private pension plan (PGBL/VGBL)
Netherlands	Dutch	EUR 45,000	World index ETF (MSCI World)	Dutch government bonds	*Pensioenfonds* through employer
Italy	Italian	EUR 45,000	World index ETF (MSCI World)	BTPs	*Fondo pensione* through employer
Switzerland	German (CH)	CHF 45,000	MSCI World ETF	Swiss government bonds	*Pensionskasse* (second pillar) plus *Säule 3a*
Poland	Polish	PLN 200,000	World index fund (MSCI World ETF)	Polish government bonds	PPK plan through employer
South Korea	Korean	KRW 70,000,000	Global equity index fund (KODEX or TIGER world ETF)	KTBs	Retirement pension (*toejik yeongeum*) plus national pension (*gukmin yeongeum*)
China	Chinese (simplified)	CNY 350,000	QDII global index fund (MSCI World)	Chinese government bonds	Basic pension insurance (*jiben yanglao baoxian*) plus enterprise annuity
Mexico	Spanish (MX)	MXN 1,000,000	Global index fund (MSCI World ETF)	CETES	Afore retirement account
Turkey	Turkish	TRY 2,000,000	Global equity index fund (MSCI World ETF)	Turkish government bonds	BES private pension account

Notes: The upper block lists the six English-prompting countries, which have no Version B-cross cell, and the lower block the fifteen local-language countries. The investment amount is the round local-currency equivalent of $50,000, identical across Versions B, C, and B-cross within a country, and rendered in the prompt in the native number format (e.g. "500 000 kr" in Sweden, "45.000 Euro" in Germany). The three context columns give the country-specific vocabulary that Version C substitutes for the generic Version B wording ("a broad stock market index fund," "government bonds," "a standard pension plan"), condensed from the exact prompt sentences. In most countries the Version C equity instrument is a global or MSCI World-type tracker. India is the exception, naming a domestic Nifty index fund, so for India the Version C step also shifts the underlying equity exposure and not only its label. Pension terms that appear in Japanese, Korean, or Chinese script in the prompts are shown here in romanized form. The full prompt texts for every country, version, and question are provided in the replication package.

Table A2: Query accounting by experimental arm

Arm	Version	Countries	Question–age blocks	Cells	Queries
Country/language	B	21	Q1 at ages 30/40/55	63	40,950
	C	21	Q1 at ages 30/40/55	63	40,950
	B-cross	15	Q1 at ages 30/40/55	45	29,250
	B	21	Q2 and Q3 at age 40	42	27,300
	C	21	Q2 and Q3 at age 40	42	27,300
	B-cross	15	Q2 and Q3 at age 40	30	19,500
Stated investor	B$\gamma\rho\tau$ (six cells)	21	Q1 at age 40	126	81,900
Total				411	267,150

Notes: A cell is one country × version × question × age combination, and for the stated-investor arm one country × treatment cell. Every cell is queried by all thirteen models fifty times, so each cell contributes $13 \times 50 = 650$ queries. Version B-cross exists only for the fifteen non-English countries. The two pre-registered collection-robustness cells (temperature bookends at 0 and 1.0, and Version B with the investment amount stated in U.S. dollars) added 8,450 queries outside this grid and are reported in Section 4.6. The exploratory re-collection of the Version B, Question 1, age-40 cell with the Instructions block translated (Section 4.6, Appendix Table A18) added 5,460 queries, twenty per model and country, on October 29, 2026. The exploratory re-collection with the stake restated at purchasing-power parity (Section 4.6, Appendix Tables A13 and A14) added 13,325 queries on the same day, twenty-five per model and cell, of which 6,825 re-collect Version B unchanged as a same-day reference.

Table A3: Advice by model: level, dispersion, and stability

				Stability across repetitions	
Model	Vendor	Mean	SD	Within-cell SD	Identical cells
Gemini 3.1 Pro Preview	Google	0.37	0.08	0.037	0.05
GPT-5-mini	OpenAI	0.38	0.08	0.043	0.00
GPT-5.5	OpenAI	0.41	0.05	0.032	0.00
DeepSeek V4 Flash	DeepSeek	0.45	0.12	0.110	0.00
DeepSeek V4 Pro	DeepSeek	0.48	0.12	0.105	0.00
Grok 4.3	xAI	0.54	0.09	0.065	0.00
Gemini 3 Flash Preview	Google	0.56	0.07	0.000	0.62
Gemini 3.5 Flash	Google	0.57	0.06	0.020	0.24
Claude 4.8 Opus	Anthropic	0.58	0.04	0.000	0.52
Claude 4.6 Sonnet	Anthropic	0.63	0.04	0.000	0.67
Claude 4.6 Opus	Anthropic	0.65	0.07	0.000	0.95
Mistral Medium 3.5	Mistral	0.70	0.04	0.024	0.29
Claude 4.5 Haiku	Anthropic	0.71	0.04	0.007	0.48

Notes: Recommended equity share by model on the primary cell (Question 1, Version B, age 40), pooled across the twenty-one countries, roughly 1,050 responses per model. Within-cell SD is the median, across a model's twenty-one country cells, of the standard deviation over the fifty repetitions in the cell. Identical cells is the share of those cells whose fifty repetitions return one identical number, despite sampling at temperature 0.7. A variance decomposition of the primary cell attributes 69.5 percent of the total variation to model identity, 5.7 percent to the country, 9.5 percent to their interaction, and 15.3 percent to within-cell sampling.

Table A4: Primary-instrument recommendation on the inflation question by country (Question 3, Version B, age 40)

Country	Equities	Infl.-linked	Gold	Real estate	Other
Brazil	13.1	86.7	–	–	0.2
Turkey	17.5	66.0	16.3	–	0.2
Mexico	41.9	58.1	–	–	–
Poland	61.1	38.6	–	–	0.3
South Africa	82.0	16.3	–	–	1.7
India	88.5	7.8	3.7	–	–
Italy	93.2	6.8	–	–	–
South Korea	94.0	6.0	–	–	–
Spain	96.4	3.6	–	–	–
USA	96.8	2.9	–	–	0.3
Canada	97.2	2.6	–	–	0.2
France	97.8	2.2	–	–	–
United Kingdom	98.0	2.0	–	–	–
Australia	98.3	1.5	–	–	0.2
Switzerland	95.2	0.5	–	4.3	–
Netherlands	99.5	0.3	–	–	0.2
Japan	99.7	0.2	–	–	0.2
Sweden	99.8	0.2	–	–	–
China	96.3	0.2	–	–	3.5
Germany	88.6	0.2	–	11.2	–
Finland	100.0	–	–	–	–

Notes: Share of responses (in percent) recommending each instrument category as the primary way to protect savings against inflation (Question 3, Version B, age 40), pooled across the thirteen models. Each response is assigned a single primary instrument by a deterministic multilingual keyword classifier, and the five categories sum to 100; "Other" collects mixed, bond, and money-market funds and the rare response with no reported instrument. Countries are ordered by the combined inflation-protection share (inflation-linked bonds plus gold), descending. A dash denotes a category recommended in under 0.05 percent of responses. This is the categorical companion to the numeric outcome of Table A11, discussed in Section 4.5.

Table A5: Summary statistics on the stated-investor arm by treatment cell

Treatment cell	Mean	SD	p10	Median	p90	*N*
Risk-averse, limited pension	0.30	0.08	0.20	0.30	0.40	13646
Risk-averse, generous pension	0.29	0.07	0.20	0.30	0.40	13647
Risk-tolerant, limited pension	0.77	0.10	0.60	0.80	0.85	13643
Risk-tolerant, generous pension	0.79	0.09	0.70	0.80	0.90	13648
Horizon 1 year	0.36	0.24	0.00	0.30	0.70	13645
Horizon 20 years	0.76	0.05	0.70	0.75	0.80	13647

Notes: Distribution of the recommended equity share in the stated-investor arm (Version B$\gamma\rho\tau$, Question 1, age 40, English), pooled across the twenty-one countries and thirteen models. The upper block reports the four crossed cells that state the investor's risk attitude and pension generosity at a common five-year horizon, and the lower block the two cells that state only the investment horizon with no risk or pension statement. Each cell is queried by all thirteen models fifty times in each of the twenty-one countries. These summaries describe the sample behind the pre-registered stated-investor tests (H11–H13) in Section 6: the near-vertical response to stated risk attitude, the flat response to stated pension generosity, and the large response to horizon that the CRRA benchmark predicts should be absent.

Table A6: Stated-investor arm: estimated effects of the risk, pension, and horizon statements

	Contrast	Estimate	95% CI	p (wild)
H11	Risk-tolerant vs. risk-averse	46.3	[41.1, 51.4]	<0.001
H12	Generous vs. limited pension	−1.0	[−2.3, 0.2]	0.107
–	Risk × pension interaction	3.6	[2.0, 5.2]	<0.001
H13	20-year vs. 1-year horizon	39.9	[25.7, 53.4]	<0.001

Notes: Coefficients from the pre-registered stated-investor regressions (Version B$\gamma\rho\tau$, Question 1, age 40), estimated with country and model fixed effects on the sample summarized in Table A5. β_R, β_P, and the interaction come from equation (7) on the four crossed risk–pension cells, with the risk-averse, limited-pension cell as the omitted baseline. β_H comes from the horizon analog on the two horizon cells, with the one-year horizon as the baseline. Estimates and confidence intervals are in percentage points of the recommended equity share. p (wild) is the wild cluster bootstrap p-value (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$), and the confidence interval comes from the same procedure. The risk–pension interaction is descriptive and not a pre-registered test. These are the estimates behind hypotheses H11–H13 in Table A16 and Section 6.

Table A7: Exploratory balance-sheet probes: pension composition, currency hedging, and the liquid share

Contrast	Estimate	95% CI	p (wild)
Pension bond-like annuity (Merton: +)	4.8	[0.1, 9.6]	0.042
Pension equity-invested (Merton: −)	−16.9	[−21.8, −12.1]	<0.001
Fund currency-hedged (at mean σ_c)	0.4	[−1.5, 2.2]	0.659
Fund unhedged (at mean σ_c)	−1.3	[−3.2, 0.4]	0.149
Hedged $\times\sigma_c$ (Merton: +)	23.8	[6.1, 42.1]	0.010
Unhedged $\times\sigma_c$ (Merton: −)	36.0	[15.8, 55.5]	0.002
Stake = all liquid wealth (US only)	−23.3	[−30.4, −16.1]	<0.001
Stake = half of wealth (US only)	1.4	[−0.4, 3.3]	0.109
Stake = a third of wealth (US only)	2.5	[−0.1, 4.9]	0.057
Linear B/L slope, per step (US only)	7.6	[5.4, 9.8]	<0.001

Notes: Exploratory estimates from the three balance-sheet probes added to the stated-investor arm after the pre-registration was filed (Section 6.1). Each probe regresses the recommended equity share on indicators for its probe cells against the no-statement Version B and B-cross reference rows (Question 1, age 40, statements in English), with country and model fixed effects. Estimates and confidence intervals are in percentage points, except the two σ_c-interaction rows, which are in percentage points per unit of local-currency volatility σ_c (the annualized local-currency standard deviation of the world fund, spanning roughly 0.18 to 0.29 across the panel, so panel-range effects are about one tenth of the printed coefficients). The hedged and unhedged rows report the cell effects at the treated-sample mean σ_c, with the σ_c main effect absorbed by the country dummies. The pension-composition and hedging probes pool all twenty-one countries (N = 37,447 and 37,565, fifty repetitions per cell in the United States and twenty-five elsewhere), and the wealth ladder is U.S.-only (N = 2,595). The linear B/L row is the per-rung slope over the stated safe savings beside the stake, $D/W = 0, 1, 2$ in the notation of Section 6.1 (the stake stated as all, half, or a third of liquid wealth). Row labels carry the benchmark-implied sign where the benchmark makes a directional prediction. p (wild) is the two-sided wild cluster bootstrap p-value (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$), and the confidence interval comes from the same procedure. These estimates are post-filing exploratory deviations recorded in the design lock, in no Holm family, with probe cells collected in August 2026 against reference cells from the July 2026 main collection.

Table A8: Stated-investor effects by country

	Pre-registered cells			Exploratory probes		
Country	Risk tolerance	Horizon	Generous pension	Pension in stocks	Pension annuity	Hedged − unhedged
Japan	48.9	36.9	0.5	−20.0	1.7	0.5*
Turkey	49.7	35.2	0.3	−16.3†	6.3*	−2.4*
South Korea	48.2	34.6	0.6	−19.9	3.0	0.3
Spain	48.0	41.8	1.0	−16.2†	4.9	1.3
Poland	48.2	44.5*	0.9	−13.5†	6.1	3.1
Italy	47.8	42.5	1.4	−14.2†	5.7	3.0
China	48.8	34.1	0.6	−19.0	5.3	3.1
Brazil	48.3	37.6	1.0	−13.8†	7.5*	4.0
Mexico	49.9	34.3	0.9	−19.7*	3.9	1.2
France	47.1	44.9*	1.2	−15.4†	5.1	2.0
Germany	47.7	43.8*	0.5	−14.8†	5.1	0.9*
India	47.5	38.0	0.8	−18.9*	4.1	2.7
South Africa	46.7*	38.5	−0.1	−19.0	4.9	2.1
Switzerland	47.4	44.2†	0.6	−16.5†	5.8*	2.4
Netherlands	47.1	41.5*	0.9	−15.0†	5.2	1.9*
Canada	47.3*	41.1	1.2	−20.6	3.9	0.9*
United States	48.2	35.8	0.9	−21.7	3.5	3.1
United Kingdom	48.3	43.1*	0.9	−15.4†	3.4	1.2*
Australia	47.9	36.5	0.9	−19.9	3.4	1.2*
Finland	48.2	42.2*	1.0	−14.5†	4.7	2.7
Sweden	48.5	46.0†	0.6	−10.8†	3.9	0.8*
Mean of country effects	48.1	39.9	0.8	−16.9	4.6	1.7
Range	46.7 to 49.9	34.1 to 46.0	−0.1 to 1.4	−21.7 to −10.8	1.7 to 7.5	−2.4 to 4.0
Median CI half-width	±5.4	±14.4	±1.4	±5.3	±5.0	±1.6
Countries differing from U.S. (Holm)	0 of 20	2 of 20	0 of 20	12 of 20	0 of 20	0 of 20

Notes: Effect of each stated-investor treatment on the recommended equity share, in percentage points, estimated separately for each country (Question 1, age 40, statements in English). Risk tolerance is the risk-tolerant minus risk-averse difference pooled over the two pension statements, and generous pension the generous minus limited difference pooled over the two risk statements, both from the four crossed cells of equation (7). Horizon is the twenty-year minus one-year difference on the two horizon cells. Pension in stocks and pension annuity are the probe cells of Section 6.1 against the country's own no-statement reference, and hedged − unhedged is the difference between the two hedging cells, estimated on those cells alone. Each column comes from one regression that interacts the treatment with country indicators and absorbs model fixed effects, so each entry is the treatment effect within that country. † marks a country whose effect differs from the U.S. effect at the five percent level after Holm correction within the column's family of twenty contrasts, and * a difference significant at five percent before correction only. The summary rows give the unweighted mean and range of the twenty-one country effects, the median half-width of the per-country 95 percent confidence intervals, and the number of Holm-significant contrasts. All inference is the wild cluster bootstrap (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$). No joint test of equal effects across countries is reported, because the model-clustered covariance of twenty interaction coefficients has rank at most twelve with thirteen clusters. The per-country breakdown is a post-filing exploratory deviation recorded in the design lock.

Table A9: Country effects on the recommended equity share (Question 1, Version B, age 40)

Country	Estimate	95% CI	p (wild)	p (Holm)
Brazil	−11.0	[−15.0, −6.9]	<0.001	0.002
Italy	−8.6	[−11.9, −5.3]	<0.001	<0.001
Poland	−7.4	[−10.8, −3.9]	<0.001	0.011
France	−5.4	[−7.5, −3.4]	<0.001	<0.001
China	−4.4	[−8.8, −0.2]	0.041	0.371
Spain	−4.4	[−6.3, −2.5]	<0.001	<0.001
Switzerland	−4.3	[−7.6, −1.1]	0.016	0.200
Netherlands	−4.2	[−6.8, −1.7]	0.004	0.049
Germany	−4.0	[−7.7, −0.4]	0.034	0.335
Turkey	−3.1	[−6.5, −0.1]	0.044	0.371
Mexico	−2.7	[−4.2, −1.2]	<0.001	0.003
Finland	−2.4	[−5.0, 0.4]	0.086	0.602
India	−1.9	[−3.3, −0.4]	0.015	0.200
Canada	−1.1	[−2.1, −0.1]	0.029	0.317
South Africa	−0.9	[−2.4, 0.5]	0.219	1.000
Sweden	−0.2	[−3.7, 3.4]	0.931	1.000
Australia	0.6	[−0.7, 1.9]	0.343	1.000
United Kingdom	0.7	[−0.8, 2.2]	0.352	1.000
Japan	1.2	[−2.1, 4.8]	0.492	1.000
South Korea	1.4	[−2.8, 5.7]	0.496	1.000

Notes: Country fixed effects γ_c from equation (1), estimated on the primary cell (Question 1, Version B, age 40, $N = 13{,}639$) with model fixed effects and the United States as the omitted baseline. Estimates and confidence intervals are in percentage points of the recommended equity share, and countries are ordered by point estimate. p (wild) is the wild cluster bootstrap p-value (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$), and the 95 percent confidence interval comes from the same procedure. p (Holm) applies the pre-registered Holm step-down correction across the family of twenty contrasts. Figure 2 plots these estimates.

Table A10: Country effects on the mortgage question (Question 2, Version B, age 40)

Country	Estimate	95% CI	p (wild)	p (Holm)
Japan	−28.6	[−42.5, −15.0]	<0.001	0.002
China	−22.7	[−32.5, −13.7]	<0.001	0.002
Switzerland	−21.6	[−29.8, −14.0]	<0.001	0.002
Germany	−21.1	[−30.0, −12.0]	<0.001	0.002
Spain	−20.4	[−28.6, −12.3]	<0.001	0.002
Italy	−19.5	[−26.7, −12.5]	<0.001	0.002
Finland	−19.1	[−27.1, −11.3]	<0.001	0.002
South Korea	−17.3	[−24.4, −10.1]	<0.001	0.002
Australia	−17.1	[−24.8, −9.8]	<0.001	0.003
Poland	−16.2	[−21.0, −11.5]	<0.001	0.002
Netherlands	−16.0	[−23.2, −9.9]	<0.001	0.002
France	−15.7	[−24.6, −6.6]	<0.001	0.003
United Kingdom	−14.3	[−21.5, −8.1]	<0.001	0.003
Canada	−12.3	[−18.4, −6.6]	<0.001	0.003
Sweden	−10.0	[−16.5, −4.1]	0.001	0.006
Premise-acceptance group				
India	−3.8	[−6.7, −0.8]	0.015	0.060
South Africa	−2.4	[−4.3, −0.7]	0.002	0.008
Mexico	−1.9	[−5.3, 1.7]	0.262	0.787
Brazil	−1.7	[−6.2, 3.0]	0.452	0.904
Turkey	−1.6	[−8.4, 5.3]	0.628	0.904

Notes: Country fixed effects on the recommended fraction to invest in a broad stock market index fund rather than prepay a 4 percent mortgage (Question 2, Version B, age 40, $N = 13{,}619$), estimated with the specification of equation (1) with model fixed effects and the United States as the omitted baseline. Estimates and confidence intervals are in percentage points, and countries are ordered by point estimate within group. Inference is the model-level wild cluster bootstrap (Webb six-point weights, 9,999 replications, $G = 13$), and p (Holm) applies the step-down correction across the twenty contrasts, which form their own family separate from Question 1 as pre-registered. The five countries below the separator are the pre-specified premise-acceptance group, whose prevailing mortgage rate at the filing date exceeded twice the prompted 4 percent, so their contrasts test whether the model accepts or corrects the premise rather than whether it calibrates to local fundamentals. Figure 5 plots these estimates alongside those of the other two questions.

Table A11: Country effects on the inflation question (Question 3, Version B, age 40)

Country	Estimate	95% CI	p (wild)	p (Holm)
Japan	−11.3	[−13.7, −8.9]	<0.001	<0.001
China	−7.2	[−9.6, −4.7]	<0.001	<0.001
South Korea	−6.9	[−10.3, −3.5]	0.001	0.020
Netherlands	−6.5	[−8.9, −4.2]	<0.001	0.002
Switzerland	−5.0	[−7.3, −2.7]	<0.001	0.006
Italy	−5.0	[−7.3, −2.8]	<0.001	0.002
France	−4.3	[−6.9, −1.8]	0.003	0.041
Spain	−4.0	[−6.6, −1.3]	0.007	0.091
Finland	−2.5	[−5.2, −0.1]	0.041	0.473
Germany	−2.0	[−4.1, −0.1]	0.039	0.473
Poland	−1.8	[−5.3, 1.6]	0.269	1.000
United Kingdom	−1.5	[−3.2, 0.1]	0.064	0.636
Turkey	−0.9	[−4.5, 2.7]	0.596	1.000
Australia	−0.2	[−2.2, 1.9]	0.862	1.000
Sweden	−0.1	[−3.1, 2.7]	0.930	1.000
Mexico	0.1	[−3.3, 3.4]	0.973	1.000
India	0.8	[−2.5, 3.9]	0.618	1.000
Canada	1.6	[−0.4, 3.6]	0.100	0.799
South Africa	2.3	[−0.3, 4.9]	0.082	0.739
Brazil	2.6	[−1.0, 5.9]	0.139	0.972

Notes: Country fixed effects on the recommended fraction of savings to move out of a 1 percent savings account in the face of 3 percent inflation (Question 3, the registration's Q4, Version B, age 40, N = 13,637), estimated with the specification of equation (1) with model fixed effects and the United States as the omitted baseline. Estimates and confidence intervals are in percentage points, and countries are ordered by point estimate. Inference is the model-level wild cluster bootstrap (Webb six-point weights, 9,999 replications, G = 13), and p (Holm) applies the step-down correction across the twenty contrasts, which form their own family. The numeric fraction is the primary Question 3 outcome because the collection produced no refusals and the pre-specified refusal contingency never bound (Section 3.7). Figure 5 plots these estimates alongside those of the other two questions, and Table A4 reports the categorical companion.

Table A12: The age gradient and its interaction with pension generosity (Question 1, Version B)

	(1) No interaction	(2) Continuous $\tilde{\rho}$	(3) Median split
Age 30 (vs. 40)	6.24	6.24	6.24
	(0.57)	(0.57)	(0.57)
Age 55 (vs. 40)	-11.13	-11.13	-11.34
	(1.90)	(1.90)	(1.96)
× std. replacement rate (ψ)		-0.02	
		(0.33)	
× above-median pension			0.44
			(0.61)
Country fixed effects	Yes	Yes	Yes
Model fixed effects	Yes	Yes	Yes
Wild p: Age 30	0.000	0.000	0.000
Wild p: Age 55	0.000	0.000	0.000
Wild p: interaction		0.964	0.489
Observations	40,910	40,910	40,910
Model clusters (G)	13	13	13

Notes: Estimates of equation (4) on Question 1, Version B, pooled across the three prompted ages (30, 40, 55), the twenty-one countries, and the thirteen models. The outcome is the recommended equity share in percentage points, and every column includes country and model fixed effects with the United States and age 40 as the omitted baselines. Column (1) fits the age effects alone. Column (2) adds the pre-registered interaction of the age-55 indicator with the standardized OECD net pension replacement rate $\tilde{\rho}_c$ (the coefficient ψ of hypothesis H6). Column (3) replaces it with the coarser interaction of age 55 with an above-median replacement-rate indicator. Parentheses report CR1 cluster-robust standard errors clustered by model. The wild cluster bootstrap p-values (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$) are the inference of record and are reported in the lower panel.

Table A13: Collection-robustness cells: temperature bookend, numeral magnitude, and real stake

Panel A: Country contrasts vs. the U.S. at the temperature bookends

	$T = 0$		$T = 0.7$ (design)		$T = 1.0$	
Country	Estimate	95% CI	Estimate	95% CI	Estimate	95% CI
Brazil	−10.0	[−14.2, −5.8]	−11.0	[−15.0, −6.9]	−11.4	[−15.1, −7.6]
Germany	−3.6	[−7.4, 0.2]	−4.0	[−7.7, −0.4]	−4.2	[−7.7, −0.7]
Sweden	−0.3	[−3.7, 3.1]	−0.2	[−3.7, 3.4]	−0.7	[−4.0, 2.5]
Japan	2.5	[−2.0, 7.0]	1.2	[−2.1, 4.8]	0.2	[−3.4, 4.0]
Mean within-cell SD	0.035		0.038		0.041	

Panel B: Stating the amount as $50,000 instead of the native-currency figure

Country	Mean (Version B)	Mean (B-USD)	Shift	95% CI	p (wild)
Germany	0.53	0.54	1.2	[−0.4, 2.7]	0.130
Japan	0.58	0.55	−2.8	[−5.4, −0.4]	0.010
Sweden	0.57	0.58	1.2	[−0.8, 3.4]	0.255

Panel C: Stating the stake at purchasing-power parity (exploratory)

Country	Ratio	Mean (B)	Mean (B-PPP)	Shift	95% CI	p (wild)	p (Holm)
India	4.00	0.54	0.55	1.7	[0.6, 2.9]	0.003	0.062
Turkey	3.33	0.54	0.55	0.9	[−1.6, 3.6]	0.457	1.000
South Africa	2.40	0.56	0.58	2.4	[0.4, 4.7]	0.016	0.264
Brazil	2.00	0.45	0.46	1.8	[0.5, 3.1]	0.007	0.133
Mexico	2.00	0.53	0.54	1.1	[−1.2, 4.3]	0.759	1.000
Poland	2.00	0.48	0.50	2.3	[−0.5, 5.1]	0.108	1.000
China	1.94	0.50	0.52	1.9	[0.3, 3.6]	0.023	0.373
South Korea	1.75	0.57	0.58	0.1	[−1.7, 1.7]	0.927	1.000
Spain	1.61	0.50	0.52	2.1	[0.8, 3.4]	0.004	0.072
Japan	1.58	0.58	0.59	1.4	[−1.3, 4.3]	0.311	1.000
Italy	1.50	0.47	0.48	1.2	[−0.4, 3.8]	0.346	1.000
France	1.32	0.50	0.51	0.5	[−0.9, 1.8]	0.486	1.000
Germany	1.29	0.50	0.51	0.2	[−0.5, 0.9]	0.545	1.000
Netherlands	1.23	0.52	0.52	0.3	[−0.5, 1.1]	0.462	1.000
Canada	1.22	0.55	0.55	0.2	[−0.6, 1.2]	0.663	1.000
United Kingdom	1.21	0.57	0.57	−0.4	[−1.3, 0.6]	0.382	1.000
Finland	1.20	0.52	0.54	1.5	[−0.3, 4.5]	0.247	1.000
Sweden	1.18	0.57	0.56	−0.8	[−2.6, 0.2]	0.656	1.000
Australia	1.10	0.57	0.58	1.0	[−0.6, 2.8]	0.250	1.000
Switzerland	0.93	0.52	0.52	0.6	[−0.5, 1.8]	0.275	1.000
Pooled, 20 countries				1.0	[0.5, 1.7]	<0.001	
Per doubling of the real stake				−1.2	[−2.0, −0.6]	<0.001	
net of a common shift				−0.9	[−1.4, −0.4]	0.003	

Notes: Panels A and B report the two pre-registered collection-robustness cells (Section 4.6), re-collected on July 6, 2026 alongside the main corpus (Section 3.7). Panel A re-estimates the country fixed effects of equation (1) separately at each sampling temperature on the five bookend countries (Question 1, Version B, age 40, United States omitted), in percentage points of the recommended equity share. The $T = 0.7$ column uses the primary-collection cells and the bookend columns use the re-collected cells, so the $T = 0.7$ estimates differ from Appendix Table A9 only in the estimation sample being restricted to these five countries. The dispersion row reports the mean across country–model cells of the within-cell standard deviation of the recommended share, in fraction units. Panel B compares Version B, whose stake is a round native-currency amount, with an otherwise identical prompt stating the amount as $50,000, for the three pre-registered countries with country and language held fixed. The mean columns are recommended equity shares, and the shift is the B-USD minus Version-B difference in percentage points, estimated with model fixed effects. Panel C reports the exploratory re-collection of October 29, 2026, which restates the stake in the twenty non-U.S. Version B prompts at purchasing-power parity (World Bank, 2024) and compares it with Version B re-collected the same day, twenty-five repetitions per cell. The ratio is the filed over the restated stake. The shift has model fixed effects and is Holm-corrected across the twenty countries. The pooled row adds country fixed effects, and the last two rows give the slope on the log real stake per doubling, without and with an indicator for the restated prompt. All p-values and confidence intervals come from the wild cluster bootstrap (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$). The pooled estimates quoted in Section 4.6 average over the countries shown here.

Table A14: Country contrasts with the stake at purchasing-power parity

Panel A: Country contrasts vs. the U.S., percentage points

Country	Stake ratio	Main collection (July)	Same-day re-collection		PPP − filed
			Filed stake	PPP stake	
Brazil	2.00	−11.0†	−11.5†	−9.7†	1.8
Italy	1.50	−8.6†	−8.9†	−7.6	1.2
Poland	2.00	−7.4†	−8.4†	−6.1†	2.3
France	1.32	−5.4†	−6.1†	−5.6†	0.5
China	1.94	−4.4	−6.3†	−4.4	1.9
Spain	1.61	−4.4†	−6.0†	−3.9†	2.1
Switzerland	0.93	−4.3	−4.3	−3.7	0.6
Netherlands	1.23	−4.2†	−4.4†	−4.1†	0.3
Germany	1.29	−4.0	−5.7	−5.6	0.1
Turkey	3.33	−3.1	−1.7	−0.8	0.9
Mexico	2.00	−2.7†	−3.2	−2.2	1.1
Finland	1.20	−2.4	−3.7	−2.1	1.5
India	4.00	−1.9	−2.6†	−0.9	1.7
Canada	1.22	−1.1	−1.5	−1.2	0.2
South Africa	2.40	−0.9	−0.2	2.2	2.4
Sweden	1.18	−0.2	0.7	−0.0	−0.8
Australia	1.10	0.6	0.5	1.6	1.0
United Kingdom	1.21	0.7	0.9	0.5	−0.4
Japan	1.58	1.2	1.5	2.9	1.5
South Korea	1.75	1.4	1.3	1.4	0.1

Panel B: Calibration slope on each set of contrasts

Contrasts	$\hat{\kappa}$	95% CI	p ($\kappa = 0$)	p ($\kappa = 1$)
Main collection (July)	0.13	[−0.14, 0.33]	0.317	<0.001
Same day, filed stake	0.10	[−0.18, 0.34]	0.541	<0.001
Same day, PPP stake	0.07	[−0.22, 0.29]	0.661	<0.001

Notes: Panel A reports the country fixed effects of equation (1) (Question 1, Version B, age 40, United States omitted, model fixed effects) in three samples. The main-collection column reproduces Appendix Table A9 (July 2026, fifty repetitions per cell). The two same-day columns come from the exploratory re-collection of October 29, 2026 (twenty-five repetitions per cell), with the filed stake and with the stake restated at purchasing-power parity. Both use the same-day U.S. rows as the baseline, because the U.S. stake does not change. The last column is their difference, which equals the within-country shift in Appendix Table A13, Panel C. The stake ratio is the filed stake divided by the restated stake. † marks a contrast significant at the five percent level after Holm correction across the twenty countries, using the wild cluster bootstrap (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$). Panel B regresses each column's twenty contrasts on the benchmark deviations at the baseline calibration, as in equation (16), with the wild bootstrap over countries. The benchmark does not depend on the size of the stake, so the regressor is the same in all three rows. Post-filing exploratory deviation recorded in the design lock.

Table A15: Country-specific calibration inputs for the normative benchmark

Country	σ_c (%)	r_c (%)	z_c (std)	ρ_c (%)	$\bar{w}_c$ (000s)	R_c (age)	L_c (yrs)	g_c (%)	τ_c^m	π_c (%)	α_c^S ($\bar{\gamma}$=3)
USA (USD)	18.4	0.4	0.12	51.3	87	67	18	1.1	0.15	2.5	0.52
United Kingdom (GBP)	19.1	-0.3	0.05	54.2	47	68	17	0.3	0.00	2.9	0.49
Sweden (SEK)	19.2	-0.3	0.05	66.3	546	70	16	1.0	0.30	2.1	0.48
Finland (EUR)	18.5	-0.1	-0.28	65.7	51	68	17	0.5	0.00	2.0	0.52
Germany (EUR)	18.5	-0.6	-0.04	53.3	55	67	18	0.8	0.00	2.2	0.52
France (EUR)	18.5	0.2	-0.03	70.0	44	65	22	0.6	0.00	1.9	0.52
Spain (EUR)	18.5	0.9	-0.16	86.3	35	65	22	0.5	0.00	2.0	0.52
Japan (JPY)	20.4	-0.1	-0.36	42.4	5,186	65	22	-0.2	0.10	0.9	0.43
Brazil (BRL)	27.5	0.8	-0.25	97.5	56	64	20	1.2	0.00	5.5	0.23
Australia (AUD)	19.0	0.9	0.14	53.0	105	67	20	0.7	0.00	2.7	0.49
Canada (CAD)	18.5	0.3	0.18	45.1	86	65	21	0.9	0.00	2.2	0.52
India (INR)	19.6	1.0	-0.28	44.6	273	58	23	4.9	0.10	6.4	0.46
South Africa (ZAR)	19.0	3.7	0.97	8.9	339	60	20	0.1	0.00	5.5	0.49
Netherlands (EUR)	18.5	-0.5	0.19	96.0	61	70	15	0.1	0.35	2.4	0.52
Italy (EUR)	18.5	1.3	-0.09	79.0	35	70	16	-0.3	0.00	2.0	0.52
Switzerland (CHF)	18.7	0.2	-0.02	47.5	98	65	22	0.7	0.20	0.5	0.51
Poland (PLN)	22.5	0.7	-0.07	40.6	101	62	21	2.6	0.00	3.7	0.35
South Korea (KRW)	18.4	1.0	-0.04	38.9	53,916	65	21	1.3	0.05	2.2	0.53
China (CNY)	19.1	1.4	-0.02	103.6	106	60	22	6.8	0.05	2.2	0.48
Mexico (MXN)	21.7	2.9	-0.14	79.6	277	65	18	0.1	0.00	4.4	0.38
Turkey (TRY)	29.1	-4.2	0.02	96.4	743	64	18	11.5	0.00	19.4	0.21

Notes: Actual country-level inputs to the normative benchmark of Appendix C, one row per country. σ_c is the annualized volatility of a world equity index measured in local currency, and r_c is the real safe rate, local government yields less expected inflation, taken as a trailing twenty-year mean. The equity premium $\mu_c - r_c$ is held common across countries at 5.32 percent, the arithmetic mean of the net MSCI World index in U.S. dollars over the U.S. long-term rate (1999–2025), so under common preferences the cross-country variation in α_c^S enters through σ_c alone. z_c is the country-mean GPS willingness to take risks, standardized at the individual level (Falk et al., 2018). ρ_c is the OECD net pension replacement rate for the average earner and $\bar{w}_c$ is the average annual wage in thousands of local currency (OECD, 2023). R_c is the normal retirement age, L_c the years of life expectancy at retirement, and g_c the real wage growth. τ_c^m is the assumed effective mortgage-interest deductibility for the mortgage question, which is income- and itemization-dependent rather than point-sourced, so α_c^{Q2} is reported over a $\tau^m \in [0, 0.5]$ band. π_c is expected inflation from central-bank targets and consensus forecasts. The final column is the derived standalone benchmark share α_c^S at the baseline grid point ($\bar{\gamma} = 3$, $\eta = 0$, world index in local currency), which reproduces the benchmark column of Table A20 and ranges from 0.21 in Turkey to 0.53 in South Korea. Return and wage moments are unhedged local-currency values, and Table A20 reports the currency-hedged variant. Standard errors are not separately estimated for each input. The released benchmark output instead spans the full sensitivity grid $\bar{\gamma} \in \{2, 3, 5, 10\}$, $\eta \in \{0, 0.3, 0.6\}$, $\zeta \in \{0, 0.2\}$, and the thin-data group (Brazil, China, India, Turkey, Mexico, South Africa, Poland) relies on shorter MSCI and local-bond series that carry wider bands, with DFBETA influence diagnostics pre-specified for the calibration regression and reported in Appendix C.6. Sources are DMS (Dimson, Marsh and Staunton, 2002) and JST (Jordà, Knoll, Kuvshinov, Schularick and Taylor, 2019) for long-run return robustness, MSCI local-currency series for the baseline moments, Falk et al. (2018) for z_c, and OECD Pensions at a Glance and Average Annual Wages for the pension and wage inputs. The prompt stake W is the round local-currency equivalent of $50,000.

Table A16: The pre-registered hypotheses, their tests, and their outcomes

	Statement	Test and correction	Status	Result
H1	Equity recommendations differ systematically across countries (country fixed effects in Eq. (1)).	Joint wild-bootstrap Wald test of all twenty contrasts, with the per-country contrasts corrected as a single Holm family.	Primary	Supported. $F(20, 12) = 20.56$, $p = 0.006$. Seven of twenty contrasts survive the Holm correction, all negative, from Brazil (-11.0 pp) to Mexico (-2.7 pp).
H2	Advice does not move with country fundamentals (the U.S.-default anchor): $\kappa = 0$ in the calibration regression (16).	Wild bootstrap over the twenty non-U.S. countries, HC2 standard errors alongside.	Second altitude[a]	Descriptive read consistent with compression. $\hat{\kappa} = 0.13$ (HC2 se 0.11) at the baseline grid point, ranging 0.07–0.43 across the sensitivity grid and always far below one. The inference of record is the wild bootstrap over countries, which does not reject $\kappa = 0$ ($p = 0.32$) and rejects full calibration $\kappa = 1$. The two-stage bootstrap that propagates only the first-stage model-sampling uncertainty, holding the country panel fixed, bounds the slope in $[0.07, 0.19]$, but this reflects a narrower source of variation than the primary procedure and is reported as secondary. The registered takeaway is no movement with fundamentals detectable under the primary test, and at most a small positive slope far below the unit slope of full calibration (Appendix C.6).
H3	The U.K. contrast is equivalent to zero within a ± 5 pp margin, consistent with shared Anglo-American norms.	Two one-sided tests: supported when the 90 percent wild-bootstrap CI for γ_{UK} lies inside the margin.	Primary	Supported. $\hat{\gamma}_{UK} = +0.7$ pp with 95 percent CI $[-0.8, +2.2]$, so the 90 percent interval of the two one-sided tests is inside the margin as well.
H4	Asking in the local language rather than in English shifts advice, holding the country name fixed ($\lambda \neq 0$ in Eq. (3)).	Single pre-specified coefficient, reported uncorrected.	Primary	Supported. $\hat{\lambda} = -1.3$ pp, $p = 0.007$, CI $[-2.3, -0.4]$.
H5	Naming local instruments and pension vehicles shifts advice beyond the country name alone ($\phi \neq 0$ in Eq. (2)).	Single pre-specified coefficient, reported uncorrected.	Primary	Not supported at the 5 percent level. $\hat{\phi} = -0.7$ pp, $p = 0.069$, CI $[-1.5, +0.1]$.
H6	The age-55 tilt toward bonds is smaller where pension replacement rates are higher (age-55 $\times$ standardized replacement rate in Eq. (4)).	Single pre-specified interaction, reported uncorrected.	Secondary	Not supported. -0.02 pp, $p = 0.964$. The coarser above-median split gives $+0.4$ pp, $p = 0.489$.
H7	Country effects covary positively with the country's GPS risk-taking score.	Bivariate slope over the twenty non-U.S. countries, wild bootstrap over countries.	Second altitude[a]	Descriptive read unsupported. Slope $+0.028$ (HC2 se 0.021), wild $p = 0.272$.
H8	Country effects covary positively with the country's financial literacy score.	Bivariate slope over the twenty non-U.S. countries, wild bootstrap over countries.	Second altitude[a]	Descriptive read unsupported. Slope $+0.055$ (HC2 se 0.040), wild $p = 0.178$.
H9	The rate at which the explanation text invokes a given rationale differs across countries (Section 7).	Country fixed effects per admitted category, one Holm family per category.	Second altitude[b]	Supported where the design gives it variation. Local-context references are elevated in all twenty non-U.S. countries ($+30$ to $+78$ pp, nineteen of twenty Holm-surviving) and inflation reasoning is most elevated in the five high-inflation countries (Turkey $+56$ pp, South Africa $+33$ pp, both Holm-surviving). Pension-as-exposure rates are flat everywhere (all $\lvert$estimates$\rvert \leq 8.5$ pp, none survive). Horizon

Table A17: Model-tracking waves: stability of the advice across model generations

	All models		New models	Ca
	July	September	September	July
Panel A: Level and composition of the variance				
Models	13	21	14	7
Responses kept	13,645	11,011	7,339	7,347
Repetitions per cell	50	25	25	50
Mean equity share (%)	54.9	49.4	45.8	55.4
Variance share, model (%)	78.0	70.5	42.9	73.9
Variance share, country (%)	2.8	4.7	14.8	1.5
Variance share, model × country (%)	4.3	6.7	10.6	2.1
Variance share, within cell (%)	14.9	18.1	31.7	22.5
Range of country means, pooled over models (pp)	9.4	10.3	13.1	7.0
SD of country means net of noise, average over models (pp)	3.2	3.5	3.8	2.2
Non-U.S. cells below the model's own U.S. cell (%)	67	77	79	62
Slope of country means on α_c^S	0.118	0.081	0.080	0.073
(robust standard error)	(0.091)	(0.113)	(0.143)	(0.070)

Panel B: Carried-over models, same identifier

	Mean, July	Mean, September	Change (se)	Δ SD of countr
DeepSeek V4 Flash	46.3	54.8	+8.46 (1.08)	+2.47
Grok 4.3	54.5	57.0	+2.43 (0.37)	−0.70
GPT-5-mini	37.2	37.5	+0.22 (0.26)	+0.09
Claude Haiku 4.5	72.5	72.6	+0.06 (0.12)	+0.02
Mistral Medium 3.5	70.1	70.0	−0.07 (0.05)	−0.08
Gemini 3 Flash	59.4	59.0	−0.44 (0.57)	+2.53
DeepSeek V4 Pro	47.4	45.0	−2.31 (0.52)	+0.44

Panel C: Slope on release date, percentage points per year (27 models, 34 points)

	Estimate	Cluster se	p (wild)
SD of country means net of noise	+0.98	0.53	0.19
with wave fixed effects	+0.92	0.59	0.30

Notes: Question 1, age 40, one English prompt per country (Version B in the six English-prompting countries, Version B-cross in the oth July columns are the main collection of July 3–4, 2026 cut to this slice, and the September columns the tracking wave of September 19, 20 September models absent from the paper panel. Carried-over models are the seven paper-panel models re-queried under the same identifier. of squares relative to the total on response-level data, as in Section 4.1. With 25 rather than 50 repetitions per cell, sampling noise raises the amount. The SD of country means net of noise subtracts the average sampling variance of the cell means, s^2_{mc}/n_{mc}, from the variance of a r The benchmark slope regresses the twenty-one country means, pooled over the models in the column, on the standalone benchmark of Apper effects, so it is a descriptive analog of κ and its changes across columns are compositional. Panel B: mean recommended equity shares in pe the twenty-one within-country changes in cell means, with its standard error across countries. The SD change and the largest absolute cell c countries at a given mean. Seven comparisons, uncorrected. Panel C: each model-by-wave outcome regressed on the model's OpenRouter list entering once per wave at the same date, standard errors clustered by model, and wild cluster bootstrap p-values (Webb weights, 9,999 rep collinear with the wave, so only the wave-fixed-effects row separates vintage from collection date. Post-filing exploratory deviation recorded in

Table A18: Translating the Instructions block: the primary cell re-collected with local-language instructions

	Estimate	95% CI	p (wild)	p (Holm)
Panel A: Translated minus English instructions, pooled				
All twenty-one countries	−1.2	[−2.7, 0.2]	0.095	–
English-prompting countries (identical prompts)	−0.6	[−1.8, 0.4]	0.257	–
Non-English countries	−1.4	[−3.2, 0.1]	0.084	–
Language component (non-English minus English)	−0.8	[−1.8, 0.2]	0.130	–
Panel B: Translated minus English instructions, by country				
Non-English countries (instructions translated)				
Finland	−4.6	[−7.1, −2.2]	<0.001	0.006
Mexico	−3.7	[−6.8, −0.8]	0.013	0.250
China	−3.6	[−8.0, 0.5]	0.098	1.000
Italy	−2.0	[−4.8, 0.6]	0.127	1.000
Poland	−1.8	[−5.7, 1.7]	0.368	1.000
South Korea	−1.1	[−3.7, 1.6]	0.396	1.000
Japan	−1.1	[−3.8, 1.8]	0.461	1.000
Netherlands	−1.0	[−5.1, 2.8]	0.601	1.000
Turkey	−0.9	[−2.6, 0.9]	0.310	1.000
Germany	−0.8	[−5.0, 3.1]	0.691	1.000
Spain	−0.5	[−3.7, 2.8]	0.755	1.000
France	−0.4	[−4.1, 3.0]	0.787	1.000
Switzerland	−0.4	[−4.7, 3.7]	0.867	1.000
Sweden	−0.2	[−2.4, 2.2]	0.869	1.000
Brazil	1.0	[−1.7, 3.7]	0.465	1.000
English-prompting countries (prompts identical in both arms)				
Canada	−1.5	[−3.6, 0.3]	0.208	1.000
Australia	−1.2	[−2.5, 0.0]	0.057	1.000
India	−1.1	[−3.0, 0.6]	0.309	1.000
United Kingdom	−0.4	[−1.2, 0.5]	0.340	1.000
South Africa	−0.2	[−2.4, 2.3]	0.856	1.000
United States	0.5	[−2.3, 3.4]	0.750	1.000
Panel C: Translated minus English instructions, by model (descriptive)				
	English	Non-English	Component	(s.e.)
Gemini 3.5 Flash	−4.9	−9.1	−4.2	(2.3)
Gemini 3 Flash Preview	−0.3	−4.0	−3.8	(1.9)
Grok 4.3	2.6	1.2	−1.4	(1.3)
Claude 4.8 Opus	−0.5	−1.9	−1.4	(1.7)
Claude 4.6 Opus	0.1	−1.2	−1.2	(0.8)
Mistral Medium 3.5	0.0	−0.7	−0.6	(0.6)
DeepSeek V4 Pro	−3.6	−4.1	−0.5	(2.0)
Claude 4.5 Haiku	−0.1	−0.2	0.0	(0.8)
GPT-5-mini	−0.5	−0.3	0.2	(0.7)
DeepSeek V4 Flash	0.3	0.6	0.3	(4.5)
Gemini 3.1 Pro Preview	−0.5	−0.1	0.5	(1.0)
Claude 4.6 Sonnet	−0.5	0.4	0.9	(0.6)
GPT-5.5	−0.5	0.9	1.4	(0.7)

Notes: Exploratory re-collection of the primary cell (Question 1, Version B, age 40, twenty-one countries, thirteen models, twenty repetitions per cell, 5,460 queries, October 29, 2026) with the Instructions block in the local language for the fifteen non-English countries, compared with the main collection of July 2026 (fifty repetitions per cell). Entries are the October minus July difference in the recommended equity share, in percentage points, from a regression with country and model fixed effects. Panel A pools the difference over all countries, over the six English-prompting countries, whose prompts are identical in the two collections and so measure drift in the models over the interval, and over the fifteen non-English countries, and reports the language component as the non-English minus the English difference. Panel B estimates the difference for each country, with the Holm correction treating the twenty-one countries as a single family. p (wild) is the wild cluster bootstrap p-value (Webb six-point weights, 9,999 replications, clustered by model, $G = 13$), and the confidence interval is the 95 percent wild bootstrap interval. Panel C estimates the English and non-English differences and the language component separately for each model. These are descriptive. Within one model there is no model cluster to bootstrap over, and the standard error in parentheses is clustered by country ($G = 21$). Responses truncated at the token limit are excluded as in the main analysis, which removes 45 responses from the October collection, all from the two DeepSeek models.

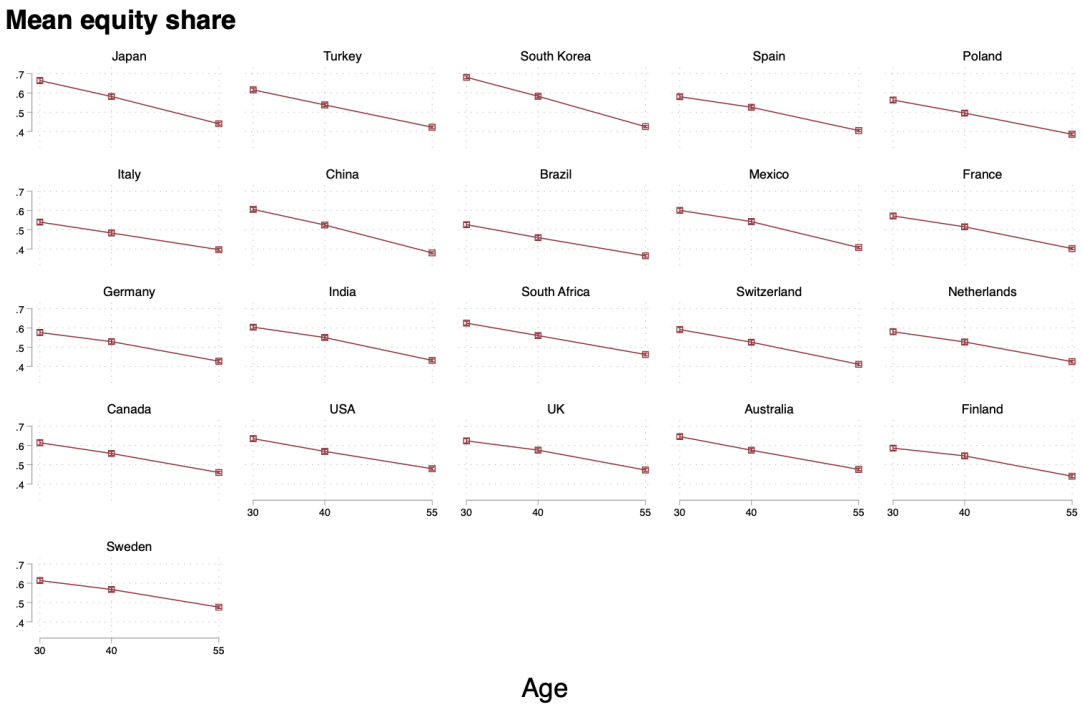
Table A19: Country effects on the provisional equity share in the elicitation probe

Country	Estimate	95% CI	p (wild)	p (Holm)
Brazil	−8.5	[−12.6, −4.3]	<0.001	0.004
Japan	−4.8	[−6.6, −3.0]	<0.001	0.002
Poland	−4.5	[−6.9, −1.9]	0.004	0.060
Italy	−4.4	[−6.4, −2.2]	<0.001	0.015
Mexico	−4.1	[−6.6, −1.6]	0.005	0.071
France	−3.6	[−5.5, −1.8]	0.001	0.016
Spain	−3.5	[−5.2, −1.8]	<0.001	0.004
China	−3.3	[−6.1, −0.6]	0.023	0.233
Switzerland	−3.1	[−5.1, −1.1]	0.005	0.071
Germany	−2.4	[−4.7, 0.1]	0.055	0.491
India	−2.1	[−3.6, −0.7]	0.005	0.071
Netherlands	−1.9	[−4.1, 0.3]	0.082	0.639
Canada	−1.8	[−2.9, −0.5]	0.006	0.071
South Africa	−0.7	[−2.5, 1.2]	0.448	1.000
Finland	−0.4	[−2.2, 1.4]	0.642	1.000
South Korea	−0.3	[−2.6, 1.9]	0.761	1.000
United Kingdom	0.3	[−1.0, 1.5]	0.647	1.000
Sweden	0.8	[−2.4, 4.1]	0.645	1.000
Australia	0.9	[−0.1, 2.0]	0.080	0.639
Turkey	1.2	[−3.1, 5.5]	0.546	1.000

Notes: Country fixed effects on the provisional equity share in the exploratory elicitation probe of Section 4.6 (October 28, 2026), in which the model lists the questions it would ask the investor before advising and states the share it would recommend provisionally with no further information (Question 1, age 40, one English prompt per country, Version B in the six English-prompting countries and Version B-cross in the other fifteen, ten repetitions per model and country, 2,730 queries). The specification is that of equation (1): model fixed effects, the United States as the omitted baseline, and standard errors clustered by model. Estimates and confidence intervals are in percentage points, and countries are ordered by point estimate. p (wild) is the wild cluster bootstrap p-value (Webb six-point weights, 9,999 replications, $G = 13$) and p (Holm) applies the Holm step-down correction to the twenty contrasts as a single family. Responses that failed to parse or were truncated at the token limit are excluded as in the main analysis. The main-arm estimates on the one-shot question are in Appendix Table A9.

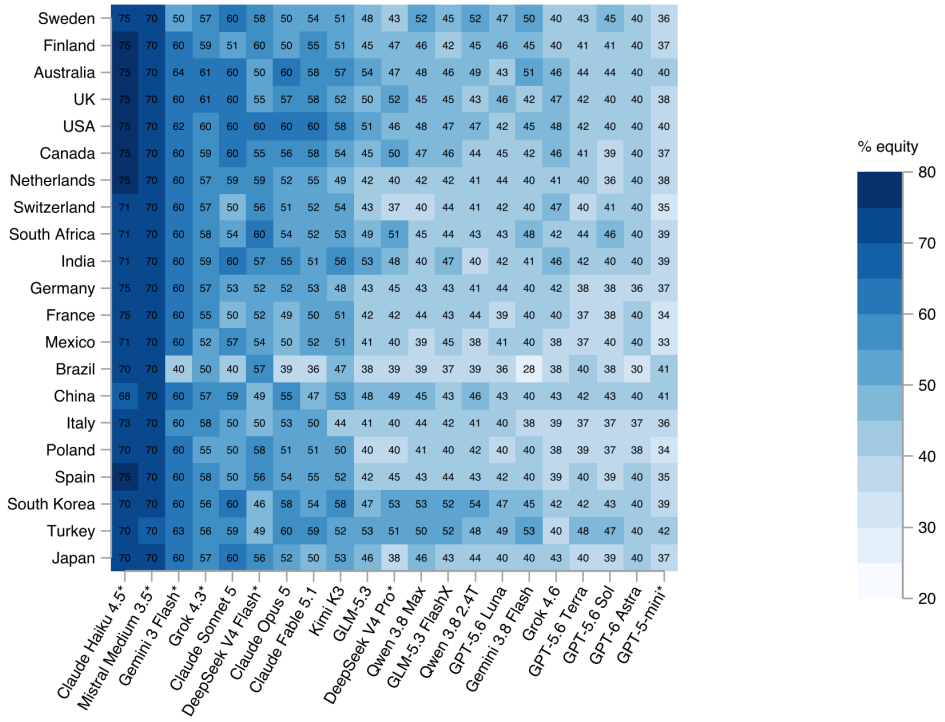
B Appendix: Figures

Figure A1: Age glide path by country (Question 1, Version B)



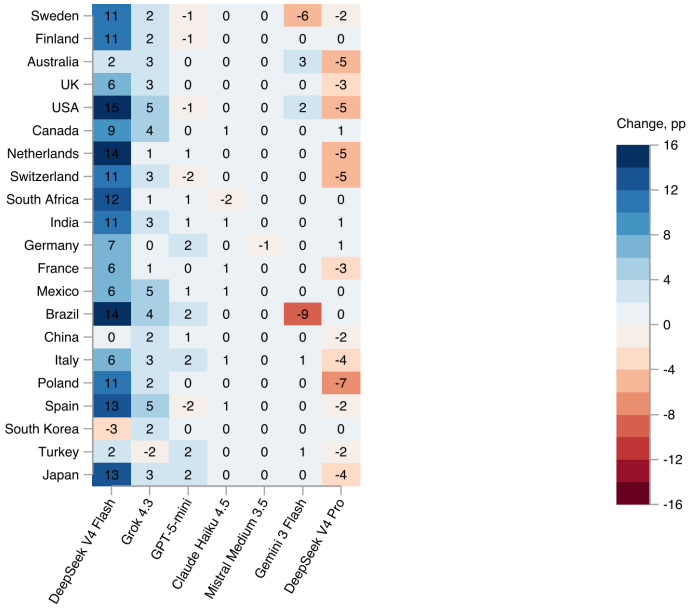
Notes: Mean recommended equity share by stated investor age (30, 40, or 55) within each country, Version B of Question 1, with 95 percent confidence intervals on the cell means. The downward glide path has nearly the same level and slope in every country, the raw pattern behind the null interaction of age with pension generosity in equation (4).

Figure A2: Advice by country and model, September 2026 tracking wave



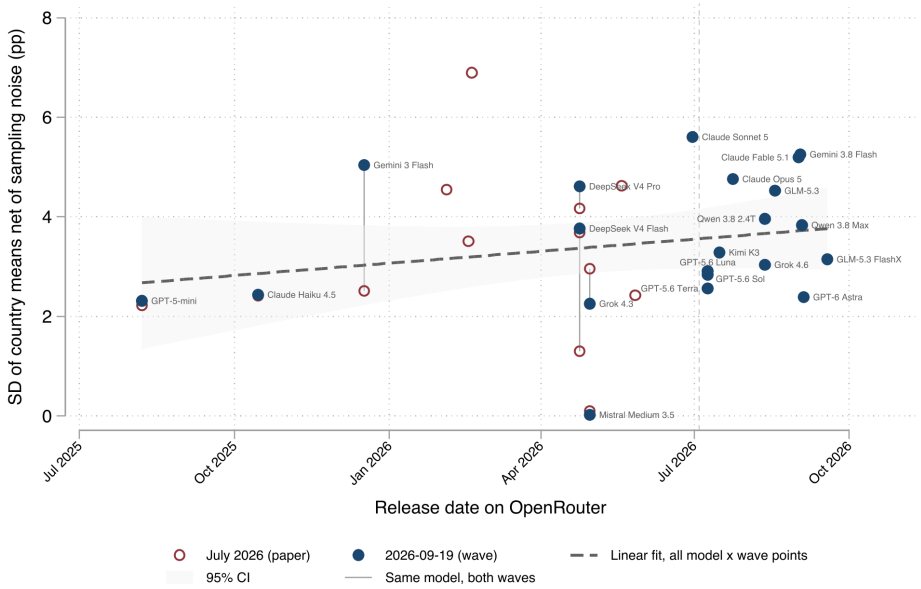
Notes: Mean recommended equity share in percent by country and model, Question 1, age 40, one English prompt per country, 25 repetitions per cell, collected September 19, 2026. Countries appear in the order of Figure 2 and models by their mean. An asterisk marks the seven models carried over from the main collection. Exploratory, outside the pre-registration.

Figure A3: Change in advice under an unchanged model identifier, July to September 2026



Notes: September 19 minus July 3–4, 2026 cell mean of the recommended equity share, in whole percentage points, for the seven models queried in both collections under the same identifier. Question 1, age 40, one English prompt per country. Models are ordered by their mean change. Exploratory, outside the pre-registration.

Figure A4: Model release date and the cross-country dispersion of advice



Notes: Each point is one model in one collection, plotting the standard deviation of its twenty-one country means net of sampling noise against its OpenRouter listing date. Hollow markers are the July collection and filled markers the September wave. Vertical segments join the two observations of each carried-over model. The dashed line is a linear fit over all 34 points with a 95 percent confidence band, and the vertical line marks the main collection. Appendix Table A17, Panel C reports the slopes. Exploratory, outside the pre-registration.

C The Normative Benchmark

This appendix develops the benchmarks summarized in Section 5. The benchmark delivers three empirical objects. First, a country-specific optimal equity share α_c^* against which the level of LLM advice can be compared. Second, a *calibration slope* that measures how much of the cross-country variation that should appear in the advice actually does appear, nesting the "U.S.-default" hypothesis (H2) as the case where the slope is zero. Third, a welfare metric that converts the gap between actual and optimal advice into a certainty-equivalent return loss, and into dollars on the $50,000 stake in the prompt. The model is deliberately static and closed form. A full life-cycle model in the tradition of Cocco et al. (2005) would add realism on several margins discussed in Section C.9, but at the cost of a large calibration apparatus (see e.g. Choukhmane et al., 2026). However, the central mechanism (background wealth crowding equity into the liquid portfolio) is captured in a simpler manner here.

C.1 The investor's problem

Consider the investor described in the prompt: age $a = 40$, liquid savings W (the local-currency equivalent of $50,000), choosing how to split W between a broad equity index fund and domestic government bonds. Let r_c denote the real return on the safe asset in country c, let the equity fund earn an expected real excess return $\mu_c - r_c$ with standard deviation σ_c, both measured in local currency, and let the investor have constant relative risk aversion (CRRA) preferences over terminal wealth with risk-aversion coefficient γ,

$$U(W_T) \;=\; \frac{W_T^{1-\gamma}}{1-\gamma}. \tag{8}$$

The CRRA assumption is the standard one in household finance and has a convenient implication used below: the investor cares about *proportional* gains and losses, so the optimal portfolio *share* does not depend on the level of wealth.

The classic result of Merton (1969) and Samuelson (1969) is that the optimal share of the portfolio held in equity is the rule of equation (5) in Section 5.1. Each of these variables maps to a country-level parameter that the data can help us discipline. The numerator is the compensation for holding equity risk: a higher local-currency equity premium raises the optimal share linearly, scaled by $1/(\gamma\,\sigma_c^2)$. The denominator is the cost of holding that risk: it rises with the variance of equity returns σ_c^2 (riskier markets warrant smaller positions) and with risk aversion γ (more risk-averse investors hold less equity). This is a convenient benchmark because of the result that the equity share is independent of the investment amount. In addition, if we assume that returns are independent over time, the equity share is also independent of the horizon. This latter result is convenient here: the prompt's five-year horizon does not require a separate model input. Time-varying expected returns would break this irrelevance, a point I return to in Section C.9.

Because the prompt restricts the answer to a fraction between zero and one, the benchmark is capped to the same interval,

$$\alpha_c^* \;=\; \min\{1,\; \max\{0,\; \theta_c^*\}\}, \tag{9}$$

which matters in practice for the integrated benchmark below.

C.2 Benchmark S: the stake as a standalone portfolio

The first benchmark takes the prompt at face value: the investor is asking about this $50,000 as a self-contained decision (the investor is "looking to invest $50,000 over a five-year horizon."). Under Benchmark S the optimal share is simply equation (5),

$$\alpha_c^S \;=\; \left[\frac{\mu_c - r_c}{\gamma_c\, \sigma_c^2}\right]_0^1, \tag{10}$$

where $[x]_0^1 \equiv \min\{1, \max\{0, x\}\}$ denotes truncation to the unit interval. Cross-country variation in α_c^S comes from two sources: local-currency equity volatility and, when preferences are allowed to vary, risk aversion. Under the world-index reading developed below, the expected excess return is common across currencies by uncovered interest parity, so it sets the level of every benchmark but not the spread across them. At the baseline $\bar{\gamma} = 3$, equation (10) produces interior optimal shares from about 0.21 in Turkey to 0.53 in South Korea, a band that overlaps the 0.46 to 0.58 range in which LLM recommendations fall, so deviations are informative in both directions. Choukhmane et al. (2026) estimate by simulated method of moments a coefficient of relative risk aversion of about 5 (5.2 under their survey prompts, 4.7 under the academic prompt) that rationalizes the advice of current-generation models, which corresponds to the $\gamma = 5$ point of the sensitivity grid, one step above the $\bar{\gamma} = 3$ baseline. At $\gamma = 2$ the U.S. benchmark rises to about 0.78, and at $\gamma = 10$ shares compress to between 6 and 16 percent across countries — below 8 percent for Turkey and Brazil — implausibly low relative to observed advice, which is itself useful information when reading the sensitivity tables.

How can we measure μ_c and σ_c in this context? The prompt says "a broad stock market index fund," which could mean the local equity market or a global fund held from country c. I take the world equity index measured in local currency as the baseline. This is a global representation of the market return, and because it is measured in local currency, the cross-country variation in α_c^S enters through local-currency volatility rather than through the equity market itself. This matches the instruments named in Version C (MSCI World trackers in most countries) and avoids attributing "bias" to any sensible reluctance to recommend concentrated exposure to a small home market. The home-market reading is reported as a robustness variant in Table A20 and in the calibration slope below.

With a world index and a common γ, essentially all cross-country variation in α_c^S comes from differences

in local-currency volatility, that is, from currency covariances and the hedged-versus-unhedged choice. The baseline uses unhedged local-currency moments, and Table A20 reports a currency-hedged variant alongside it. The equity premium is held common across countries at an excess return of about 5.3 percent, the arithmetic mean of the MSCI World index (net, in U.S. dollars, 1999–2025) over the U.S. long-term rate. For the U.S. cell the local-currency moments are an equity volatility near 18 percent and a real safe rate near 0.4 percent (trailing twenty-year mean), taken from MSCI and the Global Macro Database (Appendix Table A15). At the baseline $\bar{\gamma} = 3$ they imply an interior Merton share of about 0.52 and provide a calibration cross-check on α^S_{US}.

C.3 Benchmark I: adding pension wealth and human capital

The prompt also says "I have a standard pension plan otherwise," which invites the model to consider the investor's full balance sheet. The second benchmark therefore integrates two non-tradable assets that a 40-year-old holds outside the liquid portfolio: pension wealth P_c, the present value of future pension benefits, and human capital H_c, the present value of future labor income (Bodie, Merton and Samuelson, 1992; Campbell and Viceira, 2002; Gálvez and Paz-Pardo, 2026).

Specifically, the investor's total wealth is $W + H_c + P_c$, and by equation (5) she wants a fraction θ^*_c of *total* wealth exposed to equity. If a fraction ζ of human capital and a fraction ω of pension wealth are equity-like, the non-tradable assets already contribute $\zeta H_c + \omega P_c$ of equity exposure, and the entire remaining equity demand must be met inside the liquid account. The optimal share of the *liquid* portfolio is then

$$\alpha^I_c \;=\; \min\Big\{1,\; \max\Big\{0,\; \underbrace{\theta^*_c\,\frac{W + H_c + P_c}{W}}_{\text{scale up: equity demand on total wealth}} \;-\; \underbrace{\frac{\zeta\, H_c + \omega\, P_c}{W}}_{\text{scale down: equity held outside}} \Big\}\Big\}. \tag{11}$$

With $\zeta = \omega = 0$ this is the rule of Merton (1971) for an investor with riskless non-tradable income, in the form restated by Cocco et al. (2005, equation 12): the Merton fraction applies to total wealth, so the equity share of the liquid portfolio scales with total wealth over liquid wealth. The first term says that safe outside wealth makes the investor effectively less exposed to her liquid portfolio, so she should hold *more* equity in it. The more generous the pension system, the stronger this push. The second term works in the opposite direction when labor income co-moves with the stock market or when the pension itself is invested in stocks. In this setup, the equity-like parts of human capital and pension wealth are treated as perfectly correlated with the market and both are discounted at the safe rate. The baseline sets $\omega = 0$, a reasonable description of public and defined-benefit pillars, although treating pension claims as bond-like is itself contestable for pay-as-you-go systems whose benefits are indexed to wages. The exploratory probes of Section 6.1 state the pension's composition directly, which sets $\omega = 0$

for a guaranteed annuity and $\omega = 1$ for a pension invested in stocks.

The two balance-sheet inputs, pension wealth P_c and human capital H_c, have standard annuity expressions in the net replacement rate, the average wage, the retirement age, life expectancy, and real wage growth, with pension wealth discounted from retirement and human capital summed over the remaining working years. The two formulas, equations (12) and (13), and their calibration assumptions follow at the end of this subsection. As the persona in the prompt reveals no wage, the average earner is the natural calibration target.

Benchmark I delivers a sharp and useful prediction: because $H_c + P_c$ for an average 40-year-old is an order of magnitude larger than W, the multiplier in equation (11) pushes α_c^I to the corner of 100 percent equity in essentially every country. Under the calibration the corner is robust across the baseline region of the sensitivity grid: at $\zeta = 0$ it holds for all 21 countries through $\bar{\gamma} = 5$ and breaks at $\bar{\gamma} = 10$ only for four countries (Brazil, India, Mexico, South Africa). At $\zeta = 0.2$ it likewise holds everywhere through $\bar{\gamma} = 3$.

This benchmark has two important implications. First, to the extent LLMs reason in an integrated, full-balance-sheet way, their advice should be near 100 percent everywhere, and observed cross-country *variation* cannot be rationalized by Benchmark I at all. The integrated model converts any cross-country variation into prima facie evidence against normative calibration. Second, advice below α_c^S is conservative under the standalone reading and the baseline integrated calibration, and advice between α_c^S and one is rationalizable only with integrated reasoning. Because the capped share sits at one in every cell, the comparative statics that the stated-investor arm and the age test examine live in the *uncapped* index $\theta_c^*(W + H_c + P_c)/W - (\zeta H_c + \omega P_c)/W$. The index rises with a generous pension, falls when the pension is invested in stocks, and falls with age as human capital decays, faster where public pensions replace less. The country $\times$ age interaction in equation (4) and the stated-investor contrasts of Section 6 should be read against the ordering implied by that latent index rather than against the capped benchmark itself. Interior advice already rules out the capped benchmark in levels.

Making labor income riskier, as in Gálvez and Paz-Pardo (2026), moves some low-multiplier countries off the corner at equity loadings of human capital near one half, but the exit is knife-edge, because equation (11) differences two terms that are each an order of magnitude larger than the unit interval. I therefore do not use the high-ζ cases as the benchmark, and the extended grid is in the released code.

Pension-wealth and human-capital inputs. The two non-tradable balance-sheet inputs have simple annuity expressions. Pension wealth for an average earner is the replacement-rate-scaled wage stream

from retirement age R_c for the remaining life span L_c, discounted to age 40,

$$P_c \;=\; \rho_c\,\bar{w}_c\,(1+g_c)^{R_c-40}\,\frac{1-(1+r_c)^{-L_c}}{r_c}\,(1+r_c)^{-(R_c-40)}, \tag{12}$$

where ρ_c is the net pension replacement rate and $\bar{w}_c$ the average wage. The wage-growth factor keeps P_c consistent with equation (13), which assumes that replacement rates are defined relative to pre-retirement earnings rather than today's wage. Three further assumptions are that the annuity is an ordinary level annuity over L_c years whose factor tends to L_c as r_c approaches zero (relevant for Japan), that survival weighting is absorbed into L_c, and that the net replacement rate must be paired with a net wage measure when the calibration table is filled in.

Human capital is the discounted wage stream from now to retirement,

$$H_c \;=\; \sum_{t=1}^{R_c-40} \bar{w}_c\,(1+g_c)^t\,(1+r_c)^{-t}, \tag{13}$$

with g_c real wage growth.

C.4 Preferences: common versus locally calibrated risk aversion

I compute every benchmark under two preference regimes. Under the *common-preference* regime, $\gamma_c = \bar{\gamma}$ for all countries, with $\bar{\gamma} \in \{2, 3, 5, 10\}$ as the sensitivity grid. Cross-country variation in the benchmark then reflects institutions and asset markets only. Under the *preference-calibrated* regime, risk aversion varies with the country's measured willingness to take risks from the Global Preferences Survey (Falk et al., 2018). I map the GPS risk-taking index into risk aversion as

$$\gamma_c \;=\; \bar{\gamma}\,\exp\!\big(-\eta\,(z_c - z_{US})\big), \tag{14}$$

where z_c is the GPS risk-taking score, which is standardized at the individual level and aggregated to the country mean by Falk et al. (2018). $\eta \geq 0$ governs how strongly measured preferences translate into the benchmark. Concretely, η is the semi-elasticity of risk aversion with respect to the standardized GPS score ($d\ln\gamma_c/dz_c = -\eta$), so a one-standard-deviation increase in measured willingness to take risks scales risk aversion by the constant factor $e^{-\eta}$ wherever the country sits. The functional form keeps γ_c positive and makes the U.S. benchmark invariant to η.

The mapping is admittedly ad hoc, as there is no consensus conversion from survey risk measures to CRRA coefficients. I therefore treat η as a transparent sensitivity parameter on a grid such as $\{0, 0.3, 0.6\}$, where $\eta = 0$ recovers the common-preference regime exactly. At $\eta = 0.6$ the implied U.S.–Japan risk-aversion ratio is roughly 1.3, a moderate magnitude relative to documented preference

heterogeneity. The comparison then shows which benchmark the LLM advice tracks: if the advice tracks the preference-calibrated benchmark but not the common-preference one, models are responding to cultural risk attitudes.

C.5 Taking the benchmark to the data

The benchmark is computed from external country-level inputs and then confronted with the experimental estimates using three measures: the advice gap, the calibration slope and the welfare cost of the gaps. Appendix Table A15 reports every country-specific input the benchmark requires, with the common parameters and sources in the table notes. It lists the eight country-varying portfolio inputs, the equity premium being common across countries, the two mortgage-question inputs of deductibility and expected inflation, and the resulting standalone benchmark share α_c^S. Three common parameters complete the calibration: baseline risk aversion $\bar{\gamma}$ (grid $\{2, 3, 5, 10\}$), the preference-mapping intensity η (grid $\{0, 0.3, 0.6\}$), and the equity loading of human capital ζ (baseline 0, robustness 0.2).

Advice gaps. For each country, the gap between mean LLM advice and the benchmark,

$$\Delta_c \;=\; \bar{y}_c - \alpha_c^S, \tag{15}$$

where $\bar{y}_c$ is the mean recommended fraction in the primary cell (Version B, Q1, age 40). The gap Δ_c is a descriptive level comparison. The wild-cluster-bootstrap confidence interval inherited from the country fixed effects describes the advice deviation from the U.S. baseline, not the level gap itself.

The calibration slope. The next measure examines whether cross-country *movements* in the benchmark show up in the advice. Let $\hat{b}_c$ denote the estimated country fixed effect from equation (1) (the LLM advice deviation from the U.S. baseline, written γ_c there but renamed here to avoid collision with risk aversion) and let $\delta_c^* = \alpha_c^S - \alpha_{US}^S$ denote the benchmark's prescribed deviation. The regression

$$\hat{b}_c \;=\; \kappa_0 + \kappa\, \delta_c^* + e_c \tag{16}$$

summarizes calibration in a single number. Full normative calibration implies $\kappa = 1$: advice moves one for one with what local fundamentals prescribe. The U.S.-default hypothesis implies $\kappa = 0$. Note that the converse does not hold, since large country effects orthogonal to fundamentals also produce a zero slope. I therefore report the gap table and the slope together. Intermediate values measure partial calibration.

Welfare cost of the gaps. CRRA preferences yield a closed-form price tag for deviating from the optimum. The certainty-equivalent return of holding share α is quadratic, $ce(\alpha) = r_c + \alpha(\mu_c - r_c) -$

$\frac{1}{2}\gamma_c\alpha^2\sigma_c^2$, so the annual loss from holding α instead of the benchmark is computed directly as

$$\mathcal{L}_c \;=\; ce(\alpha_c^*) - ce(\alpha) \;=\; \tfrac{1}{2}\,\gamma_c\,\sigma_c^2\left[\left(\theta_c^* - \alpha\right)^2 - \left(\theta_c^* - \alpha_c^*\right)^2\right], \tag{17}$$

which reduces to the familiar $\frac{1}{2}\gamma_c\sigma_c^2(\alpha-\alpha_c^*)^2$ whenever the benchmark is interior ($\alpha_c^* = \theta_c^*$). This formula remains correct when the cap binds, as it does throughout Benchmark I, where the simpler expression would understate losses materially.

Two features are worth emphasizing when reading the results. The loss is locally quadratic, so small advice gaps are nearly free while large ones are disproportionately costly. Around an interior optimum it is symmetric, so excessive conservatism and excessive aggression are priced on the same scale. Evaluated at the prompt's stake, the dollar cost over the five-year horizon is approximately $W\left[(1+\mathcal{L}_c)^5-1\right] \approx 5\,W\mathcal{L}_c$. At plausible calibrations a 10 percentage point advice gap costs single-digit basis points per year, on the order of $100 over five years on $50,000. This is still a relatively small number, and hence the more informative welfare object is the cross-country *incidence* of losses rather than their absolute size.

C.6 Advice gaps and the calibration slope

The benchmark prescribes large differences in advice across this panel. At the baseline grid point ($\bar{\gamma} = 3$, $\eta = 0$), the standalone benchmark α_c^S ranges from 0.21 in Turkey and 0.23 in Brazil, whose local-currency volatilities are the highest in the panel, to 0.53 in South Korea and 0.52 in the United States, a prescribed spread of thirty-two percentage points. Mean advice spans twelve points, from 0.46 to 0.58 (Table 1), so the advice covers roughly a third of the spread that local fundamentals warrant. In addition, Turkey, whose prescribed share is the lowest in the panel, receives mean advice of 0.54, statistically indistinguishable from the U.S. baseline and near the top of the realized band.

Table A20 reports the per-country results. The gaps are positive nearly everywhere, and they are largest where fundamentals depart most from the U.S. baseline: 33 percentage points in Turkey, 23 in Brazil, 17 in Mexico, and 15 in Japan and Poland. Across most of western Europe, whose prescribed shares sit close to the U.S. value, the gaps are within a few points of zero, with France exact to within a fifth of a point.

The hedged columns of Table A20 replace the local-currency volatility with the U.S.-dollar volatility of the index, which a currency-hedged world fund delivers in every currency to first order. With the premium common across countries, the hedged benchmark is 0.52 in every country, and mean advice lies within 6.2 percentage points of it everywhere, with Brazil furthest below. The thirty-two-point spread of the unhedged benchmark, and with it every non-zero prescribed deviation δ_c^* below, therefore comes from exchange-rate risk alone.

Table A20: Advice gaps against the standalone benchmark and their welfare cost

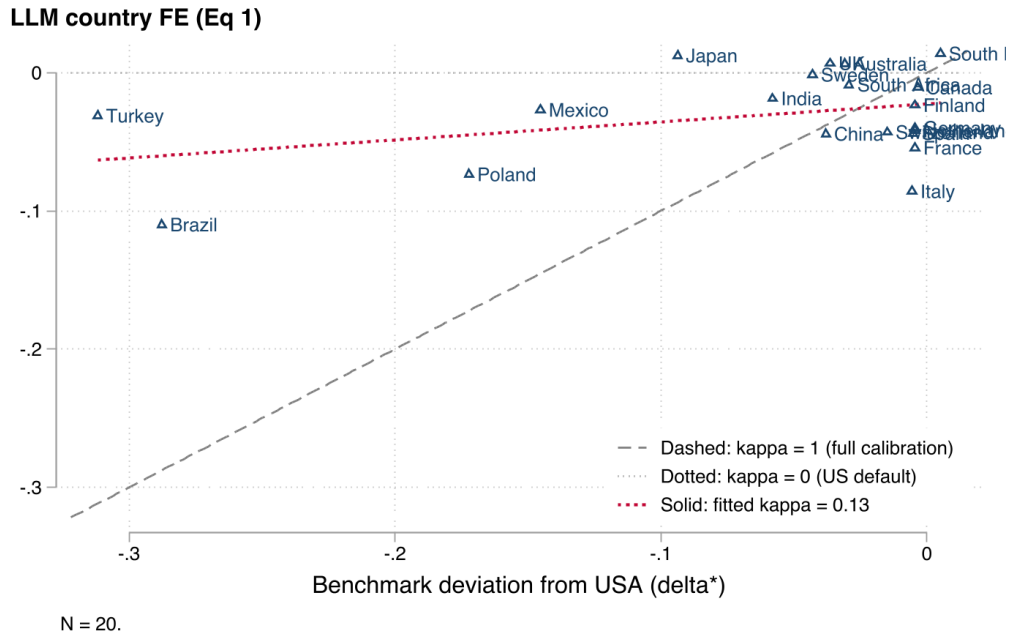
	Mean	Unhedged		Hedged		Home index		Loss	$ over
Country	advice	α_c^S	Gap Δ_c	α_c^S	Gap	α_c^S	Gap	(bp/yr)	5y
Turkey	0.54	0.21	0.33	0.52	0.02	0.21	0.33	136.7	3,513
Brazil	0.46	0.23	0.23	0.52	−0.06	0.38	0.08	57.8	1,461
Mexico	0.54	0.38	0.17	0.52	0.02	—	—	19.5	489
Poland	0.50	0.35	0.15	0.52	−0.03	0.14	0.35	16.3	408
Japan	0.58	0.43	0.15	0.52	0.06	0.51	0.07	14.7	368
United Kingdom	0.58	0.49	0.09	0.52	0.05	0.57	0.00	4.5	113
Sweden	0.57	0.48	0.09	0.52	0.05	0.44	0.13	4.4	110
India	0.55	0.46	0.09	0.52	0.03	0.38	0.17	4.3	109
Australia	0.58	0.49	0.08	0.52	0.05	0.74	−0.17	3.9	98
South Africa	0.56	0.49	0.07	0.52	0.04	0.58	−0.02	2.5	63
South Korea	0.58	0.53	0.06	0.52	0.06	0.42	0.16	1.6	40
USA	0.57	0.52	0.05	0.52	0.05	0.68	−0.11	1.2	29
China	0.52	0.48	0.04	0.52	0.00	—	—	0.9	23
Canada	0.56	0.52	0.04	0.52	0.04	0.68	−0.12	0.8	21
Italy	0.48	0.52	−0.03	0.52	−0.04	0.13	0.35	0.5	14
Finland	0.55	0.52	0.03	0.52	0.02	0.08	0.47	0.4	11
Switzerland	0.53	0.51	0.02	0.52	0.00	0.46	0.06	0.2	5
Germany	0.53	0.52	0.01	0.52	0.01	0.35	0.18	0.1	2
Netherlands	0.53	0.52	0.01	0.52	0.01	0.43	0.09	0.1	1
Spain	0.53	0.52	0.01	0.52	0.00	0.34	0.19	0.0	1
France	0.52	0.52	−0.00	0.52	−0.01	0.31	0.20	0.0	0

Notes: Mean recommended equity share by country in the primary cell (Question 1, Version B, age 40) against the standalone benchmark α_c^S at the baseline grid point ($\bar{\gamma} = 3$, $\eta = 0$, world index in local currency). The gap is $\Delta_c = \bar{y}_c - \alpha_c^S$. The hedged columns replace σ_c with the U.S.-dollar volatility of the index, which a currency-hedged world fund carries in every currency to first order, and report the benchmark and the gap under that assumption. The home-index columns replace both moments with those of the country's own MSCI country index in local currency (net total return, 2001–2025, premium over the Global Macro Database long rate) at the same γ_c, an exploratory post-registration variant. A dash marks the two countries without a country series (China, Mexico). The annual loss prices the unhedged gap with the certainty-equivalent formula (17) at the same grid point, in basis points of the liquid stake per year, and the last column converts it to dollars over the prompt's five-year horizon on the $50,000-equivalent stake. Countries are ordered by annual loss.

The home-index columns take the opposite reading of the prompt's "broad stock market index fund": each country's own MSCI country index in local currency, with the premium and the volatility estimated over 2001 to 2025 and the same γ_c. Where the hedged benchmark removes the cross-country spread, the home-index benchmark widens it. The prescribed share runs from 0.08 in Finland and 0.13 in Italy, whose realized premia over the window are below two percent, to 0.74 in Australia and 0.68 in the United States, a spread of sixty-six points. The advice does not follow. Regressing the country effects on the home-index deviations over the eighteen non-U.S. countries with a series gives a slope of 0.11 (HC2 standard error 0.03, $R^2 = 0.33$). Full calibration is rejected ($p < 0.001$), and at this tighter fit the U.S.-default anchor is rejected as well ($p = 0.01$). The world-index slope on the same eighteen countries is 0.14. Under either reading the advice transmits about a tenth of the cross-country variation the benchmark prescribes. Realized twenty-five-year premia are noisy inputs, which is why the baseline uses one global premium, so the home-index variant is a check on that choice rather than a competing benchmark.

Figure A5 plots the estimated country effects against the benchmark's prescribed deviations and fits the calibration regression (16). The slope is $\hat{\kappa} = 0.130$ (standard error 0.112). The primary wild bootstrap

Figure A5: The calibration slope: advice deviations against prescribed deviations



Notes: Estimated country fixed effects $\hat{b}_c$ from equation (1) plotted against the benchmark's prescribed deviations $\delta_c^* = \alpha_c^S - \alpha_{US}^S$ at the baseline grid point ($\bar{\gamma} = 3$, $\eta = 0$), for the twenty non-U.S. countries. The dashed 45-degree line is full calibration ($\kappa = 1$), the dotted horizontal line is the U.S.-default anchor ($\kappa = 0$), and the solid line is the fitted regression (16) with slope $\hat{\kappa} = 0.13$. Inference of record is the wild bootstrap over countries reported in the text.

over countries rejects full calibration decisively ($p < 0.001$ against $\kappa = 1$) and does not reject the U.S.-default anchor ($p = 0.317$ against $\kappa = 0$, 95 percent confidence interval $[-0.145, 0.328]$). Taken at face value, the advice transmits about an eighth of the cross-country variation that local fundamentals prescribe.

The pre-registered influence diagnostics flag two of the seven countries with thin benchmark data. Brazil and Turkey, whose currencies are the most volatile in the panel, have DFBETA values on the slope above the conventional threshold of $2/\sqrt{N}$ in absolute value. Dropping Brazil lowers $\hat{\kappa}$ to 0.03, dropping Turkey raises it to 0.21, and dropping any other country leaves it between 0.11 and 0.15. No leave-one-out estimate comes close to full calibration, but the point estimate depends on which of the two high-volatility countries is in the sample.

The slope's distance from full calibration also survives treating the benchmark regressor as measured with error, the concern that historical parameter uncertainty in δ_c^* attenuates $\hat{\kappa}$ toward zero. Because δ_c^* is a deterministic function of the local-currency return volatility σ_c, estimated from the same twenty-seven annual return years (1999–2025), classical measurement error is the right frame. Attenuation biases the forward regression down, so the structural slope is bracketed between $\hat{\kappa} = 0.13$ and the reciprocal of the reverse regression of δ_c^* on the country effects. That Frisch bound is $[0.13, 0.97]$: even its upper end, which attributes the entire off-line scatter of the country effects to mismeasured fundamentals, stays

below the unit slope of full calibration. The bracket is wide because the advice barely moves. The estimated country effects span twelve percentage points against the thirty-two the benchmark prescribes ($R^2 \approx 0.14$), so the distance from full calibration is a fact about the advice rather than an artifact of a precisely measured regressor. A complementary check isolates the volatility-sampling channel directly: resampling the twenty-seven return years with replacement and recomputing every δ_c^* on each draw, holding the year set common across countries so the shared world-index return cancels in the difference, shifts the slope only from 0.13 to a resampled center of 0.10. That exercise prices in volatility sampling error alone. Uncertainty in the common equity premium enters δ_c^* through the ratio $(\mu - r)/\gamma$ and so maps onto the common risk-aversion grid of the next paragraph, where full calibration is rejected at every grid point whose prescribed levels overlap the realized advice.

The slope is also not an artifact of the baseline grid point. Across the full $(\bar{\gamma}, \eta)$ grid, $\hat{\kappa}$ ranges from 0.07 to 0.43 and is never near one. The rise with $\bar{\gamma}$ is mechanical, since higher common risk aversion compresses the prescribed deviations that form the regressor. At $\bar{\gamma} = 10$ the point estimate reaches 0.43 with an interval too wide to reject full calibration, but that grid point prescribes equity shares below 16 percent in every country (as low as 6 percent in Turkey and Brazil), implausibly far below any advice observed in the data (Section C.2). At every grid point whose prescribed levels overlap the realized advice, $\hat{\kappa}$ is at most 0.22 and full calibration is rejected with $p < 0.001$. Allowing risk aversion to vary with measured preferences does not raise the slope either: at $\bar{\gamma} = 3$ the preference-calibrated values are 0.12 at $\eta = 0.3$ and 0.09 at $\eta = 0.6$, slightly below the common-preference estimate.

C.7 The welfare cost of the country gaps

For the median country the gap costs 1.6 basis points of the liquid stake per year, about $40 over the prompt's five-year horizon on the $50,000-equivalent stake, and sixteen of the twenty-one countries lie below five basis points per year (Table A20). The loss is locally quadratic, so the moderate gaps that dominate the panel are nearly free. This is the sense in which the compression finding, taken on its own, is individually cheap.

However, the incidence of these losses differs widely. The countries where the gap is expensive are those whose prescribed shares sit farthest below the U.S. value. Turkey pays 137 basis points per year, about $3,500 over five years or seven percent of the stake, Brazil pays 58 basis points, Mexico 19, Poland 16, and Japan 15, while no western European country pays more than five. Of course, these figures assume the user adopts the advice, that the standalone benchmark is the right target, and that the allocation is static, so they price the advice rather than realized behavior. They also rest on the unhedged benchmark, which treats local government bonds as riskless in real terms. In high-inflation economies such as Turkey and Brazil local bonds carry substantial real risk and an unhedged foreign equity fund partly hedges it,

so the benchmark likely understates the optimal equity share there and the losses for these countries are upper bounds. Under the hedged benchmark the loss is below two basis points a year in every country.

C.8 The mortgage benchmark

This subsection derives the benchmark of Section 5.3 for Question 2 (invest versus prepay a 4 percent mortgage). Prepaying a mortgage is a risk-free investment whose return is the after-tax mortgage rate. Because every return in this section is real, the prompt's quoted nominal rate m must be deflated, so the relevant safe-asset return is

$$r_c^m \;=\; m\,(1-\tau_c^m)-\pi_c,$$

where τ_c^m is the share of mortgage interest that is effectively deductible in country c and π_c is expected inflation.

Two country-specific channels operate here. Deductibility differs sharply across the sample (broadly deductible in the United States for itemizers, capital-income offset in Sweden, no owner-occupier deduction in Germany), and expected inflation differs too, so a fixed nominal mortgage is a better real investment precisely where inflation is low, as in Japan. The Merton logic then applies with r_c^m as the safe rate,

$$\alpha_c^{Q2} \;=\; \min\Big\{1,\; \max\Big\{0,\; \frac{\mu_c - r_c^m}{\gamma_c\,\sigma_c^2}\Big\}\Big\}, \tag{18}$$

with the corner at zero now economically relevant: full prepayment is optimal wherever the expected equity return μ_c falls below the after-tax real mortgage rate r_c^m. Cross-country differences in Q2 advice are therefore a gap only *net of* the cross-country differences in α_c^{Q2}, which come from inflation, the local real rate, and currency volatility as well as from deductibility. Because effective deductibility depends on income, itemization, and loan size rather than on a single statutory rate, I treat τ_c^m as a calibration assumption rather than a point-sourced number and report α_c^{Q2} across a $\tau^m \in [0, 0.5]$ band. The baseline sets τ_c^m to zero in thirteen countries and to between 0.05 and 0.35 in the other eight (Table A15). Widening the band from no deductibility to a fifty percent effective deduction raises α_c^{Q2} by at most about twenty percentage points, too little to overturn the comparison that follows, which places the advice above the benchmark in nearly every country even at the equity-favorable end of the band.

C.9 What the simple model leaves out

I note five omissions from the simple model and the likely direction of each. First, the model is static, so with mean-reverting returns longer horizons would justify higher equity shares for the younger investors

Table A21: The mortgage question: advice against the country-specific benchmark

Country	Mean advice	Benchmark α_c^{Q2}	Band ($\tau^m \in [0, 0.5]$)	Gap Δ_c^{Q2}
USA	0.83	0.47	[0.41, 0.61]	0.36
Japan	0.54	0.20	[0.16, 0.32]	0.34
France	0.67	0.33	[0.33, 0.53]	0.34
Sweden	0.73	0.39	[0.28, 0.46]	0.34
Switzerland	0.61	0.27	[0.20, 0.39]	0.34
Canada	0.70	0.37	[0.37, 0.56]	0.33
Germany	0.61	0.28	[0.28, 0.48]	0.33
United Kingdom	0.68	0.35	[0.35, 0.54]	0.33
Finland	0.63	0.32	[0.32, 0.51]	0.32
Poland	0.66	0.38	[0.38, 0.51]	0.29
Netherlands	0.67	0.44	[0.31, 0.50]	0.22
Spain	0.62	0.41	[0.41, 0.60]	0.21
Australia	0.65	0.45	[0.45, 0.64]	0.20
Italy	0.63	0.45	[0.45, 0.64]	0.18
South Korea	0.65	0.47	[0.45, 0.65]	0.18
China	0.60	0.47	[0.45, 0.63]	0.13
Brazil	0.81	0.34	[0.34, 0.42]	0.47
Mexico	0.81	0.61	[0.61, 0.75]	0.20
Turkey	0.81	0.65	[0.65, 0.73]	0.16
India	0.79	0.79	[0.76, 0.93]	−0.00
South Africa	0.80	0.98	[0.98, 1.00]	−0.17

Notes: Mean recommended index-fund share on the mortgage question (Question 2, Version B, age 40) against the country-specific benchmark α_c^{Q2} from Equation (18) at the baseline grid point ($\bar{\gamma} = 3$, $\eta = 0$), which uses the after-tax real mortgage rate as the safe return. The band reports α_c^{Q2} at the two ends of the effective-deductibility range, $\tau^m = 0$ and $\tau^m = 0.5$. The gap is $\Delta_c^{Q2} = \bar{y}_c^{Q2} - \alpha_c^{Q2}$, so a positive value means the model recommends more equity, and hence less prepayment, than the benchmark. Countries above the rule form the calibration group, ordered by gap. The five below form the premise-acceptance group of Section 3, whose α_c^{Q2} assumes the prompt's counterfactual 4 percent rate, so their gaps are read as premise-acceptance rather than calibration tests. $N \approx 650$ per country.

(Campbell and Viceira, 2002). Second, labor income risk is reduced to a single loading ζ, which lowers α_c^I but, in the baseline calibration, does not move it off the corner for a stake this small relative to total wealth. Only the high-ζ cases of Section C.3 do. Third, taxes on investment returns and housing wealth are excluded from the portfolio benchmarks. Fourth, the benchmark conditions on an average earner, so any country where $50,000-equivalent savings implies a very different position in the wage distribution (Brazil again) inherits calibration error in H_c and P_c. Fifth, the benchmark treats local government bonds as riskless in real terms, which overstates the case for bonds in high-inflation economies (Section C.7). Each of these would let the advice match a richer benchmark than α_c^S, but the stated-investor arm of Section 6 shows that the models act on stated balance-sheet information mainly when it signals that the investor could not absorb a loss, as with a pension invested in stocks or a stake that is all of her liquid wealth, and not in the two-sided way these richer benchmarks would require.

None of the first four caveats enters the computation of Benchmark S, which requires only return moments and preferences. Reassuringly, the one existing full life-cycle treatment of LLM advice reaches the same corner as Benchmark I: Choukhmane et al. (2026) find that a calibrated Cocco et al. (2005) model implies near-universal equity holding for young and middle-aged average earners, and that it cannot rationalize

the more conservative shares LLMs actually recommend early in life.

D The September 2026 Model-Tracking Wave

The final robustness check of Section 4.6 is exploratory and outside the pre-registration. It asks whether the results outlast the models that produced them, since the main collection is a snapshot of the models available in early July 2026. After examining the pre-registered results, I re-asked one slice of the primary cell on September 19, 2026 to a panel of twenty-one models. Fourteen of them came out after the main collection. The other seven carry over from the original panel under unchanged model identifiers, which separates the arrival of new models from changes in the model behind a fixed identifier. The slice keeps Question 1 at age 40 and uses one English prompt per country, Version B in the six English-prompting countries and Version B-cross in the other fifteen, with temperature 0.7 and 25 repetitions per cell. Table A17 and Figures A2 to A4 report the results.

Both main patterns of the July collection hold in the September wave. Model identity accounts for 70 percent of the variance and country for 5 percent, against 78 and 3 percent on the same English slice of the July collection.[14] Most countries still receive less equity than the U.S. within each model, with 77 percent of non-U.S. cells below the model's own U.S. cell against 67 percent in July (Figure A2).

The fourteen new models differ from the July panel in two respects. First, country matters more relative to model. Among the new models, model identity accounts for 43 percent of the variance and country for 15 percent. This reflects the composition of the panel rather than the collection date, since the seven carried-over models give country a 1.5 percent share in both waves. Figure A4 plots each model's cross-country dispersion against its release date. The noise-net standard deviation of a model's country means rises by 0.98 percentage points per year of release date, a slope the wild cluster bootstrap does not distinguish from zero ($p = 0.19$, Table A17, Panel C). Second, the new models agree more with one another on the level of advice, with model means between 39 and 55 percent against 36 to 73 percent in July. Their country means span 13 percentage points, from Brazil at 38 percent to Turkey at 51 percent, but the larger role of country does not bring the advice closer to the unhedged benchmark of Section 5. Turkey, where that benchmark prescribes the lowest equity share in the panel, receives the most equity.

The carried-over models also show that the advice behind an unchanged identifier can move. Four of the seven return the same average advice within half a percentage point after eleven weeks. Gemini 3 Flash keeps its average but lowers its Brazil recommendation by 9 points, and Grok 4.3 raises its average by 2.4 points (Table A17, Panel B, and Figure A3). The largest change, an 8.5-point rise for DeepSeek V4 Flash, comes from routing rather than from the model. OpenRouter sent 79 percent of its September queries to hosts that run it with the reasoning step switched off, against 28 percent in July, and these hosts

[14] These shares differ from the 69.5 and 5.7 percent of Section 4.1 only because the tracking slice replaces the local-language Version B prompt with its English counterpart in fifteen countries. Asking in the local language roughly doubles the country share, so the English slice is a conservative test for country effects.

recommend about 58 percent equity in both waves, against about 40 percent on hosts that keep the reasoning step (Section 4.6). Holding the hosts at their July shares, the September mean lies within a point of the July mean. DeepSeek V4 Pro's 2.3-point fall partly reflects the same mechanism in reverse, as its share of such hosts fell from 13 percent to under 1 percent. The Gemini shift is as large as the country contrasts of Section 4.2, which range from 2.7 to 11 points. These shifts are consistent with the within-run monitor of Section 3.7, which found no identifier rotation during the two-day main collection, because they occur between collections rather than within one. Replications of LLM-advice experiments therefore need to report the collection date alongside the model name, and for open-weight models the host that served each response.

E Technical Appendix: Translation, Collection, and Parsing

This appendix documents the execution machinery behind Sections 3 and 3.7: the translation pipeline and its quality controls, the collection infrastructure, and the parsing, refusal-detection, and validation steps. The replication package contains the pipeline code, the full prompt texts for every country, version, and question, and the translation review and provenance records.

E.1 Prompt Structure and Version Design

Each prompt consists of two blocks. The *Instructions* block specifies the JSON schema and the word limit on the explanation and is always written in English, regardless of the country/language of the rest of the prompt. The *Question* block contains the investor's stated demographics and the financial question itself and is the only part of the prompt that varies across versions. Keeping the Instructions block in English is a deliberate design choice validated in pilot: it ensures that JSON parsing succeeds at near-identical rates across all fourteen languages (thirteen non-English languages plus English, with Swiss German and Mexican Spanish as within-language variants) and removes a potential confound between language treatment and response-format compliance.

E.2 Translation pipeline and quality control

Version B and Version C question blocks were produced with the DeepL API through an automated pipeline, while Version B-cross is the original English prompt and required no translation. The pipeline translates only the Question block, never the Instructions block, protects the pinned financial terms and the locked currency amounts through the API's term-protection markup so that they survive translation verbatim, requests the formal register where the target language supports it, and appends every translation call with its raw response to a provenance log. Ages were translated as concrete values rather than placeholders after live testing showed that a protected {`age`} placeholder derailed the translation (a Swedish block came back reading "my name is {age} years"). The per-age Question 1 blocks were then collapsed, after all corrections had been applied, to a single canonical block per country and version with the age digit substituted mechanically at query time, so that the age-30, age-40, and age-55 prompts within a cell differ only in the stated age.

Machine output passed through two automated checks with documented failure modes. A pin-survival check verifies that every protected term and amount appears verbatim in the translated block, and it caught three systematic failures in which the translator dropped a locked currency amount, rendered an amount as the name of a financial product, or dropped a pension term. A complete back-translation of every unique question block to English was scored with smoothed sentence BLEU against the source, with 0.6 as the flagging threshold. The round-trip score proved too weak to serve as an acceptance gate.

It penalizes faithful translations into linguistically distant languages, flagging every Japanese, Korean, and Chinese block, and it passed at least one genuinely wrong translation, so it was used to route blocks to human review rather than to clear them.

The operative gate was native-speaker review of the Version B and Version C blocks in all fifteen translated countries, conducted in waves and prioritized where the financial vocabulary diverges most from the generic European templates (Chinese, Korean, Japanese, Turkish, and Brazilian Portuguese), where machine translation is weakest (Finnish, the panel's only Finno-Ugric language), and for the Swiss German and Mexican Spanish variants, with later waves covering the remaining European languages, Swedish (checked by the author, a native speaker), and the local-instrument wording of the English-prompting India cell. South Africa, the one country with no available native reviewer, was instead verified through a documented two-vendor LLM cross-check at temperature zero with a pre-stated escalation rule, which surfaced one instrument-naming error that was corrected through a dated design-lock amendment and re-verified on both vendors. The same cross-check was run on the UK, Australia, and Canada blocks and re-run on Mexico and Spain after their native reviews, with vendor suggestions adopted only where they did not contradict a documented native-reviewer decision. Reviewer corrections were not applied by hand to the translated files. They were recorded in a persistent override layer that the pipeline re-applies after any re-run, with the term-protection check re-executed on every corrected block, so that a later re-translation cannot silently overwrite a reviewer's fix. Before the pre-registration was filed, the complete grid passed a deterministic consistency audit of every assembled prompt with zero failures, an adversarial model-based review of all ninety unique translated blocks, and a full human read of every block, and the translations were then frozen for collection.

E.3 Collection infrastructure

The collection runner submits queries through the OpenRouter API with the design parameters of Section 3 and is built so that an interruption cannot produce a duplicate analysis record. Two append-only logs record every query. The parsed results file is the leading log: for each response the runner writes and flushes the parsed row first and the raw API response afterwards. On restart the runner reads the identifiers of completed queries from the parsed results file and submits only the remainder, so no already-recorded observation is ever queried twice. Two edge cases remain and are benign. A crash after a response returns but before its parsed row is flushed leaves that single in-flight query unrecorded, so it is re-queried on restart, at most one repeated API call per interruption. A crash between the parsed-row write and the raw-response write leaves that one recorded observation without its raw audit copy, without affecting the parsed data used in the analysis. Requests were paced below provider rate limits, retried automatically in a way that honors the provider's retry-after signal, and distributed across worker threads at a concurrency validated in throughput probes that produced no rate-limit rejections.

The main run finished with exact coverage of the design grid, every cell reaching its fifty repetitions with no duplicates and no unresolved API errors.

E.4 Parsing, refusal detection, and validation

Parsing proceeds in a fixed order of decreasing strictness. The parser first extracts the first balanced brace block in the response and attempts to read it as JSON against the schema stated in the Instructions block. If that fails, it searches for a decimal fraction, and then for a percentage, in each case only where the number appears in explicit allocation language, so that a stray number elsewhere in the response cannot be mistaken for a recommendation. The strategy that produced each observation is recorded in a parse-method field, which allows any analysis to be re-run on the strictly JSON-parsed subsample. Responses whose generation stopped at the 4,000-token ceiling are marked truncated and excluded under the pre-registered rule.

Refusal detection runs after parsing, not before. A response is classified as a refusal only when a refusal phrase matches and no numeric recommendation was parsed, which prevents advisory boilerplate such as "consult a financial advisor" on an otherwise valid answer from being misclassified. The phrase list covers all fourteen prompt languages (thirteen non-English languages plus English, with Swiss German and Mexican Spanish handled as within-language variants) and was extended at translation time as each language entered the design. The main collection produced no refusals.

Model identity is monitored at the level of the individual response. The dated model versions behind each requested identifier were pinned before launch, and every row records both the requested and the returned identifier. In the main collection, every mismatch between the two was the provider's resolution of a version alias to its dated snapshot, a one-to-one mapping that was stable across the collection window, so no model rotated versions mid-run. Staging and validation precede any estimation. A staging script reduces the raw collection to the analysis columns and verifies the absence of duplicate query identifiers, and a validation stage recomputes cell counts against the design grid, parse and refusal rates by cell, and the requested-to-returned identifier map before the first regression runs.